Welfare Effects of Buyer and Seller Power

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Abstract

We examine the welfare effects of buyer and seller power in vertical relations and introduce an empirical approach for quantifying the contributions of each to the welfare losses from market power. Our model accommodates both monopsony distortions from buyer power and double-marginalization distortions from seller power. Rather than imposing one of these vertical distortions by assumption, we let them be determined based on model primitives. We show that the relative elasticity of upstream supply and downstream demand is the key determinant of whether buyer or seller power creates a market power distortion. Applying our framework to coal procurement by power plants in Texas, we attribute 74.9% of the distortion to monopoly power of coal mines, with the remainder attributed to the monopsony power of power plants.

Keywords: Monopsony, Market Power, Vertical Relations, Bargaining
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1 Introduction

There is a growing interest in the buyer power of firms, both in labor markets (Card, Cardoso, Heining, and Kline, 2018; Berger, Herkenhoff, and Mongey, 2022; Lamadon, Mogstad, and Setzler, 2022) and in vertically-related industries (Grennan, 2013; Gowrisankaran, Nevo, and Town, 2015; Rubens, 2023). This attention is mirrored in policy circles; for instance, the Department of Justice (DOJ) has challenged mergers on the grounds of monopsony concerns (DOJ, 2022), and tackling labor market power has come to the forefront of economic policy-making (The White House, 2023; DOJ and FTC, 2023).

However, the welfare implications of buyer power vary drastically across different vertical models used in the empirical bargaining literature, which has largely relied on the 'Nash-in-Nash' bargaining protocol (Horn and Wolinsky, 1988; Collard-Wexler, Gowrisankaran, and Lee, 2019). A first class of models assumes that input prices are negotiated, while equilibrium lies on an input demand curve, because the downstream party controls how much input to buy as in Spengler (1950)'s sequential monopoly framework (Abowd and Lemieux, 1993; Crawford and Yurukoglu, 2012; Ho and Lee, 2019). In these models, *seller power* creates vertical distortions by generating upstream *markups*, and buyer power can countervail these distortions. In contrast, a second class of models features bargaining while equilibrium lies on an input supply curve as in Robinson (1933)'s monopsony framework (Angerhofer, Collard-Wexler, and Weinberg, 2025; Azkarate-Askasua and Zerecero, 2025).[1] In these models, *buyer power* creates vertical distortions by generating input price *markdowns*.

In this paper, we characterize the welfare effects of buyer and seller power in a unified framework and provide an empirical approach to quantify them. The key novelty is that we do not assume a specific type of distortion; rather, we characterize the conditions under which buyer power acts as countervailing or distortionary, as a function of the supply and demand elasticities using a sufficient statistics approach. The framework yields an empirical approach that identifies the type of vertical conduct with minimal data requirements and quantifies how much buyer and seller power each contribute to vertical distortions.

Our model is a perfect-information Nash-in-Nash bargaining game between buyers ("downstream") and sellers ("upstream") that bilaterally bargain over a linear input price, and equilibrium either lies on the input demand curve ("monopolistic vertical conduct") or on the input supply curve ("monopsonistic vertical conduct"). We consider two measures of 'buyer power': the relative bargaining ability of buyers and sellers, which we treat as

[1] Whereas Angerhofer et al. (2025) is an oligopsony model, rationing can bring equilibrium onto the labor demand curve in their model.

a policy-invariant primitive, and the relative disagreement payoffs of buyers and sellers. We avoid functional-form assumptions on the demand and cost curves and allow for both simultaneous and sequential timing. Using this model, we analyze vertical distortions, defined as the output reduction relative to joint-profit maximization in the vertical chain.

The main motivation for our paper is the observation that, under increasing upstream marginal costs and decreasing downstream demand, an equilibrium exists under both monopsonistic and monopolistic vertical conduct. This contrasts with most empirical IO models studying settings with constant marginal costs, where only monopolistic conduct is possible, and with monopsony models in labor that typically feature perfectly elastic downstream demand, where only monopsonistic conduct is possible.[2] However, in settings with both increasing marginal costs and decreasing demand, both types of vertical conduct could occur, and vertical distortions could be induced by either monopsony power of buyers or by monopoly power of sellers.

We argue that such settings are common and illustrate the application of our model in three contexts: (i) a manufacturer with increasing marginal costs bargaining with a downstream firm, (ii) a union bargaining with an employer, and (iii) a sellers' cooperative bargaining with a buyer. For each setting, we provide an empirical application that demonstrates how to determine whether the vertical distortions come from the buyer's monopsony power or the seller's monopoly power.

We begin our analysis by examining monopolistic and monopsonistic bargaining separately. We first show that buyer power has an opposite effect on output under each type of vertical conduct. Under monopolistic bargaining, greater buyer power raises output by reducing the double-marginalization problem of Spengler (1950). In contrast, under monopsonistic bargaining, buyer power lowers output due to a different form of double marginalization, in which the buyer marks down input prices in addition to marking up consumer prices (Robinson, 1933).

To understand the distortions arising from these vertical conduct types, we compare them with joint-profit maximization, in which suppliers and buyers negotiate a two-part tariff. We show that for a unique interior value of the bargaining parameter $\beta^* \in (0, 1)$, both the monopsonistic and monopolistic equilibria coincide with the joint-profit maximization outcome. Although it may not be immediately obvious, this result is intuitive: at β^*, which we call the "bilaterally-efficient bargaining parameter," the buyer's monopsony power and the seller's monopoly power exactly offset each other, leading to an outcome without vertical distortions.

[2]Recently, monopsony models have been developed in which downstream residual demand is not perfectly elastic, such as Kroft, Luo, Mogstad, and Setzler (2023), Rubens (2023), and Lobel (2024).

We show that the bilaterally-efficient bargaining parameter β^* is the key threshold determining the welfare effects of buyer and seller power. It is characterized by the relative disagreement payoffs of buyers and sellers and by the relative elasticities of upstream cost and downstream demand, as these govern the extent of downstream monopsony power (cost curve) and upstream monopoly power (demand curve). A higher cost elasticity increases the potential for monopsony power, requiring more seller power (lower β^*) to countervail it. Similarly, less-elastic demand amplifies the scope for double marginalization, necessitating greater buyer power (higher β^*) to countervail seller power.

Having shown that two vertical conduct types are possible with distinct welfare effects, we next develop a criterion that allows us to pin down whether vertical distortions are due to monopsonistic or monopolistic conduct. We impose a participation constraint within our Nash-bargaining protocol: the seller requires a nonnegative markup, and the buyer requires a nonnegative markdown in order to trade. This constraint is reasonable when frictions prevent internal transfers that would otherwise allow parties to operate marginal units at a loss (Holmstrom and Tirole, 1991; Scharfstein and Stein, 2000). For example, if the seller is a labor union, a negative markup would require transfers among members to subsidize some workers to accept wages below their reservation levels—a scenario that is implausible. However, in other settings, a party may accept an unprofitable marginal unit because inframarginal units keep its overall profits positive. For such cases, we introduce an alternative constraint that does not impose nonnegative markups and markdowns but leads to the same equilibrium.

We find that under these constraints, the nature of vertical distortions depends on how the bargaining parameter (β) compares to the bilaterally-efficient bargaining parameter (β^*). If $\beta < \beta^*$, vertical distortions arise solely from market power of sellers, which can be countervailed by increasing the buyer's disagreement payoffs ('increasing buyer power'). Conversely, if $\beta > \beta^*$, vertical distortions only arise from monopsony power of the buyer, which can be countervailed by increased sellers' disagreement payoffs ('increasing seller power'). Thus, buyer power can be either countervailing (output-increasing) or distortionary (output-decreasing), depending on how β (a policy-invariant primitive) compares with β^* (a policy-variant outcome).

The constraint also delivers an empirical criterion for identifying vertical conduct. We show that estimates of the cost and demand curves, combined with transacted input prices—but no data on quantities—suffice to determine whether vertical distortions stem from monopsony or monopoly power. The criterion compares the transacted input price to the input price that would lead to the bilaterally-efficient quantity under linear contracting. This threshold input price is unique: input prices below the threshold imply a monopsonistic distortion, whereas input prices above the threshold imply a monopolistic distortion. Because the threshold depends only on the estimated cost and demand curves, the criterion requires neither bargaining weights nor disagreement payoffs. If transacted quantities are observed in addition to input prices, the constraint can be tested.

Turning to the applications of our model, we first use our model for prospective analysis in market structure and in the contracting environment. Our model bears relevance for antitrust questions, in particular related to merger control. We show that increased buyer power from horizontal mergers between buyers can either counteract or reinforce losses in consumer surplus from market power on the downstream market, depending on the nature of vertical conduct. Thus, our model reconciles prior analyses of buyer power in merger control, with buyer power being argued to be pro-competitive in Nevo (2014); Craig, Grennan, and Swanson (2021); Sheu and Taragin (2021) but anti-competitive in Hemphill and Rose (2018); Berger, Hasenzagl, Herkenhoff, Mongey, and Posner (2023). Similarly, the model can be used to bound the welfare effects of vertical integration.

Another type of prospective analysis involves understanding the effects of moving from posted to negotiated prices, for instance due to unionization. On the one hand, negotiation can countervail buyer power, on the other hand, it could induce double marginalization due to increased seller power. We show that the bilaterally-efficient bargaining parameter β^* serves as a sufficient statistic to understand these potential benefits and costs of switching from posted to negotiated prices. We illustrate this point by collecting prior estimates of residual labor supply and demand to estimate β^* in various U.S., Brazilian, and Chinese labor markets, and argue that the welfare tradeoff between price posting and negotiation differs substantially across these settings.

Second, we apply our empirical criterion to identify the source of vertical distortions in coal procurement by power plants in the Texas ERCOT market from 2005 to 2014. The availability of detailed cost data from coal mines and power plants alongside transacted coal and electricity prices makes this an ideal setting in which both cost *and* demand curves can be estimated. We estimate coal mining cost curves using detailed unit-level production and cost data and engineering estimates, similarly to Asker, Collard-Wexler, and De Loecker (2019), and residual electricity demand and power plant marginal costs, similarly to Puller (2007). Using our estimated coal cost and demand curves and the observed transacted coal prices, we identify conduct as being either monopolistic or monopsonistic, and use this information to quantify the sources of vertical distortions. We find that overall, 74.9% of the total vertical distortion in the market comes from the monopoly power of coal mines, and the remaining 25.1% is due to the monopsony power of power plants.

Before turning to the related literature, we note two caveats. First, this paper focuses solely on the static effects of buyer and seller power while remaining agnostic about potential dynamic effects, such as those relating to innovation or investment incentives, which may be critical for welfare (Inderst and Shaffer, 2007; Inderst and Wey, 2007; Parra and Marshall, 2024). Second, we do not take a stand on how to weigh the surpluses of buyers and suppliers; rather, we examine the effects of buyer and seller power on total and consumer surplus.

Contribution to the Literature This paper contributes to the literature on Nash-in-Nash bargaining models under bilateral monopoly/oligopoly (Horn and Wolinsky, 1988; Collard-Wexler et al., 2019), which have been applied in labor to analyze wage bargaining (Manning, 1987; Abowd and Lemieux, 1993; Hosken, Larson-Koester, and Taragin, 2024; Angerhofer et al., 2025), in IO to study firm-to-firm bargaining (Crawford and Yurukoglu, 2012; Grennan, 2013; Gowrisankaran et al., 2015; Crawford, Lee, Whinston, and Yurukoglu, 2018; Ho and Lee, 2019; Cuesta, Noton, and Vatter, 2025), and in international trade to study importer-exporter bargaining (Atkin, Blaum, Fajgelbaum, and Ospital, 2024; Alviarez, Fioretti, Kikkawa, and Morlacco, 2025).[3] In these models, equilibrium is usually assumed to lie on the input demand curve, which implies that vertical distortions are due to seller power.[4][5]

Second, a distinct literature studies monopsony power in vertical relations in which equilibrium lies on the input supply curve, rather than on the input demand curve. In these models, the downstream party sets input prices while facing an upward-sloping input supply (Card et al., 2018; Lamadon et al., 2022; Azar, Berry, and Marinescu, 2022; Berger et al., 2022; Rubens, 2023).[6] A recent literature has introduced monopsony power into Nash-in-Nash bargaining models, including Angerhofer et al. (2025), which is the closest empirical paper to ours, Azkarate-Askasua and Zerecero (2025), and Rubens (2025), We contribute to these two literatures by introducing a unified framework in which we are agnostic about the source of vertical distortions (whether equilibrium lies on the input demand or supply curve), and by decomposing the vertical distortions into those induced by seller and buyer power. By doing so, we also contribute to the literature testing vertical

[3] An important difference between our paper and Alviarez et al. (2025) is that the markdown in our paper represents a wedge between the marginal revenue product of an input and the price of that input, whereas the markdown in Alviarez et al. (2025) means a negative markup of the seller.

[4] In the labor literature, equilibrium on the labor demand curve is usually obtained through a 'right to manage' of employers, which is usually assumed in wage bargaining models except in models of efficient bargaining between unions and employers, such as McDonald and Solow (1981).

[5] In a distinct class of incomplete-information bargaining models (Loertscher and Marx, 2022; Larsen and Zhang, 2021), similar results on countervailing power arise as explored in (Loertscher and Marx, 2026).

[6] These are the "neoclassical" monopsony models, as opposed to the "dynamic" monopsony models in the search-and-matching tradition (Manning, 2013).

conduct (Berto Villas-Boas, 2007; Bonnet and Dubois, 2010; Atkin et al., 2024; Duarte, Magnolfi, Sølvsten, and Sullivan, 2024).

Third, we contribute to models of countervailing power (e.g., Galbraith, 1954; Horn and Wolinsky, 1988; Snyder, 1996; Iozzi and Valletti, 2014; Loertscher and Marx, 2022; Toxvaerd, 2024), for which empirical evidence was documented in Gowrisankaran et al. (2015); Barrette, Gowrisankaran, and Town (2022); Angerhofer et al. (2025). We advance this literature by identifying the conditions under which buyer power is countervailing or distortionary.[7] In contemporaneous and complementary research, Avignon, Chambolle, Guigue, and Molina (2025) derive similar results in a bargaining model where the supplier also has monopsony power, which introduces a novel "double markdownization" phenomenon. Their model is a special case of ours when disagreement payoffs are zero, timing is sequential, and there is a single buyer and seller. The main difference is that we do not treat vertical conduct as an equilibrium outcome, and that we empirically implement our model. Also, nonzero disagreement payoffs allow us to conduct comparative statics in terms of both these disagreement payoffs and the bargaining parameter, rather than just in the bargaining parameter.

Fourth, our paper relates to the literature that analyzes bilateral bargaining when suppliers face non-constant marginal costs. Previous work in this literature has examined the effects of buyer size (Chipty and Snyder, 1999), supplier incentives (Inderst and Wey, 2007), and supplier uncertainty (Smith and Thanassoulis, 2012) under different cost function assumptions. More recently, Mukherjee and Sinha (2024) analyzed the effects of vertical mergers on output when the supplier's cost function is convex. We contribute to this literature by analyzing the different vertical distortions within a unified framework.

Finally, our paper contributes to the literature on imperfect competition in energy markets (Cicala, 2015; Hortaçsu, Luco, Puller, and Zhu, 2019; Jha, 2022; Preonas, 2023; Asker et al., 2019; Gowrisankaran, Langer, and Zhang, 2024) by studying the sources of market power in this industry.

The rest of the paper is structured as follows. In Section 2, we set up our model. Section 3 analyzes the welfare effects of buyer power while taking vertical conduct as given. Section 4 develops tools to determine the source of vertical distortions and identification of vertical conduct. In Section 5, we empirically implement our model in two calibrated applications in which input prices are unobserved and we remain agnostic about conduct, while Section 6 presents the empirical application in which input prices are observed, and in which we identify vertical conduct. All proofs are given in the Online Appendix and the Supplemental Material.

[7] See Toxvaerd (2024) for a review of results in this literature.

2 Model Setup

2.1 Costs, Demand, and Payoffs

We consider a vertical market in which a set of upstream parties $\mathcal{U}$ ('suppliers') supply inputs to a set of downstream parties $\mathcal{D}$ ('buyers'). For a given supplier–buyer pair (U, D), the supplier sells a quantity q at an input price w using a linear pricing contract. The buyer then resells this good directly to consumers without incurring any additional costs.[8] Each buyer D faces a residual inverse demand curve $p(q)$, with $p'(q) \leq 0$. Each supplier U produces output at a residual average cost curve $c(q)$, with $c'(q) \geq 0$.

U and D bargain over the input price w to maximize a Nash product with respective bargaining weights $(1 - \beta)$ and β. We denote the buyer's profit as $\pi^d(w, q) \equiv \big(p(q) - w\big)q$, the supplier's profit as $\pi^u(w, q) \equiv \big(w - c(q)\big)q$. $\pi_0^d \geq 0$ is the disagreement payoff of D, and $\pi_0^u \geq 0$ is the disagreement payoff of U if bargaining breaks down. In line with the Nash-in-Nash bargaining solution concept (Horn and Wolinsky, 1988; Collard-Wexler et al., 2019), we impose a passive-beliefs assumption, meaning that each pair U and D takes all other agreements between parties as given when bargaining. This allows us to analyze each bilateral negotiation independently, using the residual demand and supply curves.

2.2 Monopsonistic vs. Monopolistic Bargaining

Our main analysis contrasts two alternative behavioral models of vertical conduct. In the first type, the parties bargain over the input price and the transacted quantity lies on the input supply curve ($mc(q) = w$), meaning that the input price is equal to U's marginal cost $mc(q) \equiv \partial(c(q)q)/\partial q$:

$$(w^{ms}, q^{ms}) = \begin{cases} \arg\max\limits_{w} \, [(\pi^d(w, q) - \pi_0^d)^{\beta}(\pi^u(w, q) - \pi_0^u)^{1-\beta}] & (1) \\ w = mc(q) & (2) \end{cases}$$

We call this first bargaining model "monopsonistic bargaining" because the input quantity is pinned down by the input supply curve, implying that the buyer needs to pay a higher price to raise the input quantity, which is the standard definition of monopsony power (Manning, 2013). Our model nests the classical monopsony model of Robinson (1933) as a special case when the buyer has the entire bargaining weight ($\beta = 1$).

In the second type, the parties again bargain over the input price, but this time the

[8] In Online Appendix E, we extend this framework to allow downstream production to use multiple inputs, where U supplies one input and the remaining inputs are obtained from competitive markets.

transacted quantity lies on the input demand curve ($mr(q) = w$), meaning that the input price is equal to D's marginal revenue product, $mr(q) \equiv \partial(p(q)q)/\partial q$:

$$(w^{mp}, q^{mp}) = \begin{cases} \arg\max_{w} [(\pi^d(w,q) - \pi_0^d)^{\beta}(\pi^u(w,q) - \pi_0^u)^{1-\beta}] \\ w = mr(q) \end{cases} \tag{3}$$

We call this bargaining "monopolistic bargaining" because the input quantity is pinned by the buyer's input demand curve, implying that the supplier needs to negotiate a lower price to raise the input quantity it sells, which is the standard definition of upstream market power in vertical chains (Spengler, 1950). This model nests the Spengler (1950)'s successive monopoly model as a special case when the seller has the entire bargaining weight ($\beta = 0$). We assume that there are gains from trade under both types of conduct, ensuring that a solution exists in each case (Assumption OA-1).

Timing Assumptions

We consider two versions of our model that differ in timing. In both versions, U and D first observe the primitives: $c(\cdot)$, $p(\cdot)$, β, disagreement payoffs, and vertical conduct. Then, the parties negotiate and quantities are determined. The two timing assumptions differ in whether bargaining and the quantity outcome occur simultaneously or sequentially.

Definition 1 (Simultaneous Bargaining)**.** *The quantity is determined simultaneously with bargaining over the input price.*

Definition 2 (Sequential Bargaining)**.** *Parties first bargain over a input price, after which the quantities are determined.*

Both types of timing assumptions are widely used in the literature. The simultaneous model appears in Ho and Lee (2017) and Crawford et al. (2018), while the sequential model is used in Crawford and Yurukoglu (2012) and in several right-to-manage union bargaining models (Oswald, 1982; Nickell and Andrews, 1983; Abowd and Lemieux, 1993).

Assuming that the second-order conditions hold, the solution to the bargaining problem in Equation (1) is characterized by the following first-order condition (FOC):

$$\beta \frac{\partial \pi^d(w,q)}{\partial w}(\pi^u - \pi_0^u) + (1-\beta)\frac{\partial \pi^u(w,q)}{\partial w}(\pi^d - \pi_0^d) = 0 \qquad \text{(B-FOC)} \tag{4}$$

for $\beta \in (0,1)$.[9,10] The solution to monopolistic bargaining is given by Equation (3) and

[9] Appendices S.2 and B.1 present the closed-form solutions of these FOCs for the simultaneous and sequential versions of the model. Note that the FOCs characterize the solution only for $\beta \in (0,1)$. At the corner cases $\beta = 0$ and $\beta = 1$, these models must be solved as constrained optimization problems, as the nonnegative profit constraints become binding. Appendix D.1 analyzes these cases.

[10] The second-order conditions for (U-FOC) and (D-FOC) are given by $mc'(q) > 0$ and $mr'(q) < 0$. The second-

(B-FOC), while the solution to monopsonistic bargaining is given by Equation (2) and (B-FOC). In the simultaneous model, (B-FOC) simplifies to $\beta(\pi^u - \pi_0^u) = (1-\beta)(\pi^d - \pi_0^d)$, whereas under sequential timing, the parties additionally internalize how the input price w affects the output quantity ($dq/dw \neq 0$). We present the results under sequential timing in the main text and report the corresponding results under simultaneous timing in Appendix S.2.

2.3 Discussion and Empirical Examples

Our paper differs from the recent IO bargaining literature by allowing for increasing marginal costs and by allowing equilibrium to lie on either the input supply or on the input demand curve. Increasing marginal costs is necessary to consider monopsony models because otherwise the supplier would be willing to supply an arbitrarily large quantity whenever the input price exceeds constant marginal cost, implying a perfectly elastic input supply curve. We argue that allowing for increasing marginal costs is important because many industries exhibit increasing marginal costs, and highlight three vertical environments below to which our model applies. In each of these environments, either the input demand or input supply curve can be binding depending on the underlying model of firm behavior and the contracting environment.

Example 1. ***Unions:*** *Labor unions with heterogeneous reservation wages.*

A long tradition of research studies wage bargaining and labor unions (Ashenfelter and Johnson, 1969; Card, 1986; Abowd and Lemieux, 1993; Azkarate-Askasua and Zerecero, 2025; Angerhofer et al., 2025). In this literature, the supplier U is a labor union bargaining over wages w with a downstream employer D, and labor supply is upward sloping due to heterogeneity in workers' reservation wages. Monopsonistic conduct in this setting corresponds to employment being pinned down by the labor supply curve for a given negotiated wage. On the other hand, if employers have r the right-to-manage and can ration labor, employment would lie on the labor demand curve, implying monopolistic conduct. We illustrate this application in the context of U.S. construction workers in Section 5.

Example 2. ***Cooperatives:*** *Cooperatives of suppliers with heterogeneous marginal costs.*

order conditions (B-SOC) for (B-FOC) are presented in Online Appendix S.3 for simultaneous timing and in Online Appendix B.2 for sequential timing. While (B-SOC) is always satisfied under simultaneous timing, the sequential case yields a complex expression involving third derivatives of cost and demand functions. In Online Appendix B.4, we develop some sufficient conditions under which (B-SOC) holds globally under sequential timing.

When a supplier collectively bargains with a buyer on behalf of multiple suppliers, the supply curve slopes upward due to heterogeneity in suppliers' costs. Agricultural cooperatives, which are prevalent in both the U.S. and developing countries, are an example of this structure (Cook, 1995; Ito, Bao, and Su, 2012). Monopsonistic conduct in this setting occurs when a cooperative and a buyer negotiate a input price, and farmers whose costs exceed that price produce and sell to the buyer.[11] If, on the other hand, buyers determine the quantity purchased, monopolistic conduct would apply. We illustrate this setting in Section 5 in the context of Chinese tobacco markets.

Example 3. ***Firm-Level Decreasing Returns to Scale:*** *Individual suppliers with increasing marginal costs at the firm level.*

In Examples 1 and 2, the aggregation of atomistic production units generates increasing marginal costs for the supplier. Individual firms can also face increasing marginal costs due to decreasing returns to scale production technology or if they operate multiple plants with heterogeneous costs. Monopsonistic conduct in this setting corresponds to 'output contracts' (Weistart, 1973), which are common in energy markets, or appears under resale price maintenance, where the supplier affects the downstream quantity by setting downstream prices.[12] Monopolistic conduct, on the other hand, occurs if input quantities are chosen by the buyer to maximize its profits given the input price w. In our empirical application on coal markets in Section 6, we show some observed coal contracts are examples of monopsonistic conduct whereas other contracts are examples of monopolistic conduct.

2.4 Conceptualizing 'Buyer Power'

In our model, two key parameters govern how surplus is divided between the buyer and the supplier. The first is the bargaining weight β, which directly shapes the split of the surplus and is micro-founded in Collard-Wexler et al. (2019). The second is the relative size of the disagreement payoffs, π_0^d and π_0^u, which determines each party's bargaining leverage: a higher π_0^d strengthens the buyer's position, while a higher π_0^u strengthens the supplier's.

Therefore, the term 'buyer power' can be conceptualized both as the relative bargaining strength of the buyer (Grennan, 2013; Craig et al., 2021; Avignon et al., 2025) or as the

[11] For an example of this type of contract, see Peace River Seed Co-operative and Proseeds Marketing Inc. where Proseeds agreed to buy Peace River's entire two-year grass-seed output at a fixed price.

[12] An example for output contract is certain types of power purchase agreements in which the buyer purchases all the output of a power plant; see, for example, GreenLife Solar and Shine Partners Agreement and County of Santa Clara PPA.

relative disagreement payoff of the buyer (Ho and Lee, 2017; Hemphill and Rose, 2018).[13] In the remainder of our theoretical model, we will present results for both measures of buyer power unless specified otherwise. However, when we discuss applications of our model, we will mostly treat the bargaining weight β as a policy-invariant primitive and the disagreement payoffs as policy-variant.

2.5 Sources of Market Power in Vertical Relations

To quantify the distortions due to market power in the vertical chain, we consider a two-part tariff as a benchmark. Under a two-part tariff, suppliers and buyers negotiate over both the quantity and a lump-sum transfer, which corresponds to a scenario in which the parties maximize their joint profits, as in the case of vertical integration. The quantity q^* that solves this problem is simply the quantity such that upstream marginal cost equals downstream marginal revenue: $mc(q^*) = mr(q^*)$.

We define 'vertical distortion' as the extent to which the quantity realized through the linear contract is below the joint-profit maximizing quantity q^*. This distortion appears if there is a wedge between $mc(q)$ and $mr(q)$. Since we do not take a stance between monopolistic and monopsonistic conduct, this wedge can arise from two sources in our model: a seller markup and a buyer markdown. As usual, we define the markup as the wedge between the input price w and the seller's marginal costs, and the markdown as the wedge between the buyer's marginal revenue and the input price (Syverson, 2025):

$$\text{Seller Markup} : \mu^u(q) \equiv \frac{w - mc(q)}{mc(q)}, \qquad \text{Buyer Markdown} : \Delta^d(q) \equiv \frac{mr(q) - w}{mr(q)}.$$

Under monopolistic bargaining, the distortion arises from the upstream markup (since $mr(q) = w$), whereas under monopsonistic bargaining, it arises from the downstream markdown (since $mc(q) = w$). Note that a downstream markup set by the buyer over its marginal costs also creates market power, but it exists independently of vertical conduct and is also present under joint-profit maximization. Throughout the paper, we distinguish these vertical distortions by referring to seller markup as "upstream monopoly power" and buyer markdown as "downstream monopsony power."

Linear Price Contracts vs. Two-Part Tariffs

One might question why firms use a linear price contract instead of a two-part tariff, which would yield greater joint profits. We remain agnostic about firms' motivations for using linear pricing and instead focus on analyzing vertical distortions that arise from linear

[13] Similarly, 'seller power' is either the bargaining strength of the seller or its relative disagreement payoff.

contracts. Linear contracts are well documented across industries and are commonly used to model vertical relationships in the literature; see Cussen (2025) for a review. That said, in one of our results, we provide conditions under which firms unilaterally find a linear price contract more profitable than a two-part tariff, offering a rationale for linear pricing within the context of our model (Section E.2).

3 Effects of Buyer Power: Monopolistic vs. Monopsonistic Conduct

We begin our analysis by examining the existence of the monopsonistic and monopolistic equilibrium. We then characterize the effects of buyer power on output, consumer surplus, and total surplus separately under monopolistic and monopsonistic bargaining. In the next section, we unify both forms of conduct and jointly analyze their welfare effects.

3.1 Existence of Monopolistic and Monopsonistic Conduct

We establish the existence of equilibrium in monopolistic and monopsonistic bargaining by highlighting two special cases: constant marginal cost and constant marginal revenue.

Proposition 1.
(i) If the upstream marginal cost is constant, $mc'(q) = 0$, the monopsonistic bargaining problem does not have an interior solution for any β.
(ii) If the downstream marginal revenue is constant, $mr'(q) = 0$, the monopolistic bargaining problem does not have an interior solution for any β.
(iii) In all other cases, both the monopolistic and monopsonistic bargaining problems have an interior solution for $\beta \in (0, 1)$.[14]

The intuition behind Proposition 1(ii) is straightforward. Under constant upstream marginal costs, input supply is perfectly elastic, implying infinite input supply, so only monopolistic bargaining is well-defined (as in much of the IO literature). Similarly, if marginal revenue is constant, input demand is perfectly inelastic, implying infinite input demand, so only monopsonistic bargaining is well-defined (as assumed in the classical monopsony literature).[15] Yet, as shown in Proposition 1(iii), an interior equilibrium under either conduct is possible with increasing upstream marginal costs and decreasing

[14]We call a solution interior if the equilibrium input price is different from the average upstream cost $c(q)$ and downstream price $p(q)$. In the simultaneous bargaining model, a solution may fail to exist for some values of β depending on the demand and cost curves. We characterize the β range for which a solution exists for the simultaneous models in Appendix D.2, but for simplicity, we use $\beta \in (0, 1)$ for the remainder of the paper. If necessary, these bounds can be replaced with those derived in Appendix D.2.

[15]In light of Proposition 1, we assume for the remainder of this section that $mc'(q) > 0$ when analyzing monopsonistic bargaining and $mr'(q) < 0$ when analyzing monopolistic bargaining to ensure that the solutions are well-defined.

downstream demand marginal revenue. As illustrated in Examples 1–3, studies of such settings have gained prominence recently with the integration of monopsony power into IO models (Card, 2022; Azar and Marinescu, 2024), which motivates our unified approach.

3.2 Output and Buyer Power

We now characterize the relationship between equilibrium output q and buyer power under both monopsonistic and monopolistic bargaining.

Theorem 1.

(i) Under monopsonistic bargaining, the output quantity q decreases with the buyer's bargaining weight (β) and the buyer's disagreement payoff (π_0^d), and increases with the supplier's disagreement payoff (π_0^u).

(ii) Under monopolistic bargaining, the output quantity q increases with the buyer's bargaining weight (β) and the buyer's disagreement payoff (π_0^d), and decreases with the supplier's disagreement payoff (π_0^u).

Theorem 1 reveals a key distinction between two types of vertical conduct. An increase in buyer power reduces output under monopsonistic bargaining, but increases it under monopolistic bargaining. Figure 1(a) illustrates the key intuition behind this result. In monopsonistic bargaining, the equilibrium is on the input supply curve. An increase in buyer power, be it through the buyer's bargaining weight or through its disagreement payoff, leads to a movement along this input supply curve by lowering the input price. This, in turn, reduces output. In contrast, monopolistic bargaining implies that equilibrium is on the input *demand* curve. Consequently, increases in both β or π_0^d induce movements along this input demand curve, increasing output.

While similar theoretical insights have appeared in the literature under parametric assumptions or individual conduct (e.g., Chen (2003); Mukherjee and Sinha (2024); Toxvaerd (2024)), our results are, to the best of our knowledge, the first nonparametric treatment of this problem in a unified framework with nonzero disagreement payoffs, and under both simultaneous and sequential timing.[16]

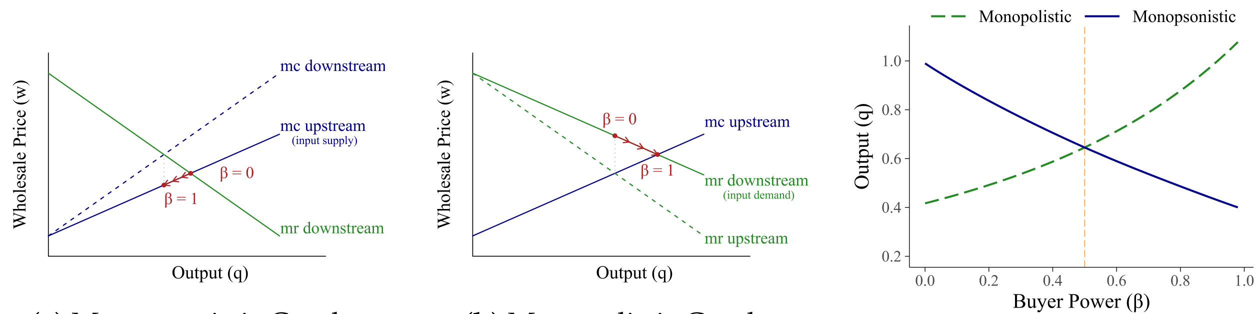
3.3 Distinguishing between "Buyer Power" from "Monopsony Power"

In the literature, the terms *buyer power* and *monopsony power* are often used interchangeably. Our results clarify the distinctions between these concepts.

Corollary 1.

[16] As discussed earlier, some of these results were also derived independently by Avignon et al. (2025).

Figure 1: Illustration of the Effects of Buyer Power on Output



(a) Monopsonistic Conduct **(b)** Monopolistic Conduct **(c)** Illustration

Notes: This figure illustrates how buyer power affects output under different vertical conduct types. Panels (a) and (b) show the theoretical intuition: in monopsonistic conduct, increasing the buyer's bargaining weight β moves output along the marginal cost curve, while in monopolistic conduct, it shifts output down the marginal revenue curve. Panel (c) presents numerical simulation results using cost curve $c(q) = q^{\psi}/(1+\psi)$ and demand curve $p(q) = q^{1/\eta}$ with parametrizations $\psi = 1/4$ and $\eta = -6$. Sequential timing is assumed; simultaneous timing results appear in Figure OA-6. For equilibrium derivations, see Appendix D.4.

(i) Under monopolistic bargaining, the downstream markdown Δ^d is always zero, so the buyer has buyer power but it does not induce a monopsony distortion. In this case, the upstream markup μ^u is decreasing with buyer power.

(ii) Under monopsonistic bargaining, the upstream markup μ^u is always zero, so the seller has seller power, but it does not induce a monopoly distortion. In this case, the downstream markdown is increasing with buyer power.

Even when the buyer's bargaining weight is nonzero ($\beta > 0$) under monopolistic bargaining, the buyer markdown is always zero, because the marginal revenue equals the input price. Hence, although there is 'buyer power' that can affect input prices, there is no monopsony distortion. Similarly, under monopsonistic bargaining, even with positive bargaining weight of the seller ($\beta < 1$), the seller markup is always zero, because upstream marginal cost equals input price. Hence, there is no double-marginalization distortion. As a result, downstream monopsony power arises only when increased *buyer* power reduces output, whereas upstream market power arises only when increased *seller* power reduces output. In all other cases, buyer and seller power are countervailing.[17]

3.4 Characterization of the Bilaterally-Efficient Bargaining Weight

Having established how output depends on buyer power under each vertical conduct type, we turn to the implications of buyer power on joint surplus, consumer surplus, and total surplus. We start with joint surplus by comparing each conduct to joint-profit

[17]See Chen (2008) for a review of the various definitions of buyer power in the literature.

maximization.

Proposition 2. *Under the assumption that disagreement payoffs are sufficiently low, there exists a unique bargaining parameter*

$$\beta^* = \frac{-p'(q^*)(q^*)^2 - \pi_0^d}{\big(c'(q^*) - p'(q^*)\big)(q^*)^2 - (\pi_0^u + \pi_0^d)} \in (0,1)$$

such that the equilibrium output from both the monopsonistic and monopolistic bargaining models corresponds to the joint-profit maximizing output. With zero disagreement payoffs, this expression simplifies to $\beta^* = \dfrac{-p'(q^*)}{c'(q^*) - p'(q^*)}$.

Proposition 2 shows that at β^*, output under the monopolistic and monopsonistic conduct coincides and is equal to the level that would be reached under joint-profit maximization. This result arises because β^* denotes the level of buyer power at which the buyer's monopsony power and the seller's monopoly power exactly offset one another, yielding an outcome with neither monopsony nor upstream monopoly distortions.[18] We denote β^* as the "bilaterally-efficient level of buyer power", which will be the key parameter when we introduce identification of vertical distortions in the next section.

The intuition for why the *same* β^* applies to both conduct models is that at q^*, the input price w has no distortionary effect on quantity. Under either conduct model, the envelope theorem eliminates the quantity-setter's response to w, while joint-surplus maximization ($mr(q^*) = mc(q^*)$) eliminates the remaining indirect effect through the other firm's profit. Both bargaining FOCs therefore reduce to the same surplus-sharing condition, $\beta\,(\pi^u - \pi_0^u) = (1-\beta)\,(\pi^d - \pi_0^d)$, yielding a unique β^* since profits at q^* are identical under both models.

The bilaterally-efficient-bargaining parameter β^* relates to the slopes of the cost and demand curves in an intuitive way, as shown in the following corollary.

Corollary 2.

(i) β^* *weakly decreases as the upstream marginal cost curve becomes steeper ($c'(q^*)$ increases) and weakly increases as the downstream inverse demand curve becomes steeper ($-p'(q^*)$ increases).*

(ii) β^* *decreases with the downstream disagreement payoff ($\partial\beta^*/\partial\pi_0^d < 0$) and increases with the upstream disagreement payoff ($\partial\beta^*/\partial\pi_0^u > 0$).*

[18] Proposition 2 has some similarity to the condition in Hosios (1990), which states that the worker's bargaining share must align with the relative ease of finding a job versus the difficulty firms face filling vacancies. Rather than balancing a congestion and market thickness externality in Hosios (1990), the bilaterally-efficient level of buyer power balances a markup and markdown distortion in our model.

The steeper the upstream cost curve, the higher the potential monopsony distortion becomes. To counterbalance this effect, the seller needs greater bargaining power, resulting in a lower value of β^*. Similarly, steeper downstream demand amplifies the potential distortion from upstream market power, requiring the buyer to have stronger bargaining power as a countervailing force. As a result, in the extreme case with perfectly inelastic supplier costs ($c'(q) \to \infty$), the joint-profit maximizing quantity is reached when the entire bargaining weight goes to the seller ($\beta^* = 0$): the potential monopsony distortion becomes so high that it needs to be fully countervailed by seller power. In contrast, if demand becomes fully inelastic ($-p'(q) \to \infty$), the entire bargaining weight needs to go to the buyer ($\beta^* = 1$), in order to countervail the double marginalization problem.

Two remarks about Proposition 2 are worth noting. First, if disagreement payoffs are sufficiently high—that is, if $\pi^d(q^*) < \pi_0^d$ or $\pi^u(q^*) < \pi_0^u$—the linear price contract cannot achieve q^*, because at q^* either the buyer's or seller's profit is below their disagreement payoff. In such cases, the same corner value of β^* yields the equilibrium output nearest to q^* under both conduct models ($\beta^* = 0$ when $\pi^d(q^*) \leq \pi_0^d$, and $\beta^* = 1$ when $\pi^u(q^*) \leq \pi_0^u$). For the remainder of the paper, we define β^* as these corner values whenever the linear contract cannot attain q^*. Second, in this section, our analysis focuses on the implications of varying the bargaining weight β by keeping disagreement payoffs fixed. While analogous results can be obtained by varying disagreement payoffs for a given β, the analysis is less tractable because, unlike β, disagreement payoffs involve two parameters (π_0^d and π_0^u), making it difficult to construct a single sufficient statistic.

3.5 Consumer and Total Surplus

In this section, we investigate at what level of buyer power consumer and total surplus are maximized. Since consumer surplus $CS(\beta) \equiv \int_0^{q(\beta)} \big(p(h) - p(q(\beta))\big)dh$ is monotonically increasing in output, it is maximized at the output-maximizing bargaining weight:

Proposition 3. *Consumer surplus is maximized at* $\beta = 1$ *under monopolistic bargaining and at* $\beta = 0$ *under monopsonistic bargaining.*

A notable implication of our results is that equilibrium output under linear contracting can exceed the output under a two-part tariff (q^*), meaning that consumers can be better off under linear contracting than under nonlinear contracting or vertical integration. As observed in Figure 1(c), this happens when $\beta < \beta^*$ in the monopsonistic model and when $\beta > \beta^*$ in the monopolistic model. This outcome arises because, in these regions, either the upstream markup or the downstream markdown becomes negative, resulting in lower downstream prices than those implied by joint-profit maximization.

We next turn to total surplus, defined as $TS = \int_0^{q(\beta)}[p(h) - mc(h)]dh$:

Proposition 4. *Under monopolistic bargaining, total surplus is maximized at some $\beta^\dagger$ in the range $\beta^* \leq \beta^\dagger \leq 1$, whereas under monopsonistic bargaining it is maximized at $\beta^\dagger = 0$.*

Proposition 4 also shows that that unlike the constant marginal cost case, total welfare is not necessarily maximized at full buyer power under monopolistic bargaining. The reason is that with increasing marginal costs, full buyer power can lead to socially inefficient overproduction, as input prices can fall below the supplier's marginal cost, while the upstream firm still earns positive profits from inframarginal production. In contrast, under monopsonistic bargaining, total surplus is maximized with full seller power. In that scenario, the seller makes a TIOLI offer to the buyer, driving the buyer's profit to zero. The downstream price p then equals the input price w, which equals upstream marginal costs. As a result, the downstream price equals the supplier's marginal cost, maximizing total welfare.

4 Determining Vertical Distortions

In Section 3, we have shown that under general conditions, vertical distortions could arise under either monopsonistic and monopolistic bargaining, with opposite welfare effects of buyer power under these two models of vertical conduct. These results are useful for understanding the impact of buyer power when conduct is known, for example, from the institutional settings or by estimating conduct using historical data. However, in many cases, vertical conduct is either not known ex ante or not easily estimable. In this section, we augment our model to analyze vertical distortions in those cases and develop ways to identify the sources of vertical distortions.

In particular, we require that the equilibrium markup and markdown are nonnegative in the bargaining problem. With this constraint, we show that vertical distortions can only arise from either monopsony *or* monopoly at any given bargaining weight. This result is particularly useful for prospective analysis of the effects of changes in buyer or seller power when conduct is ex-ante unknown, as we illustrate in Section 5. This result is also useful for identifying vertical distortions from data. We show that the sources of vertical distortions can be identified solely from input prices, without observing quantities or estimating a full bargaining model, as we illustrate in Section 6, while quantities are useful to test the nonnegative markup and markdown assumption.

4.1 Determining the Source of Vertical Distortions

We impose nonnegative markup and markdown constraints directly in the bargaining problem.

Nonnegative Markup/Markdown Constraint. *The parties bargain over w subject to the constraint that the equilibrium markup and markdown are nonnegative. Under each conduct model, the constrained bargaining problem is:*

$$\textit{Monopolistic:} \quad \begin{cases} \max_{w} \ \left(\pi^d - \pi_0^d\right)^\beta \left(\pi^u - \pi_0^u\right)^{1-\beta} \quad s.t. \quad w \geq mc(q) \\ mr(q) = w \end{cases}$$

$$\textit{Monopsonistic:} \quad \begin{cases} \max_{w} \ \left(\pi^d - \pi_0^d\right)^\beta \left(\pi^u - \pi_0^u\right)^{1-\beta} \quad s.t. \quad w \leq mr(q) \\ mc(q) = w \end{cases}$$

These constraints require that the marginal production unit is profitable for each party: the supplier's marginal cost must not exceed the input price ($w \geq mc(q)$), and the buyer's marginal revenue must not fall below the input price ($w \leq mr(q)$). If either constraint is violated, the marginal unit generates a loss, necessitating internal transfers between inframarginal and marginal units within the party. The following theorem shows that under the nonnegative markup/markdown constraint, a vertical distortion exists at any given β under exactly one of the two conducts:

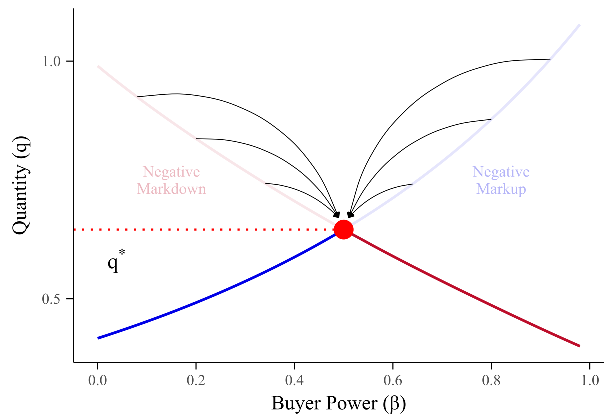
Theorem 2. *Under the nonnegative markup/markdown constraint, at any $\beta \in [0,1]$, the vertical distortion exists only under monopolistic conduct if $\beta < \beta^*$ and under monopsonistic conduct if $\beta > \beta^*$.*

The intuition is as follows, and is depicted in Figure 2.[19] Under monopolistic bargaining, the upstream markup μ^u increases in β. For $\beta \leq \beta^*$, $\mu^u \geq 0$ holds automatically, so the constraint is slack and the standard outcome prevails. For $\beta > \beta^*$, the unconstrained solution would imply $\mu^u < 0$, so the constraint $w \geq mc(q)$ binds. Together with D-FOC ($mr(q) = w$), this forces $q = q^*$ and $w = w^*$. The argument for monopsonistic bargaining is symmetric: the downstream markdown Δ^d increases in β, so the constraint $w \leq mr(q)$ binds only for $\beta < \beta^*$, again yielding $q = q^*$ and $w = w^*$.[20] Thus, for every $\beta \neq \beta^*$, the two conduct models generate distinct equilibrium outcomes: one model operates uncon-

[19] We formally characterize the constrained equilibria in Online Appendix C.8.

[20] As noted before, in the event that $\pi_d^* \leq \pi_0^d$ or $\pi_u^* \leq \pi_0^u$, the linear contract cannot reach the joint-profit maximizing solution. In this case, the monopolistic equilibrium does not exist for $\beta > \beta^*$ whereas the monopsonistic equilibrium does not exist for $\beta < \beta^*$.

Figure 2: Determining Vertical Distortions



Notes: This figure illustrates the relationship between buyer power (β) and output (q) in bargaining models under the nonnegative markup/markdown constraint, using numerical simulation under the design reported in Figure 1(c).

strained (with a distortion), while the other is constrained to the joint-profit maximizing outcome.

Discussion of the Nonnegative Markup and Markdown Constraint

Even with negative markups or markdowns, both parties can still earn positive profits from trade through their inframarginal units; a negative markup (markdown) simply means the marginal unit operates at a loss. Thus, this assumption is more plausible in settings where internal transfers are not possible. One example is when the trading party represents a collection of agents who cannot efficiently transfer surplus among themselves. For instance, in the labor union context (Example 1), it is unrealistic to expect unions to redistribute wages so that some workers accept wage offers below their reservation wage. Moreover, even within a single organization, fairness considerations, organizational constraints, or agency problems are known to prevent internal transfers (Kahneman, Knetsch, and Thaler, 1986; Holmstrom and Tirole, 1991; Scharfstein and Stein, 2000).

Moreover, the equilibrium under this constraint satisfies several desirable properties. First, the party that determines input supply or demand (the buyer under monopoly or the supplier under monopsony) prefers the constrained outcome to the unconstrained outcome.[21] Second, in cases where the supply or demand is from atomistic agents, it implies an equilibrium with excess supply (rationing) instead of excess demand (involuntary supply)—for example, an employer rationing employment rather than forcing workers to accept wages below their reservation level. Third, it generates output at or below the joint-

[21] We show this in Appendix E.2.

profit-maximizing level, implying, for example, that vertical integration never increases vertical distortions in the vertical chain.

Of course, this constraint is not always reasonable. For example, a multi-establishment firm (Example 3) could operate its marginal units even at a loss, especially if there are sunk investments or dynamic incentives. To address these cases, we develop an alternative assumption that pins down vertical distortions without the markup/markdown constraint, which we describe in Section 4.3.

4.2 Empirically Determining the Source of Vertical Distortions

We turn to discussing how the results from above can be implemented to empirically examine the nature of vertical distortions.

We start by considering settings in which transacted input prices (w) are observed, but transacted quantities (q) are not. This is relevant in settings such as collective bargaining by labor unions or cooperatives, in which wages/input prices are clear from the centrally negotiated contracts, whereas uncovering employment/quantity flows is less obvious (Angerhofer et al., 2025). We turn to settings in which both input prices and quantities are observed in Section 4.3.

We find that imposing the nonnegative markup and markdown constraint allows identifying the sources of vertical distortions by comparing the observed input price w to the input price that yields the joint-profit maximizing solution under linear contracting, $w^* \equiv mc(q^*) = mr(q^*)$:

Proposition 5. *Under the nonnegative markup/markdown constraint:*

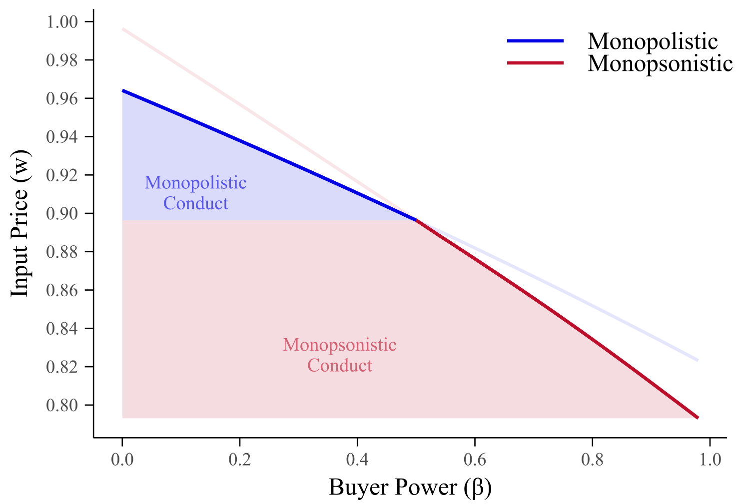
(i) *Under monopolistic bargaining, $w \geq w^*$.*

(ii) *Under monopsonistic bargaining, $w \leq w^*$.*

Therefore, vertical conduct is identified by the sign of $w - w^*$.

Figure 3 illustrates Proposition 5: w^* separates the two conduct regions. Input prices above w^* are consistent only with monopolistic conduct, while input prices below w^* are consistent only with monopsonistic conduct. The reason why input prices are useful for conduct identification whereas quantities are not goes back to Theorem 2, which shows that the vertical distortion exists only under monopolistic conduct when $\beta < \beta^*$ and only under monopsonistic conduct when $\beta > \beta^*$. Since the input price is monotone in β under both types of conduct, w^* separates the price space into two regions, which overlap with the separate vertical distortions. This is not true for q, because the monotonicity of q with

Figure 3: Conduct Identification via the Input Price



Notes: This figure illustrates Proposition 5. The horizontal dashed line marks w^*. Under monopolistic bargaining, $w \geq w^*$ (region above the dashed line), while under monopsonistic bargaining, $w \leq w^*$ (region below the dashed line). The shaded regions indicate the range of input prices consistent with each conduct model.

respect to β changes across conduct models (Theorem 1), so each equilibrium output can be generated under two different models of conduct, under different bargaining weights.

A key practical advantage is that w^* depends only on cost and demand primitives: $w^* = mc(q^*) = mr(q^*)$ and q^* solves $mc(q^*) = mr(q^*)$. Neither depends on the bargaining weights β, the disagreement payoffs π_0^d and π_0^u, or any other parameters of the bargaining model. Thus, identifying vertical conduct using input prices does not require estimating a bargaining model — it requires only estimates of the residual upstream cost and downstream demand curves. This offers important practical advantages, as estimating bargaining weights requires rich data, and disagreement payoffs require assumptions on what firms would do off the equilibrium path.

While w is sufficient to identify conduct, q is still useful for testing the nonnegative markup and markdown constraint. A direct implication from Theorem 2 is that if the constraint holds, output is bounded from above by the joint-profit maximizing quantity ($q \leq q^*$). the nonnegative markup/markdown constraint can thus be empirically tested by conducting a statistical test of whether observed quantities fall below the joint-profit maximizing quantity level. We implement this test in our main empirical application in Section 6 and provide further details on its implementation there.

4.3 Discussion and Extensions

Alternative Assumption to Determine Vertical Distortions

For settings in which the nonnegative markup and markdown assumption is not reasonable, we provide another constraint that allows determining the source of vertical distortions. In Appendix E.2, we augment our bargaining model by introducing the possibility that firms negotiate over a two-part tariff instead of over a linear input price contract, if doing so is incentive-compatible. We show that if one party has the right to choose the contract type, conditional on the primitives, it unilaterally finds a linear price contract more profitable than a two-part tariff. This alternative model leads to the same solution as the nonnegative markup and markdown constraint.

Other Ways to Empirically Identify Vertical Distortions

While we developed our results under the assumption that conduct is not known ex ante, if one has historical data on prices and quantities, conduct can also be identified directly, without even computing w^*, by testing whether the observed prices and quantities lie on the input supply or input demand curve. Therefore, imposing the nonnegative markup/markdown assumption is mainly useful when transacted quantities are not (all) observed, or when conducting prospective analysis.[22]

5 Prospective Analysis: Antitrust and Collective Bargaining

In this section, we discuss applications of our model that rely on β^* as a sufficient statistic, which the researcher can compute if there are prior cost and demand estimates. In Section 5.1, we examine the competitive effects of horizontal and vertical mergers in negotiated vertical markets. In Section 5.2, we address the question of whether labor unions and supplier cooperatives countervail monopsony power, or induce double marginalization.

5.1 Antitrust Policy

Horizontal Mergers

Our results are relevant for understanding the effects of horizontal mergers between buyers on consumer and total surplus. Such mergers decrease the relative disagreement

[22] Our results suggest additional approaches for testing vertical conduct. For instance, one can analyze how a change in the input price in reaction to a shock in disagreement payoffs that is excluded from marginal costs and marginal revenue affects quantity, since w moves in the opposite direction of q under different conduct types, as shown in Lemmas 1-1. Another possibility is to estimate markups and markdowns using a production approach and directly test Corollary 1. However, this approach typically requires behavioral assumptions, such as "buyers choose output to minimize costs," to estimate markups and markdowns using the production function (De Loecker and Warzynski, 2012; Rubens, 2025).

payoffs of the sellers, as they lose one outside option ($\pi_0^u \downarrow$).One could also let the merger alter β, as in Grennan (2013), which has similar effects as changing disagreement payoffs, as shown in our model.

First, we discuss the case in which vertical conduct is known. Under monopolistic conduct, the decrease in the seller's disagreement payoff decreases the vertical distortion, as it reduces the supplier's markup. This can potentially counteract losses in consumer surplus from downstream market power effects of the merger, which echoes the 'countervailing' merger defense in Grennan (2013), Nevo (2014), and Sheu and Taragin (2021). In contrast, under monopsonistic conduct, the decreased seller's disagreement payoff *increases* the vertical distortion, thereby lowering output. In this case, the vertical distortion and downstream market power effects on consumer surplus reinforce each other.

If the nonnegative markup and markdown constraint is satisfied, then the effects of buyer mergers on total surplus and consumer surplus are directionally the same. However, if the constraint is violated, it is possible that the increase in buyer power from a merger between buyers induces prices to falls below the supplier's marginal cost. In this case, the merger could harm total surplus even if increasing consumer surplus, due to socially-inefficient overproduction that was discussed in Proposition 4.

Second, if vertical conduct is unknown, the effects of the merger on vertical distortions are ambiguous, as is clear from the discussion above. In this case, vertical conduct needs to be identified from the data in order to know the welfare effects of horizontal mergers. Our empirical analysis in Section 6 shows that this can be accomplished by a merger authority using limited data and without having to know either disagreement payoffs or bargaining parameters.

Vertical Mergers

Our model can also help quantify the potential competitive gains from the elimination of double marginalization in vertical mergers (Chipty, 2001; Crawford et al., 2018; Luco and Marshall, 2020; Cuesta et al., 2025). The typical convention is that vertical integration eliminates double marginalization (markup/markup or markup/markdown), thereby increasing consumer surplus (abstracting from foreclosure incentives). Our model suggests that this is no longer necessarily true with increasing marginal costs and decreasing marginal revenue. As shown in Section 3, under a fixed model of vertical conduct, consumer surplus is maximized at the corner values of the bargaining weight, $\beta = 0$ or $\beta = 1$, where the vertical merger leads to output level q^*. If the original markups or markdowns are negative, this implies that vertical integration reduces output in the vertical chain even in the absence of foreclosure incentives. However, under the nonnegative markup and markdown constraint, output always weakly increases with vertical integration.[23]

5.2 Collective Bargaining

A second natural application of our model is to study collective wage bargaining institutions, as described in Example 1. With growing empirical evidence of monopsony power in labor markets (Card et al., 2018; Berger et al., 2022; Lamadon et al., 2022; Yeh, Macaluso, and Hershbein, 2022), a key question is whether labor unions can effectively countervail this power (Angerhofer et al., 2025; Azkarate-Askasua and Zerecero, 2025), or induce double marginalization instead.

Effects of Collective Bargaining

First, we consider the effects of introducing collective wage-bargaining, as opposed to wage-posting by employers. When employers unilaterally post wages to atomistic, price-taking employees, this occurs at $\beta = 1$ under monopsonistic conduct. If employees would instead be allowed to unionize, thereby bargaining collectively over wages, rather than individually, then the employer would presumably have a different bargaining ability $\beta < 1$, and conduct could be either monopolistic or monopsonistic. What would the effects of this be on vertical distortions, when we are agnostic about vertical conduct in the counterfactual?

Under our nonnegative markdown and markup constraint, the answer depends critically on β^*, which is a function of residual supply and demand elasticities.[24] If counterfactual conduct is monopsonistic, output increases as β decreases from 1 to β^*, and is flat from β^* to 0. In contrast, if counterfactual conduct is monopolistic, output is flat from 1 to β^* and decreases from β^* to 0. Therefore, the higher β^* (the more inelastic input demand and the more elastic input supply), the smaller the potential welfare gains from introducing collective bargaining by a union, and the larger the potential welfare costs.

To illustrate this point, we have collected residual input supply and demand elasticities across three settings that study monopsony power under 'pure' monopsony, in which employers/buyers set input prices. First, we collect labor supply and demand elasticities for U.S. construction workers from Kroft et al. (2023). Second, we collect labor supply and goods demand elasticities for Brazilian labor markets from Lobel (2024). Third, we consider the possible introduction of seller cooperatives, as discussed in Example 2, in the context of Chinese tobacco farmers selling to cigarette manufacturers, using supply elasticities from Rubens (2023). All used parameters are reported in Table 1.

[23] Of course, other effects of vertical integration besides elimination of double marginalization need to be taken into account as well, such as potential foreclosure, which we abstract from in our analysis.

[24] We express this in Appendix D.4.1.

Table 1: Parameters for Empirical Illustrations

Industry	Sources	ψ	η	β^*
U.S. construction workers	Kroft et al. (2023)	0.29	-7.30	0.42
Brazilian labor markets	Lobel (2024)	0.24	-1,43	0.92
Chinese tobacco farmers	Rubens (2023) Ciliberto and Kuminoff (2010)	1.90	-1.14	0.92

Notes: This table reports parameters for the inverse price elasticity of supply, ψ, and the own-price elasticity of residual downstream demand, η, as estimated in the referenced studies, for each empirical application. The final column shows the implied bilaterally-efficient level of buyer power, β^*, computed from the parameters using the log-linear approximation derived in Appendix D.4.1.

Using these cost and demand elasticities, we calculate the bilaterally-efficient levels of buyer power β^* in these three settings by setting disagreement payoffs to zero, which implies that in the event of a strike, employees do not earn anything and the firm shuts down. For U.S. construction workers, we find $\beta^* = 0.42$. This estimate suggests that collective wage bargaining requires careful consideration as a means to counteract monopsony power. If the resulting labor union has a bargaining weight above 0.58 $(1-\beta^*)$ and the equilibrium is on the labor demand curve, there would be a monopoly distortion as the union would create double marginalization. Our compilation of bargaining weights from the union literature in Appendix Table OA-3 suggests that this scenario is plausible—union bargaining power exceeds the estimated β^* threshold in roughly half of the reviewed studies.

In the Brazilian setting, we obtain a much higher parameter of $\beta^* = 0.92$, indicating that labor unions are much more likely to induce monopoly distortions and less likely to countervail monopsony distortions than in the U.S. Similarly, in the tobacco farmers' case, we find a high $\beta^* = 0.92$ despite a very inelastic farmer supply, again indicating that double marginalization from a cooperative is more likely to induce distortions than monopsony.[25] Therefore, unless the cooperative's bargaining power is extremely low—which prior estimates reported in the Appendix do not support— vertical distortions under negotiated leaf prices would either be due to double marginalization or non-existent.

Effects of Strike Prohibitions

Whereas we discussed changes in the employers' bargaining weight β above, we now turn to changes in the disagreement payoffs of unions and employers. In some industries, although wages are collectively bargained by a union, the unions are prohibited from going

[25]However, there are other possible inefficiencies due to monopsony power, such as misallocation (Rubens, 2023).

on strike. Such clauses exist, for instance, for U.S.-based airline pilots, flight attendants, and air-traffic controllers. Our model speaks to the effects of lifting such clauses on both consumer and total surplus. Given that strikes disproportionally hurt employers over employees (a pilot strike costs pilots their daily wage throughout the strike, but can cause millions of losses to the carrier), strike prohibitions decrease the relative disagreement payoff of employers, $\pi_0^d \downarrow$. Under the nonnegative markup and markdown constraint, the potential effect of such a decrease in buyer power depends, again, on the relative labor supply elasticity of pilots to the demand elasticity of airline tickets. The more inelastic airline demand is, the more likely it is that lifting strike prohibitions would lead to double marginalization. On the other hand, the more inelastic the labor supply, the more likely it is to countervail monopsony distortions.

6 Quantifying Vertical Distortions: Coal Procurement

In the examples in the previous section, we conducted prospective analysis of various changes in the contracting environment and in market structure. In this section, we turn to an 'ex-post' analysis of a vertical market, in which we decompose vertical distortions into monopsony and monopoly distortions using only cost and demand primitives. For the purpose of our empirical exercise, it is not necessary to take a stance on the parties' disagreement payoffs, nor to estimate bargaining weights.[26] We analyze coal procurements by power plants from mining firms, an industry that features (i) rich heterogeneity in cost and demand elasticities, (ii) the presence of multiple sellers and buyers, and (iii) oligopolistic competition in the downstream electricity market. In our analysis, we model mining costs by estimating individual mine marginal costs and then aggregating them to the firm level. For estimating residual electricity demand, we closely follow the seminal works in the literature (Borenstein and Bushnell, 1999; Borenstein, Bushnell, and Wolak, 2002; Puller, 2007). A key advantage of this approach is that conduct identification requires only the joint-profit maximizing quantity q^* and the input price w^* that yields q^* under linear contracting. Both of these depend solely on cost and demand primitives and not on the bargaining weights or disagreement payoffs, as established in Proposition 5.

Our empirical setting is the ERCOT (Electric Reliability Council of Texas) market, which has been previously studied in the literature (Hortaçsu and Puller, 2008; Hortaçsu et al., 2019). The ERCOT market offers three advantages: (i) it operates independently without inter-regional trade, (ii) the majority of power plants are deregulated, and (iii) hourly price and generation data are readily available. We model coal transactions between all power

[26]However, to use the model for many other counterfactual exercises, uncovering both disagreement payoffs and bargaining weights would be usually required.

plants operating in ERCOT and their coal suppliers, which may be located either within and outside the ERCOT region. We analyze the 10-year period between 2005 and 2014.

6.1 Institutional Details

Coal mining firms extract coal through underground and surface operations, with marginal costs varying significantly across geographic regions and mine types due to differences in geological conditions and local labor markets. Generally, surface mines have lower extraction costs compared to underground mines. Many mining companies operate multiple mines with different characteristics, leading to increasing marginal costs at the firm level. During our sample period, electricity generation was the primary use of coal in the U.S., accounting for 93.1% of total coal consumption in 2010 (Watson, Paduano, Raghuveer, and Thapa, 2010).

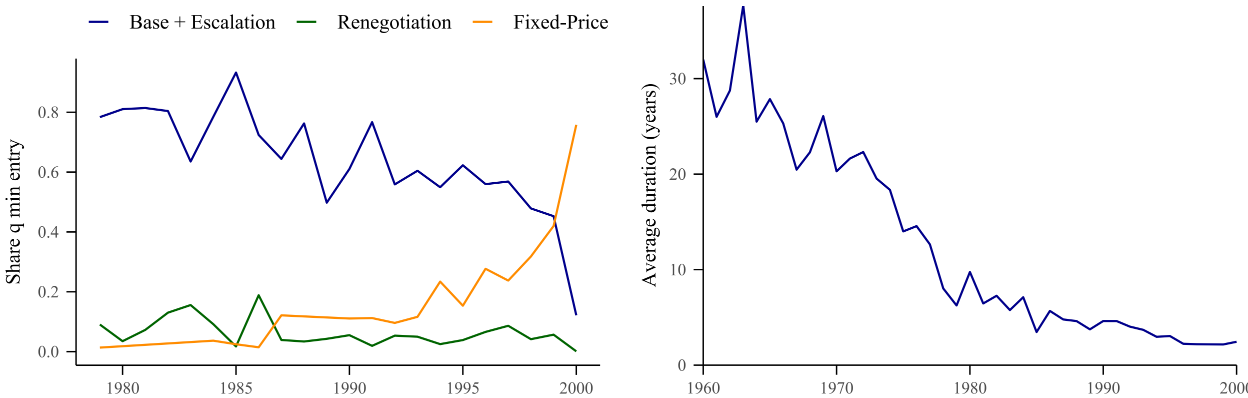
Coal-fired power plants generate electricity by burning coal to produce steam, which drives turbines. The generated electricity is injected into the transmission grid and distributed across the ERCOT region through an organized bidding market. Power plants source the majority of their coal from coal suppliers through contracts, supplementing additional requirements through spot purchases. The bulk of coal is transported from mines to power plants by rail. Additionally, coal quality varies considerably, depending on attributes such as heat content (measured in Btu/lb), sulfur content, and ash levels. Higher heat content directly increases electricity output per ton of coal, whereas sulfur and ash content primarily affect compliance with environmental regulations.

In recent years, coal-fired power plants and coal mines have undergone significant structural changes, with many facilities closing due to stricter emissions regulations and falling natural gas prices driven by the shale gas boom (Davis, Holladay, and Sims, 2022). To isolate our analysis from these confounding factors, we restrict our study to the period 2005–2014, during which coal's share of electricity generation in ERCOT remained relatively stable at 35%–39%. This timeframe also largely precedes the substantial wave of mine closures initiated in the early 2010s.[27]

Contract Types and Vertical Conduct The characteristics of the coal procurement market make it an ideal empirical setting to apply our framework. First, the industry structure aligns closely with our modeling assumptions: empirical evidence supports the

[27] We specifically conclude our sample in 2014, when the average capacity factor for coal power plants dropped notably from 78% to 65%. As noted by Gowrisankaran et al. (2024), this decline was largely due to the rise of fracking, resulting in increased cycling and ramping costs and consequently accelerating coal plant retirements. Modeling ramping costs and exit decisions requires a dynamic framework (Borrero, Gowrisankaran, and Langer, 2024), which is beyond the scope of our static model. Thus, we select a period when ramping considerations were less critical and abstract away from these dynamics.

Figure 4: Coal Contract Characteristics by Signing Year



(a) Contract Type by Signing Year **(b)** Avg. Contract Duration by Signing Year

Notes: The data in this figure come from the EIA's Coal Transportation Rate Database for the years 1979–2000, as described in Section G.1 of the Appendix. Panel (a) shows the share of coal quantity shipped by year and contract type, representing the three main types in the dataset. "Fixed-Price Contracts" are linear price contracts that remain constant throughout the contract's duration. Panel (b) presents the average duration of contracts (in years) by the year they were signed. Contracts signed prior to 1979 appear due to their overlap with the data period.

widespread use of linear price contracts, the literature and antitrust enforcers have expressed concerns about both monopsony and monopoly power, and existing contracts do not necessarily specify which party sets quantities. Second, this market provides a uniquely data-rich environment: we observe detailed transaction-level data, including prices, quantities, and coal characteristics, along with detailed production and cost data for both power plants and coal mines.

While our data do not include contract types during the sample period that we study, historical data from 1980–2000 do provide such information. In this dataset, coal contracts take several forms, including base price plus escalation, market-indexed pricing, cost-plus contracts, and linear price contracts (Kozhevnikova and Lange, 2009). The data indicate that the share of linear price contracts increased from just 3% in 1979 to over 75% by 2000, as shown in Figure 4(a). Assuming that this trend was not reversed after 2000, linear price contracts likely represent the majority of contracts in our sample period.

There is also no clear ex-ante indication as to which vertical conduct prevails in this industry, as concerns about market power have been expressed in the context of both monopsony power of power plants and monopoly power of coal mines. For example, the FTC highlighted increased market power of coal mines as a potential harm when challenging the proposed joint venture between Peabody Energy and Arch Coal (Federal Trade Commission, 2020), whereas several empirical studies have documented evidence of monopsony power exercised by utility firms (Atkinson and Kerkvliet, 1986, 1989; Wilson, O'Boyle, and Lehr, 2020; Kellogg and Reguant, 2021).[28] Moreover, limited evidence from publicly available contracts suggests that both types of quantity-setting arrangements (by buyer or seller) are possible in this industry. Some contracts take the "requirements" structure, which allows downstream firms to determine quantities for a given linear price, whereas others use "take-or-pay" contracts, which set quantities at contract signing and require the downstream buyer to accept the specified volume (Baruya, 2015).[29]

Finally, it is also worth discussing the hold-up problem and coal specificity in this industry, which have been studied by the seminal works of Paul Joskow (Joskow, 1985, 1987, 1988). Since Joskow's papers from the 1980s, coal markets have undergone significant changes that have reduced asset specificity through technological advancements and regulatory reforms. For example, the widespread adoption of scrubbers has made coal more homogeneous from the perspective of power plants, thereby mitigating hold-up concerns (Kacker, 2014). Environmental regulations and reforms in railroad transportation have further diminished supplier specificity by incentivizing boiler upgrades and expanding access to diverse coal markets (Ellerman, Joskow, Schmalensee, Montero, and Bailey, 2000).[30] Collectively, these developments have markedly reduced the reliance on long-term contracts, as illustrated in Figure 4(b).

6.2 Data Sources and Summary Statistics

Our analysis combines data from four sources: Velocity Suite, CostMine, the BLS, and the Mine Safety and Health Administration (MSHA). We describe these data sources in Section G.1 of the Appendix and provide an overview below.

Our coal mine data are obtained from multiple sources. First, we utilize datasets from MSHA, which include quarterly data on production and employment (Form 8) as well as technical characteristics of mines, such as openings and vein thickness (Form 10). Mine ownership details are sourced from Velocity Suite. Mine-level cost data, including detailed operating and capital expenses, come from the 2019 Coal Cost Guide published by CostMine Intelligence, which has been previously used in the mining engineering literature (Shafiee and Topal, 2012) and other academic studies (World Bank, 2017). This guide classifies costs by mine characteristics, such as mining technology, daily capacity,

[28]For other examples of concerns of monopsony power, see Pacific Power (2023) and Neuburger (2024).

[29]For an example of a requirement contract see Coal Supply Agreement Mill Creek and for an example of take-or-pay contract see Armstrong Energy Inc. Contract #598018. In these take-or-pay contracts, the available information typically does not specify which party makes the quantity decision.

[30]The 1990 Clean Air Act Amendment led power plants to adopt technologies accommodating lower-sulfur coal, increasing their fuel flexibility (Ellerman et al., 2000). Supporting this observation, Kacker (2014) finds that during Phase I of the Amendment, plants required to switch technology were more likely to adopt shorter-term and fixed-price contracts compared to unaffected plants. Moreover, the 1980 Railroad Reform expanded the coal market for power plants by reducing transportation costs.

and vein thickness. For wage information, we rely on data from the Quarterly Census of Employment and Wages provided by BLS, calculated as annual averages at the state level and converted to hourly wages.

Power plant characteristics, costs, and generation data are compiled from Velocity Suite, which integrates information from various public sources such as EIA, EPA, NERC, and proprietary research. The datasets include detailed monthly generator characteristics, hourly generation and emissions data from the EPA's Continuous Emissions Monitoring System (CEMS), EIA-923, ERCOT hourly load data, and ERCOT's 60-Day SCED Disclosure Reports. Lastly, coal transaction data are sourced from Velocity Suite, which combines information on transaction details, prices, transportation costs, and contractual arrangements, primarily based on EIA-923 forms supplemented with internal research and railroad waybills.

Table 2 presents summary statistics for our empirical sample, which includes all mining firms supplying coal to ERCOT power plants and all electricity-generation firms (power firms) operating coal-fired plants in ERCOT. The sample consists of nine upstream mining firms operating 25 mines and two downstream power firms operating 19 coal-fired generators. Mining firms deal with 22.1 partners on average, including those outside of ERCOT, while power firms engage with an average of 2.55 partners, suggesting that sourcing from multiple trade partners is common. The average share of the largest trading partner is about half in both upstream and downstream. We provide the list of upstream and downstream firms in Table OA-4 of the Appendix.

Panel B of Table 2 provides summary statistics on transactions. The transactions typically take the form of medium-term contracts, with an average duration of 1.22 years and an average price of $0.84 per MMBtu. Most transactions occur through contracts, with spot market transactions accounting for 3% of total transactions.

6.3 Model Primitives

We examine coal price negotiations between mining and power firms. We begin by estimating the model's primitives: the cost curve of mining firms, the cost curve of power firms, and the residual electricity demand faced by power firms. Although we perform these estimations annually, we omit year subscripts for notational clarity. Section G of the Appendix provides the details of the estimation procedures, and Table OA-2 summarizes the model's notation.

Table 2: Summary Statistics

	Upstream	Downstream
	Panel A. Firm Characteristics	
Number of units (mine or generator)	25	19
Number of firms	9	2
Avg. number of units per firm	2.51	3.30
Avg. number of trade partners	22.15	2.55
Avg. share of largest partner	0.42	0.53
	Panel B. Transaction Characteristics	
Avg. FOB price (USD per MMBtu)	-	0.84
Avg. Contract duration (years)	-	1.22
Share of spot-market transactions	-	0.03

Notes: This table presents the summary statistics for the sample used in our empirical analysis. Panel A reports the characteristics of mining firms (upstream) and power firms (downstream), whereas Panel B provides information on transaction details. The sample includes all mining firms supplying coal to ERCOT power plants and all power firms operating coal-fired plants in ERCOT between 2005 and 2014.

Cost Curve of Mining Firms

Each upstream mining firm u operates a portfolio of n_u mines indexed by i. Each mine has capacity $\bar{q}^c_{iu}$ and constant marginal cost c_{iu}, determined by mine characteristics and labor costs.[31] To estimate marginal cost, we first specify the mine production function. Mine i produces q^c_{iu} short tons of coal using l_{iu} labor hours and m_{iu} units of intermediate inputs according to a Leontief production function:

$$q^c_{iu} = \min\{l_{iu}\gamma_{\theta(iu)}, m_{iu}\}\omega_{iu}$$

The Leontief specification reflects the limited input substitution possibilities in short-run coal production (Byrnes, Färe, Grosskopf, and Knox Lovell, 1988), assuming perfect complementarity between labor and intermediate inputs, and has been applied for other resource-extracting industries, such as oil drilling (Asker et al., 2019). Two key parameters capture technological heterogeneity in the production function: (i) $\gamma_{\theta(iu)}$, which determines the labor-materials ratio based on mine type θ (characterized by the combination of capacity bin, vein thickness, and mining technology), and (ii) ω_{iu}, which accounts for mine-specific productivity differences.[32]

[31] The EIA collects data on coal mine capacity, which have been used in prior research. However, the EIA no longer makes these data available to researchers. Consequently, we infer mine capacity from production data. For each year, we define a mine's capacity as the maximum historical production observed at that mine up to that year. This approach makes mine capacity time-varying, as it reflects changes in production over time.

[32] The Coal Cost Guide provides 69 different mine types θ, of which 16 occur in our dataset.

Given hourly wages w^l_{iu} and material costs p^m_{iu}, the Leontief production function implies the following marginal cost function:

$$c_{iu} = w^l_{iu} \frac{l_{iu}}{q^c_{iu}} \left(1 + \gamma_{\theta(iu)} \frac{p^m_{iu}}{w^l_{iu}}\right) \quad \text{if} \quad q^c_{iu} \leq \bar{q}^c_{iu}. \tag{5}$$

While we can estimate unit labor costs by multiplying wages (w^l_{iu}) and mine-level output-per-labor (l_{iu}/q^c_{iu}), unit material costs are not directly estimable because our data do not include materials expenditures at the mine level. To address this, we utilize the Coal Cost Guide, published by the industry research firm CostMine, which provides engineering estimates of both labor and material unit costs for different mine types θ. From these data, we calculate the materials-to-labor cost ratio, equal to $\gamma_{\theta(iu)}(p^m_{iu}/w^l_{iu})$ for mines of type θ, enabling us to recover marginal costs in Equation (5).

Coal's value in electricity generation depends primarily on its heat content (measured in millions of British thermal units, or MMBtu) rather than its weight. To convert between these measures, we define a mine-specific conversion factor λ_{iu}, which equals heat content per short ton for mine i. This factor transforms weight-based quantities into heat-based quantities through $q_{iu} = \lambda_{iu} q^c_{iu}$. The conversion factor λ_{iu} varies by coal type and mining area, representing an important source of heterogeneity across mines. Following this conversion, we express all coal quantities and mine capacities in MMBtu and coal costs per MMBtu for the remainder of the paper.

Using mine marginal costs, we construct firm-level cost curves by ordering each firm's mines from lowest to highest marginal cost and calculating cumulative production cost. This aggregation yields the following firm-level cost function:

$$C_u(Q, c_u, \bar{q}_u) = \begin{cases} \sum_{i=1}^{n_u} c_{iu} \max\left\{0, \min\left[\bar{q}_{iu}, Q - \sum_{l=1}^{i-1} \bar{q}_{lu}\right]\right\}, & \text{if } 0 \leq Q \leq \sum_{i=1}^{n_u} \bar{q}_{iu}, \\ \infty, & \text{if } Q > \sum_{i=1}^{n_u} \bar{q}_{iu} \end{cases}$$

where the vector $c_u := \{c_{iu}\}_{i=1}^{n_u}$ is such that $c_{1u} \leq c_{2u} \leq \cdots \leq c_{n_u u}$ and $\bar{q}_u$ is the vector of mine capacities.

Cost Curve of Power Firms

Each downstream power firm d operates a portfolio of n_d generation assets. Asset j is characterized by a constant marginal cost c_{jd} and a capacity k_{jdt}. The capacity can vary over time due to intermittency in renewable resources across seasons and hours of the day. We therefore define hourly capacity for each time type t, representing specific combinations of month, hour, and weekend/weekday status.[33]

The marginal cost of a fossil-fuel generator depends on fuel prices and its efficiency, which is measured by the heat rate. We define marginal costs as follows:

$$c_{jd} = \begin{cases} (w_d + \kappa_{jd})h_{jd} + m_{jd} & \text{if coal} \\ w^{gas} h_{jd} + m_{jd} & \text{if gas} \\ 0 & \text{if nuclear and renewables} \end{cases}$$

where h_{jd} represents generator j's (inverse) heat rate, w^{gas} is the natural gas price common to all gas generators, w_d is firm d's weighted average FOB coal price, and m_{jd} represents the maintenance cost. For coal generators, we also add the per MMBtu average transportation cost κ_{jd} to generator j.[34] We assume constant heat rates within a year for each generator, calculated as total heat input divided by total electricity generation.[35]

To determine capacity k_{jdt}, we cannot rely solely on nameplate capacity, due to maintenance downtime in fossil-fuel units and intermittency in renewables. Instead, we calculate an "effective" capacity by multiplying the nameplate capacity by a capacity factor. For fossil-fuel units, we obtain annual, fuel-type-specific capacity factors from the Generating Availability Data System (GADS) maintained by the Federal Energy Regulatory Commission (FERC). For renewable units, we derive a time-varying capacity factor for each unit as the ratio of average hourly generation in each hour type t to the nameplate capacity.[36]

Using unit-level capacity and cost data, we construct firm d's cost function in hour type t by ordering all units in ascending order of marginal cost and calculating their cumulative production cost. The resulting cost curve takes the following form:

$$C_{dt}(Q, c_d, k_{dt}) = \begin{cases} \sum_{j=1}^{n_d} c_{jd} \max\left\{0, \min\left[k_{jdt}, Q - \sum_{l=1}^{j-1} k_{ldt}\right]\right\}, & \text{if } 0 \le Q \le \sum_{j=1}^{n_d} k_{jdt}, \\ \infty, & \text{if } Q > \sum_{j=1}^{n_d} k_{jdt} \end{cases}$$

where $c_d := \{c_{jd}\}_{j=1}^{n_d}$ is the vector of marginal costs ordered such that $c_{1d} \le c_{2d} \le \cdots \le c_{n_d d}$, and k_{dt} is the vector of generator capacities.

[33] For example, 8 a.m. on a weekday in January represents one hour type t in our model. Even though capacity does not vary meaningfully across weekdays and weekends, we include it to capture variation in demand.

[34] We treat transportation costs κ_{jd} as exogenous and do not model railroad firms as separate agents in the value chain. Prior research shows that railroad companies could have significant market power in coal procurement markets (Preonas, 2023). In our model, upstream agents can be interpreted as jointly representing coal firms and railroad operators. For instance, if coal firms and railroad companies bargain efficiently, the upstream entity can be viewed as maximizing their joint profits. Thus, double marginalization attributed to coal firms can alternatively be interpreted as arising from railroad companies' market power.

[35] We calculate the marginal cost using coal prices from the transaction data, transportation costs, and the heat rate. This calculation assumes the coal is blended without a specific order if there are multiple coal suppliers.

[36] The capacity factors are, on average, 92, 90, 29, and 21% for coal, gas, wind, and solar, respectively.

Downstream Electricity Demand and Profit

We model competition in the electricity market using a Cournot framework, following the prior literature (Borenstein and Bushnell, 1999; Borenstein et al., 2002; Puller, 2007). In this model, regulated firms and small firms (< 5% market share) act as price takers, while larger firms behave strategically. Although only two power firms operate coal-fired generators, our demand model includes all power firms with generation capacity in ERCOT.[37] Since both demand and capacity fluctuate hourly, we estimate a separate Cournot model for each hour type t. The expected demand curve faced by strategic firms is given by

$$Q_t(P) = \bar{Q}_t^D - Q_t^{\text{fr}}(P),$$

where $\bar{Q}_t^D$ denotes the expected inelastic demand during hour type t, and $Q_t^{\text{fr}}(P)$ denotes the quantity supplied by the competitive fringe firms at a price P. We assume that firms have rational expectations, so $\bar{Q}_t^D$ equals the mean inelastic demand in hour type t.[38] Let $P_t(Q)$ denote the expected inverse demand curve faced by strategic firms. The profit function of firm d in hour type t is given by

$$\pi_t^d(Q_{dt}, C_{dt}) = P_t(Q_{-dt} + Q_{dt})Q_{dt} - C_{dt}(Q_{dt}),$$

where Q_{-dt} denotes the total production of all strategic firms except firm d. The annual profit function is obtained by summing over hourly profits:

$$\pi^d(Q_d, C_d) = \sum_t f_t \pi_t^d(Q_{dt}, C_{dt}). \tag{6}$$

Here, f_t is the number of occurrences of hour type t in a year, $Q_d = \{Q_{dt}\}_{t=1}^{n_t}$ is the vector of quantities, and $C_d = \{C_{dt}\}_{t=1}^{n_t}$ denotes the set of cost functions of firm d.

Upstream Profit

Each mining firm u faces a set of coal buyers D_u, where quantity q_{ud} is traded with each buyer d at price w_{ud}. The mining firm's profit function is the total revenue from these transactions minus the total production cost:

$$\pi^u(w_u, q_u) = \sum_{d \in D_u} w_{ud} q_{ud} - C_u\left(\sum_{d \in D_u} q_{ud}\right). \tag{7}$$

[37] Five strategic and 247 fringe firms are present in our data. In Section G.6 of the Appendix, we present the details of how we implement the Cournot model.

[38] The realized demand is given by $Q_{\tau t} = \bar{Q}_t^D + \epsilon_{\tau t}$, where τ is an hour within an hour type t and $\epsilon_{\tau t}$ is the error term. We compute the expected demand by taking the average of realized demand within t as $\mathbb{E}_t[\bar{Q}_t^D + \epsilon_{\tau t}]$.

Here, w_u and q_u represent the vector of all prices and quantities for firm u.

6.4 Joint-Profit Maximizing Quantity Calculation and Conduct Identification

After constructing the cost and demand primitives and firms' profit functions, the next step is conduct identification. For this, we use Proposition 5, which requires the joint-profit maximizing quantity q^*_{ud} and the associated input price w^*_{ud} for each trading pair. To find q^*_{ud}, we first write the gains from trade. If bargaining between u and d breaks down, upstream obtains a disagreement payoff π^u_0 and downstream obtains π^d_0. Using the profit functions defined in Equations (7) and (6), the gains from trade are:

$$\mathrm{GFT}^u_{ud} = \pi^u(w_u, q_u) - \pi^u_0, \qquad \mathrm{GFT}^d_{ud} = \pi^d(Q_d, C_d) - \pi^d_0$$

Since the disagreement payoffs are constants that do not depend on q_{ud}, the quantity that maximizes the joint gains from trade also maximizes the sum of profits:

$$q^*_{ud} = \underset{q_{ud}}{\operatorname{argmax}} \left\{GFT^u_{ud} + GFT^d_{ud}\right\} = \underset{q_{ud}}{\operatorname{argmax}} \left\{\pi^u + \pi^d\right\} \tag{8}$$

By Proposition 2, the joint-profit maximizing quantity q^*_{ud} and the associated input price $w^*_{ud} = mc_{ud}(q^*_{ud}) = mr_{ud}(q^*_{ud})$ are the same under both monopsonistic and monopolistic conduct, and depend only on cost and demand primitives, not on the bargaining weights or the disagreement payoffs.

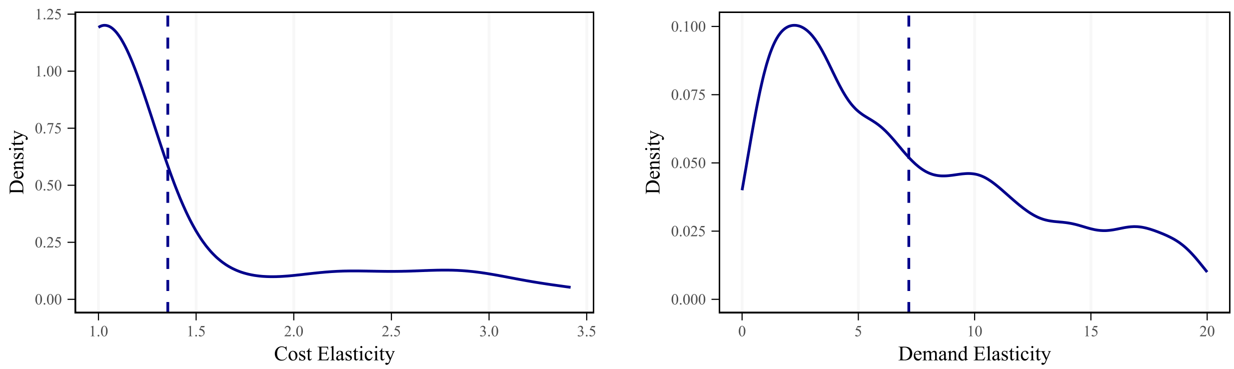
6.5 Estimation

We estimate the model separately for each contracting pair (mining and power firms) by year. The estimation proceeds in two steps: we first estimate the cost and demand primitives, and then use them to compute w^*_{ud} and q^*_{ud} for each pair-year. Conduct is then identified by comparing the observed coal price w_{ud} to w^*_{ud}, following Proposition 5.

To estimate the primitives, we first construct the residual demand curve. We assume total electricity demand is fully inelastic in the short run and compute this demand by averaging observed demand within each hour type. To estimate fringe supply, we derive the marginal cost curve for each fringe firm and aggregate these curves to the industry level. We then assume fringe firms supply quantities in each hour such that the market price equals their marginal cost. Subtracting this fringe supply curve from the inelastic total demand yields the residual industry demand curve faced by strategic firms, which we use in the estimation of the Cournot model.

Using the estimated demand and cost functions, we compute q^*_{ud} for each contracting pair using Equation (8), and w^*_{ud}, at which the marginal cost of upstream production (from

Figure 5: Distribution of Cost and Demand Elasticities



(a) Upstream Cost Estimates **(b)** Downstream Demand Estimates

Notes: Panel (a) presents a kernel density estimates of the distribution of the elasticity of the cost function of mining firms (which is equal to $1 + \frac{dmc(q)}{dq}\frac{q}{mc(q)}$). A cost elasticity of one implies constant marginal costs. Panel (b) presents a kernel density estimates of the distribution of the absolute value of the residual elasticity of demand of power firms (the elasticity of $p^{-1}(q)$ in our notation). Each observation corresponds to a mining firm-year in Panel (a) and an hour-type power firm in Panel (b). Since cost and demand functions are estimated nonparametrically, we report the elasticities at the observed quantity levels in Panel (a) and Panel (b). The dashed vertical line indicates the average in each panel.

Equation (7)) equals the marginal revenue of downstream sales (from Equation (6)). We then classify vertical conduct for each pair-year by comparing the observed coal price to w^*: $w_{ud} > w^*_{ud}$ indicates monopolistic conduct and $w_{ud} < w^*_{ud}$ indicates monopsonistic conduct.

The only source of uncertainty in our model is the inelastic demand curve in the market in a given hour type. To account for this uncertainty, we implement a bootstrap procedure, resampling the demand in each hour within hour type with replacement. Section G of the Appendix provides details on the estimation procedure.

6.6 Results

Cost and Demand Elasticity Estimates

Figure 5(a) presents the distribution of estimated cost elasticities by mine-year. The estimates cluster around one, indicating that many mines operate under approximately constant marginal costs; however, several mines exhibit elasticities ranging from 1.5 to 3.5. This variation primarily arises from geographic differences. For instance, mines in the Powder River Basin (Wyoming and Montana) are characterized by large-capacity surface operations with relatively flat marginal cost curves, while mines in the Gulf region tend to be smaller underground facilities with more heterogeneous marginal cost structures. We find an average materials-to-labor cost ratio $\gamma_{\theta(iu)}(p^m_{iu}/w^l_{iu})$ of 2.32, which indicates that material costs are on average more than twice as high as labor costs. However, there is large heterogeneity across mines, with this ratio being merely 0.31 at the 10th percentile of mines, but 5.53 at the 90th percentile.

Figure 5(b) presents the distribution of estimated residual demand elasticities, with each observation corresponding to an hour-firm pair. The estimates exhibit considerable variation, though most are concentrated between 1 and 4. This variability primarily arises from changes in the shape of the fringe firms' supply curve across different times of day and seasons. Additionally, we estimate the average market-level demand elasticity faced by strategic firms to be -0.84, which is broadly consistent with Puller (2007), who reports an average demand elasticity of -1.24 in the California electricity market.

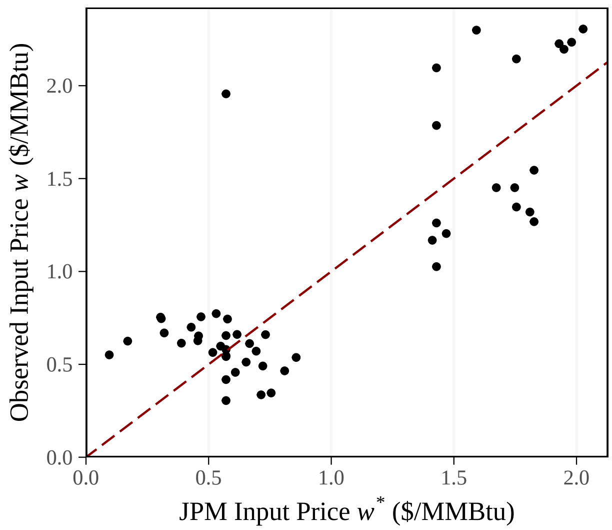
Conduct Identification

We identify vertical conduct for each contracting pair by comparing the observed coal price w_{ud} to w^*_{ud}, following Proposition 5. Figure 6(a) makes this comparison for each buyer-seller-year combination. Observations above the 45-degree line ($w_{ud} > w^*_{ud}$) indicate monopolistic conduct, while observations below ($w_{ud} < w^*_{ud}$) indicate monopsonistic conduct. The majority of observations lie above the 45-degree line, indicating that monopolistic conduct prevails in most trading relationships. This result implies that most output distortion in this setting originates from seller market power rather than buyer power, leading primarily to double marginalization.

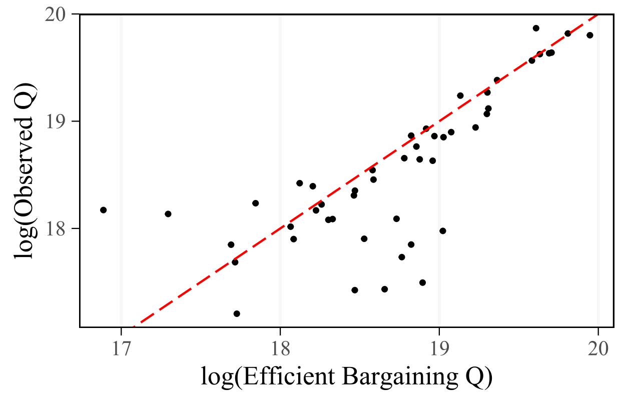
We also test the nonnegative markup/markdown constraint using Proposition 4.2, which states that observed output quantities should not exceed the joint-profit maximizing quantity q^*. Figure 6(b) illustrates this comparison, showing observed quantities relative to joint-profit maximizing quantities for each trading relationship. With only a few exceptions, which can be attributed to statistical noise, the observed output is consistently lower than the joint-profit maximizing output, lending empirical support to the nonnegative markup/markdown constraint. To test this relationship formally, we rely on moment inequalities using the implication that $E[\log(q)] < E[\log(q^*)]$. By constructing these moments from the data, we find that the average log quantity is statistically significantly larger than the average log observed quantity (difference coef. = 0.069, s.e. = 0.003).

As a model validation exercise, our dataset includes transactions between a vertically integrated buyer and seller. We find substantially lower output distortion for the vertically integrated pair compared to other transactions (3.51% vs. 14.7%), consistent with the prediction that vertical integration reduces double marginalization.

Figure 6: Conduct Identification and Model Testing



(a) Observed vs. Bilaterally-Efficient Coal Price



(b) Observed vs. Bilaterally-Efficient Quantity

Notes: Panel (a) plots w_{ud} against w^*_{ud} for each buyer-seller-year combination. Points above the 45-degree line (dashed red) indicate monopolistic conduct ($w > w^*$), and points below indicate monopsonistic conduct ($w < w^*$). Panel (b) compares the logarithm of observed coal-transaction quantities $\log(q)$ to the logarithm of the joint-profit maximizing quantity $\log(q^*)$. Under the nonnegative markup/markdown constraint, all observations should lie below the 45-degree line.

Decomposing Vertical Distortions: Monopolistic and Monopsonistic Conduct

We use the estimated model to quantify the total vertical distortion and decompose it into components attributable to monopsony power and monopoly power. Since short-run electricity demand is inelastic, market power distortions in electricity markets arise primarily from allocative inefficiency rather than lost output (Borenstein et al., 2002). Specifically, monopoly and monopsony distortions induce strategic firms to produce less than they would in the absence of market power, shifting production to higher-cost fringe firms. Accordingly, we measure the vertical distortion as the electricity generation allocated to higher-cost fringe firms from strategic firms due to double marginalization or monopsony power.

To decompose the total vertical distortion into monopsony and monopoly components, we first calculate the difference between the observed output and joint-profit maximizing output for each trading pair, separately for trades classified as monopsonistic and monopolistic using the w vs. w^* criterion. This gives us both the total underproduction of strategic firms compared to the joint-profit maximizing benchmark and a decomposition of this amount into monopsonistic and monopolistic components.

Table 3: Decomposing Vertical Distortions

	Estimates
	Panel A. Total Quantity Distortion (%)
Percent Misallocated Quantity	8.03 (0.67)
	Panel B. Distortion Decomposition (%)
Due to Monopsony Power	25.10 (4.97)
Due to Monopoly Power	74.90 (4.97)

Notes: This table reports the share of total misallocated quantity (additional output produced by fringe firms compared to a coal market with no vertical distortion) and decomposes it into its monopsony and monopoly power components. Numbers in parentheses are bootstrapped standard errors reported for the percentages.

The results are presented in Table 3. We estimate that the total misallocation in the ERCOT market corresponds to 8.03% of total output from coal transactions. In other words, the observed quantities are 8.03% lower than what would be achieved under joint-profit maximization. This suggests that total efficiency losses are modest, consistent with the observation in Figure 6(a) that most observed coal prices lie close to the bilaterally-efficient coal price.

In terms of sources, 74.9% of the vertical distortion is attributed to double marginalization resulting from the monopoly power of coal mining firms, while the remaining portion is due to the monopsony power of power companies. Therefore, in this market, an increase in buyer power is likely to be countervailing, whereas an increase in seller power could be further distortionary.

7 Concluding Remarks

Vertical relationships between buyers and sellers are studied across a variety of settings, ranging from labor unions to healthcare markets, to quantify market distortions under monopsony/oligopsony and double-marginalization settings. In this paper, we introduce a unified framework that nests both monopsonistic and monopolistic (double marginalization) vertical conduct. We first show that with increasing upstream marginal costs and decreasing downstream marginal revenues, an equilibrium exists under both conduct types with distinct welfare implications. We then provide a method to determine which type of vertical distortions emerge based on the relative bargaining positions of buyers and sellers and the underlying primitives of the cost and demand functions.

We illustrate our model with three empirical applications: labor unions, farmer cooperatives, and sellers facing decreasing returns to scale. In our main empirical application, we use the model to quantify the sources of vertical distortions in coal procurement by power plants in Texas. We find that inefficiencies mainly come from double marginalization due to mining firms' monopoly power rather than from the monopsony power of power plants.

Our results provide several insights into antitrust policy. For horizontal mergers, we characterize the conditions under which changes in the bargaining power of upstream and downstream firms are distortionary or countervailing. Under monopolistic conduct, increased buyer power countervails double-marginalization distortions and increases welfare, whereas under monopsonistic conduct, increased buyer power increases monopsony distortions and reduces welfare. For vertical mergers, our framework provides an approach to evaluate potential efficiencies from eliminating double marginalization using the distance between the actual and bilaterally-efficient levels of buyer power.

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Welfare Effects of Buyer and Seller Power

Mert Demirer, Michael Rubens

Online Appendix

Contents

A Existence of Equilibrium and the Bilaterally-Efficient Bargaining Weight

This section establishes the conditions under which (i) an equilibrium with positive trade exists and (ii) the joint-profit maximizing quantity q^* is achievable. We also characterize what happens when the joint-profit maximizing quantity cannot be reached.

A.1 Gains-from-Trade Assumption

Under each conduct model, the input price is pinned down by q (via $w = mc(q)$ under monopsonistic conduct or $w = mr(q)$ under monopolistic conduct), so the profit of each party becomes a function of q alone:

Monopsonistic conduct ($w = mc(q)$):

$$\pi^u(q) = c'(q)\, q^2, \tag{OA.1}$$
$$\pi^d(q) = \big(p(q) - c(q) - c'(q)\, q\big)\, q. \tag{OA.2}$$

Monopolistic conduct ($w = mr(q)$):

$$\pi^d(q) = -p'(q)\, q^2, \tag{OA.3}$$
$$\pi^u(q) = \big(p(q) + p'(q)\, q - c(q)\big)\, q. \tag{OA.4}$$

We impose the following assumption:

Assumption OA-1 (Gains from Trade)**.** *Under each conduct model, the set of quantities at which both participation constraints are satisfied is nonempty. Specifically, define the feasible set under each conduct model as:*
Under monopsonistic conduct ($w = mc(q)$):

$$\mathcal{Q}^{ms} \equiv \left\{ q > 0 : (p(q) - c(q) - c'(q)q)q \geq \pi_0^d \ \textit{and} \ c'(q)q^2 \geq \pi_0^u \right\} \neq \emptyset.$$

Under monopolistic conduct ($w = mr(q)$):

$$\mathcal{Q}^{mp} \equiv \left\{ q > 0 : -p'(q)q^2 \geq \pi_0^d \ \textit{and} \ (p(q) + p'(q)q - c(q))q \geq \pi_0^u \right\} \neq \emptyset.$$

This assumption ensures that under each conduct model, there is a nonempty set of output levels at which both parties prefer trade to their disagreement payoffs.

A.2 When Is the Joint-Profit Maximizing Quantity Achievable?

Proposition OA-1 (Conditions for $\beta^* \in (0, 1)$)**.** *The joint-profit maximizing quantity q^* is achievable under both conduct models if and only if each party's profit at q^* (Lemma OA-10)* *exceeds its disagreement payoff:*

$$\pi_0^d < \pi^{d*} \equiv -p'(q^*)(q^*)^2, \tag{OA.5}$$
$$\pi_0^u < \pi^{u*} \equiv c'(q^*)(q^*)^2. \tag{OA.6}$$

These conditions require that each party's profit at the joint-profit maximizing quantity q^* (under the input price $w^* = mc(q^*) = mr(q^*)$) exceeds its disagreement payoff. When both conditions hold, there exists a unique $\beta^* \in (0, 1)$ such that the equilibrium output equals q^* under both conduct models.

Proof. Necessity: At q^* under monopsonistic conduct, $\pi^u(q^*) = c'(q^*)(q^*)^2 = \pi^{u*}$ and $\pi^d(q^*) = -p'(q^*)(q^*)^2 = \pi^{d*}$. For the equilibrium to achieve q^*, both parties must prefer trade at q^* to their disagreement payoffs: $\pi^u(q^*) > \pi_0^u$ and $\pi^d(q^*) > \pi_0^d$, which gives conditions (OA.5)–(OA.6).

Under monopolistic conduct, the same profit values obtain at q^* (since $mc(q^*) = mr(q^*)$ implies the same w^* and hence same profits), so the same conditions are necessary.

Sufficiency: When both conditions hold, Proposition 2 in Section C.4 derives the formula for β^* and verifies $\beta^* \in (0, 1)$. □

A.3 When the Joint-Profit Maximizing Quantity Is Not Achievable

When conditions (OA.5)–(OA.6) fail but Assumption OA-1 holds, trade still occurs but the joint-profit maximizing quantity q^* cannot be reached. We show that in these cases, the equilibrium output nearest to q^* is achieved at a corner value of β.

The proof relies on four facts: (i) by Assumption OA-1 and Proposition 1(iii), an equilibrium exists for every $\beta \in [0, 1]$, so $q(\beta)$ is defined on the full interval; (ii) $q(\beta)$ is continuously differentiable in β (Lemma OA-11); (iii) by Theorem 1, $q(\beta)$ is strictly monotone ($dq^{ms}/d\beta < 0$ and $dq^{mp}/d\beta > 0$); (iv) when $\beta^* \notin (0, 1)$, then q^* is not in the range of $q(\beta)$, so by continuity and the intermediate value theorem, the entire range lies on one side of q^*.

Proposition OA-2. *If $\pi_0^d > \pi^{d*}$ (downstream disagreement payoff too high) and Assumption OA-1 holds, then for every $\beta \in [0, 1]$, the equilibrium output satisfies $q(\beta) < q^*$ under monopsonistic conduct and $q(\beta) > q^*$ under monopolistic conduct. The output nearest to q^* is achieved at $\beta = 0$ under both conduct models.*

Proof. Since $\pi^d(q^*) = \pi^{d*} < \pi_0^d$, the equilibrium at any β cannot achieve exactly q^* (both parties must earn strictly above their disagreement payoffs at the Nash bargaining optimum, but downstream earns $\pi^{d*} < \pi_0^d$ at q^*). Therefore $q(\beta) \neq q^*$ for any β.

Monopsonistic conduct: Since $q^{ms}(\beta)$ is strictly decreasing in β and $q^{ms}(\beta) \neq q^*$ for any β, the entire range of q^{ms} lies on one side of q^*. At $\beta = 1$ (classical monopsony), $q^{ms}(1)$ is the minimum output, which satisfies $q^{ms}(1) < q^*$ (since at $\beta = 1$, upstream earns $\pi^u > \pi_0^u$ while downstream is driven toward its disagreement payoff, and the resulting output is below q^* when $\pi_0^u < \pi^{u*}$). Since q^{ms} is decreasing and $q^{ms}(1) < q^*$, we have $q^{ms}(\beta) < q^*$ for all β (if $q^{ms}(0) \geq q^*$, by the intermediate value theorem there would exist β with $q^{ms}(\beta) = q^*$, contradicting the above). Therefore $q^{ms}(\beta) < q^*$ for all β. The maximum output (nearest to q^*) is at $\beta = 0$.

Monopolistic conduct: Since $q^{mp}(\beta)$ is strictly increasing in β and $q^{mp}(\beta) \neq q^*$ for any β, the entire range lies on one side of q^*. At $\beta = 0$ (Spengler double marginalization), the upstream firm has full bargaining power and the downstream participation constraint $\pi^d(q) = -p'(q)q^2 \geq \pi_0^d$ requires $q \geq \underline{q}^d$ where $\underline{q}^d \geq q^*$ (since $\pi^d(q^*) = \pi^{d*} < \pi_0^d$ and π^d is increasing). Since q^{mp} is increasing and $q^{mp}(0) \geq q^*$, we have $q^{mp}(\beta) \geq q^*$ for all β. The minimum output (nearest to q^*) is at $\beta = 0$. □

Proposition OA-3. *If $\pi_0^u > \pi^{u*}$ (upstream disagreement payoff too high) and Assumption OA-1 holds, then for every $\beta \in [0, 1]$, the equilibrium output satisfies $q(\beta) \geq q^*$ under monopsonistic conduct and $q(\beta) \leq q^*$ under monopolistic conduct. The output nearest to q^* is achieved at $\beta = 1$ under both conduct models.*

Proof. The argument is symmetric to Proposition OA-2 with the roles of upstream and downstream interchanged.

Monopsonistic conduct: The upstream participation constraint $\pi^u(q) = c'(q)q^2 \geq \pi_0^u$ requires $q \geq \underline{q}^u$ where $\underline{q}^u \geq q^*$ (since $\pi^u(q^*) = \pi^{u*} < \pi_0^u$ and π^u is increasing). At $\beta = 0$, $q^{ms}(0) \geq q^*$. Since q^{ms} is decreasing and no β achieves q^*, the entire range satisfies $q^{ms}(\beta) \geq q^*$. The minimum output (nearest to q^*) is at $\beta = 1$.

Monopolistic conduct: At $\beta = 1$ (buyer's TIOLI), $q^{mp}(1)$ is the maximum output, which satisfies $q^{mp}(1) < q^*$ when $\pi_0^u > \pi^{u*}$. Since q^{mp} is increasing and $q^{mp}(1) < q^*$, the entire range satisfies $q^{mp}(\beta) \leq q^*$. The maximum output (nearest to q^*) is at $\beta = 1$. □

Summary. In both cases, the corner value of β that achieves the output nearest to q^* is the same under both conduct models: $\beta^* = 0$ when $\pi_0^d > \pi^{d*}$ (Case 2) and $\beta^* = 1$ when $\pi_0^u > \pi^{u*}$ (Case 3). We adopt this as the definition of β^* whenever the joint-profit maximizing quantity is not achievable.

A.4 Illustration

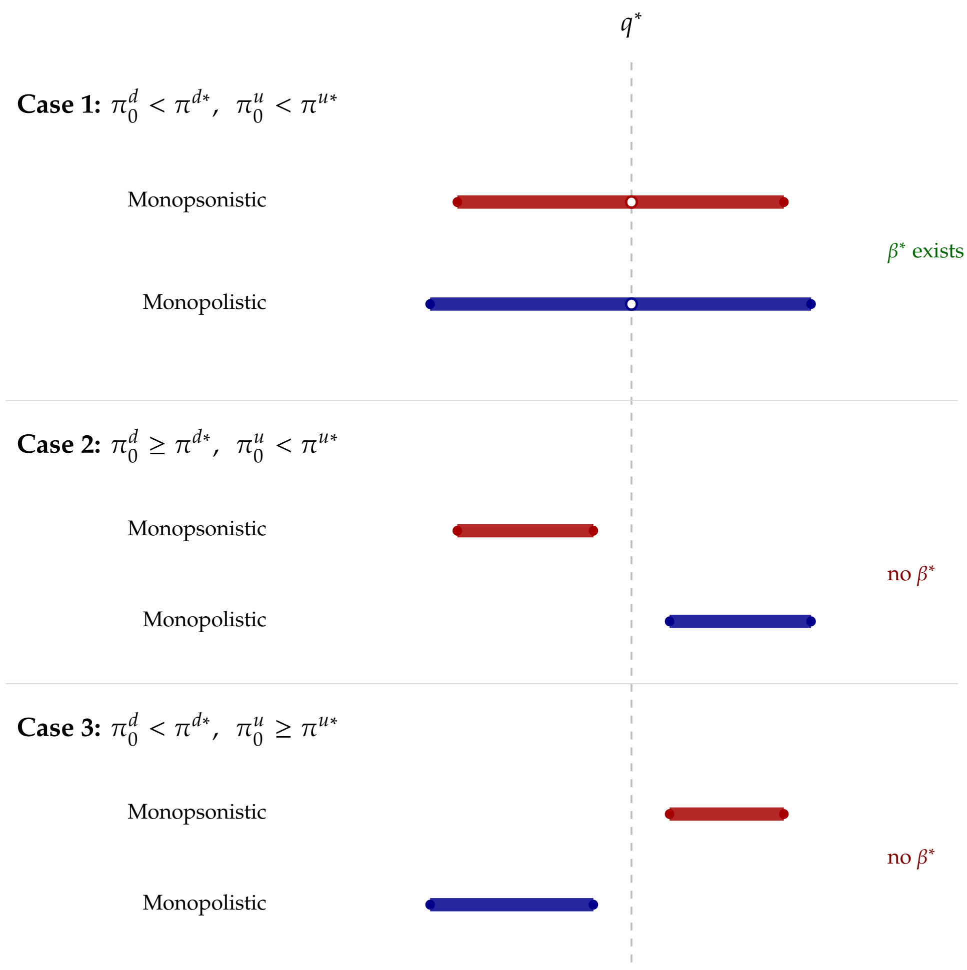


Figure OA-1: Summary of feasible quantity sets across all cases and conduct models.

Notes: Each bar represents the feasible quantity set under the indicated conduct model and disagreement-payoff regime. The dashed vertical line marks q^*; open circles indicate $q^* \in \Omega$. In Case 1, β^* exists. In Cases 2 and 3, q^* is not feasible and the feasible set lies on one side of q^*. Red bars: monopsonistic; blue bars: monopolistic.

B First- and Second-Order Conditions

This section provides the first- and second-order conditions of monopolistic and monopsonistic bargaining, their derivations, and sufficient conditions for the second-order conditions to hold.

B.1 First-Order Conditions

Under the sequential bargaining models with disagreement payoffs $\pi_0^d \geq 0$ and $\pi_0^u \geq 0$, the maximization problems are given by:

$$\begin{cases} \max\limits_{q^d} \; p(q^d)\, q^d - w\, q^d & \text{(Downstream's problem)} \\ \max\limits_{q^u} \; w\, q^u - c(q^u)\, q^u & \text{(Upstream's problem)} \\ \max\limits_{w} \; \left[\left(p\left(q^d(w)\right) q^d(w) - w\, q^d(w) - \pi_0^d \right)^{\beta} \left(w\, q^d(w) - c\left(q^d(w)\right) q^d(w) - \pi_0^u \right)^{1-\beta} \right] & \text{(MP bargaining)} \\ \max\limits_{w} \; \left[\left(p\left(q^u(w)\right) q^u(w) - w\, q^u(w) - \pi_0^d \right)^{\beta} \left(w\, q^u(w) - c\left(q^u(w)\right) q^u(w) - \pi_0^u \right)^{1-\beta} \right] & \text{(MS bargaining)} \end{cases}$$

For $\beta \in (0,1)$ and under Assumption OA-1, the equilibrium satisfies $\pi^d(q) > \pi_0^d$ and $\pi^u(q) > \pi_0^u$ (both parties earn strictly above their disagreement payoffs) and is characterized by the following first-order conditions, derived in Section B.3[39]:

$$\begin{cases} w = p'(q)q + p(q) = mr(q) & \text{(D-FOC)} \\ w = c'(q)q + c(q) = mc(q) & \text{(U-FOC)} \\ \beta \left(\dfrac{-q + (p'(q)q + [p(q) - w])\,(dq^d/dw)}{[p(q) - w] \cdot q - \pi_0^d} \right) + (1-\beta) \left(\dfrac{q + ([w - c(q)] - c'(q)q)\,(dq^d/dw)}{[w - c(q)] \cdot q - \pi_0^u} \right) = 0 & \text{(D-B-FOC)} \\ \beta \left(\dfrac{-q + (p'(q)q + [p(q) - w])\,(dq^u/dw)}{[p(q) - w] \cdot q - \pi_0^d} \right) + (1-\beta) \left(\dfrac{q + ([w - c(q)] - c'(q)q)\,(dq^u/dw)}{[w - c(q)] \cdot q - \pi_0^u} \right) = 0 & \text{(U-B-FOC)} \end{cases}$$

where $dq^d/dw = 1/(2p'(q) + p''(q)q)$ and $dq^u/dw = 1/(2c'(q) + c''(q)q)$. The numerators represent the total derivatives of each party's profit with respect to w (accounting for the indirect effect through $q(w)$), while the denominators represent the net gains from trade for each party.

Substituting the conduct FOCs and simplifying, the equilibrium-quantity conditions

[39] We use q instead of q^u and q^d whenever there is no ambiguity to simplify the notation.

are:

$$\begin{cases} \beta\left(\dfrac{-q}{-p'(q)q^2-\pi_0^d}\right)+(1-\beta)\left(\dfrac{q+(p(q)-c(q)+p'(q)q-c'(q)q)\cdot\dfrac{1}{2p'(q)+p''(q)q}}{[p(q)-c(q)+p'(q)q]\,q-\pi_0^u}\right)=0 & \text{(MP-Q-FOC)} \\ (1-\beta)\left(\dfrac{q}{c'(q)q^2-\pi_0^u}\right)+\beta\left(\dfrac{-q+([p(q)-c(q)]+[p'(q)q-c'(q)q])\,\dfrac{1}{2c'(q)+c''(q)q}}{[p(q)-c(q)-c'(q)q]q-\pi_0^d}\right)=0 & \text{(MS-Q-FOC)} \end{cases}$$

B.2 Second-Order Conditions

Quantity-setting SOCs. The second-order conditions for the Stage 2 quantity-setting problems are:

$$\underbrace{2p'(q)+p''(q)\,q}_{=mr'(q)}<0 \quad \text{(D-SOC)}, \qquad \underbrace{2c'(q)+c''(q)\,q}_{=mc'(q)}>0 \quad \text{(U-SOC)}.$$

These ensure that the downstream profit function $(p(q)-w)q$ is concave in q (yielding D-FOC) and the upstream profit function $(w-c(q))q$ is concave in q (yielding U-FOC). Equivalently, marginal revenue is strictly decreasing and marginal cost is strictly increasing.

Bargaining SOCs. The second-order condition of monopsonistic bargaining under sequential timing is given by:

$$\beta\left\{\frac{1}{qG^d}\left[\frac{N'(q)T(q)-N(q)T'(q)}{T(q)^3}-\frac{1}{T(q)}\right]-\left[\frac{N(q)-qT(q)}{qG^dT(q)}\right]^2\right\}+(1-\beta)\left\{\frac{1}{T(q)qG^u}-\frac{1}{q(G^u)^2}\right\}<0.$$

where

$$D(q)\equiv p(q)-c(q)-qc'(q), \qquad N(q)\equiv p(q)-c(q)+q[p'(q)-c'(q)],$$
$$T(q)\equiv 2c'(q)+qc''(q), \qquad G^d\equiv D(q)\cdot q-\pi_0^d, \quad G^u\equiv c'(q)q^2-\pi_0^u.$$

Here $G^d=\pi^d-\pi_0^d$ and $G^u=\pi^u-\pi_0^u$ represent the net gains from trade for the downstream and upstream firms, respectively.

The second-order condition of the monopolistic bargaining under sequential timing is given by:

$$(1-\beta)\left\{\frac{1}{q\tilde{G}^u}\left[\frac{\tilde{N}'(q)\tilde{T}(q)-\tilde{N}(q)\tilde{T}'(q)}{\tilde{T}(q)^3}+\frac{1}{\tilde{T}(q)}\right]-\left[\frac{\tilde{N}(q)-q\tilde{T}(q)}{q\tilde{G}^u\tilde{T}(q)}\right]^2\right\}+\beta\left\{\frac{1}{\tilde{T}(q)q\tilde{G}^d}-\frac{1}{q(\tilde{G}^d)^2}\right\}<0.$$

where

$$\begin{aligned}
&\tilde{D}(q)\equiv p(q)+qp'(q)-c(q), && \tilde{N}(q)\equiv p(q)-c(q)+q[p'(q)-c'(q)],\\
&\tilde{T}(q)\equiv 2p'(q)+qp''(q), && \tilde{G}^d\equiv -p'(q)q^2-\pi_0^d,\quad \tilde{G}^u\equiv \tilde{D}(q)\cdot q-\pi_0^u.
\end{aligned}$$

B.3 Derivation of FOCs Under Sequential Bargaining

B.3.1 U-B-FOC

Proof. We differentiate the logarithm of the objective with respect to w:

$$\beta\left(\frac{1}{[p(q)-w]\,q-\pi_0^d}\cdot\frac{d}{dw}\left([p(q)-w]\,q\right)\right)+(1-\beta)\left(\frac{1}{[w-c(q)]\,q-\pi_0^u}\cdot\frac{d}{dw}\left([w-c(q)]\,q\right)\right)=0.$$

Note the following intermediate derivatives:

$$\frac{d}{dw}\left([p(q)-w]\,q\right)\;=\;\left(\frac{d}{dw}[p(q)-w]\right)q+[p(q)-w]\frac{dq}{dw}\;=\;-q+\left(p'(q)q+[p(q)-w]\right)\frac{dq}{dw}$$

$$\frac{d}{dw}\left([w-c(q)]\,q\right)\;=\;\left(\frac{d}{dw}[w-c(q)]\right)q+[w-c(q)]\frac{dq}{dw}\;=\;q+\left([w-c(q)]-c'(q)q\right)\frac{dq}{dw}$$

Differentiating both sides of U-FOC, $w=c(q)+c'(q)\,q$ with respect to w, we find

$$\frac{dq}{dw}=\frac{1}{2\,c'(q)+c''(q)\,q}.$$

Substituting the expressions above into the FOC yields:

$$\beta\left(\frac{-q+\left(p'(q)q+[p(q)-w]\right)\dfrac{dq}{dw}}{[p(q)-w]\,q-\pi_0^d}\right)+(1-\beta)\left(\frac{q+\left([w-c(q)]-c'(q)q\right)\dfrac{dq}{dw}}{[w-c(q)]\,q-\pi_0^u}\right)=0,$$

and plugging in dq/dw:

$$\beta\left(\frac{-q+\left(p'(q)q+[p(q)-w]\right)\dfrac{1}{2c'(q)+c''(q)q}}{[p(q)-w]\,q-\pi_0^d}\right)+(1-\beta)\left(\frac{q+\left([w-c(q)]-c'(q)q\right)\dfrac{1}{2c'(q)+c''(q)q}}{[w-c(q)]\,q-\pi_0^u}\right)=0.$$

From U-FOC, substitute $w = c(q) + c'(q)q$ and simplify to obtain:

$$\beta \left(\frac{-q + ([p(q) - c(q)] + [p'(q)q - c'(q)q]) \dfrac{1}{2c'(q) + c''(q)q}}{[p(q) - c(q) - c'(q)q]\, q - \pi_0^d} \right) + (1-\beta) \left(\frac{1}{c'(q)q - \pi_0^u/q} \right) = 0.$$

□

B.3.2 D-B-FOC

Proof. Calculate dq/dw using $w = p(q) + p'(q)q$:

$$\frac{dq}{dw} = \frac{1}{2p'(q) + p''(q)q}.$$

Substitute dq/dw and $w = p(q) + p'(q)q$ into the FOC:

$$\beta \left(\frac{-q + (p'(q)q + [p(q) - w]) \cdot \frac{1}{2p'(q)+p''(q)q}}{[p(q) - w] \cdot q - \pi_0^d} \right) + (1-\beta) \left(\frac{q + ([w - c(q)] - c'(q)q) \cdot \frac{1}{2p'(q)+p''(q)q}}{[w - c(q)] \cdot q - \pi_0^u} \right) = 0.$$

From (D-FOC), substitute $w = p(q) + p'(q)q$ and simplify to obtain:

$$\beta \left(\frac{-q}{-p'(q)q^2 - \pi_0^d} \right) + (1-\beta) \left(\frac{q + (p(q) - c(q) + p'(q)q - c'(q)q) \cdot \dfrac{1}{2p'(q) + p''(q)q}}{[p(q) - c(q) + p'(q)q]\, q - \pi_0^u} \right) = 0.$$

□

B.4 Analysis of Second-Order Conditions

In this section, we will analyze the second-order conditions given in Section B.2. Since the second-order conditions are complex, there are no simple primitive conditions that guarantee that they hold. However, we develop two sufficient conditions under which the second order conditions hold separately under the nonnegative markup/markdown constraint given in Section 4 and also globally.

Lemma OA-1. *The second-order conditions for the sequential monopsonistic bargaining model hold if $mc''(q) \geq 0$ and $\Delta^d \geq 0$. The second-order conditions for the sequential monopolistic bargaining hold if $mr''(q) \leq 0$ and $\mu^u \geq 0$.*

The second-order conditions of the sequential monopsonistic bargaining model are given by:

$$\beta\Bigg\{\underbrace{\frac{1}{qG^d}}_{(+)}\left[\frac{\overbrace{N'(q)T(q)}^{(-)}-N(q)\overbrace{T'(q)}^{(+)}}{\underbrace{T(q)^3}_{(+)}}-\underbrace{\frac{1}{T(q)}}_{(+)}\right]-\underbrace{\left[\frac{N(q)-qT(q)}{qG^dT(q)}\right]^2}_{(+)}\Bigg\}+(1-\beta)\underbrace{\left\{\frac{1}{T(q)G^u}-\frac{q^2}{(G^u)^2}\right\}}_{(-)}$$

where $G^d \equiv D(q)\cdot q-\pi_0^d$ and $G^u \equiv c'(q)q^2-\pi_0^u$ are the net gains from trade, and

$$\begin{aligned}
D(q) &= p(q)-c(q)-qc'(q)\geq 0, & D'(q) &= p'(q)-2c'(q)-qc''(q)<0\\
N(q) &= p(q)+qp'(q)-[c(q)+qc'(q)], & N'(q) &= [2p'(q)+qp''(q)]-[2c'(q)+qc''(q)]<0\\
T(q) &= 2c'(q)+qc''(q)>0, & T'(q) &= 3c''(q)+qc'''(q)
\end{aligned}$$

Therefore, we need to show that $N(q)\geq 0$. Note that $N(q)=mr(q)-mc(q)$. Since under monopsonistic bargaining $mc(q)=w$, $mr(q)\geq w$ implies that $N(q)\geq 0$.
The second-order conditions of the sequential monopolistic bargaining model are given by:

$$(1-\beta)\Bigg\{\underbrace{\frac{1}{q\tilde{G}^u}}_{(+)}\left[\frac{\overbrace{\tilde{N}'(q)\tilde{T}(q)}^{(+)}-\tilde{N}(q)\overbrace{\tilde{T}'(q)}^{(-)}}{\underbrace{\tilde{T}(q)^3}_{(-)}}+\underbrace{\frac{1}{\tilde{T}(q)}}_{(-)}\right]-\underbrace{\left[\frac{\tilde{N}(q)-q\tilde{T}(q)}{q\tilde{G}^u\tilde{T}(q)}\right]^2}_{(+)}\Bigg\}+\beta\underbrace{\left\{\frac{1}{\tilde{T}(q)\tilde{G}^d}-\frac{q^2}{(\tilde{G}^d)^2}\right\}}_{(-)}<0.$$

where $\tilde{G}^d \equiv -p'(q)q^2-\pi_0^d$ and $\tilde{G}^u \equiv \tilde{D}(q)\cdot q-\pi_0^u$ are the net gains from trade, and

$$\begin{aligned}
\tilde{D}(q) &\equiv p(q)+qp'(q)-c(q)\geq 0 & \tilde{D}'(q) &= -c'(q)+2p'(q)+qp''(q)<0\\
\tilde{N}(q) &\equiv p(q)+qp'(q)-[c(q)+qc'(q)] & \tilde{N}'(q) &= [2p'(q)+qp''(q)]-[2c'(q)+qc''(q)]<0\\
\tilde{T}(q) &\equiv 2p'(q)+qp''(q)<0 & \tilde{T}'(q) &= 3p''(q)+qp'''(q)<0
\end{aligned}$$

Therefore, we need to show that $\tilde{N}(q)\geq 0$. Note that $\tilde{N}(q)=mr(q)-mc(q)$. Since under monopolistic bargaining $mr(q)=w$, and positive markup implies that $mc(q)\leq w$, it follows that $\tilde{N}(q)\geq 0$.

Remark. Under the distortion selection result from (Theorem 2), the conditions $\Delta^d\geq 0$ (monopsonistic) and $\mu^u\geq 0$ (monopolistic) are automatically satisfied by Constraint 4.1. Therefore, the SOC holds under the nonnegative markup/markdown constraint whenever $mc''(q)\geq 0$ and $mr''(q)\leq 0$.

Lemma OA-2. *The second-order conditions of the sequential monopsonistic bargaining model are* *satisfied for all β if $(1/mc(q))'' \geq 0$ and $mc''(q) \geq 0$.*[40]

Proof. As shown in the proof of Lemma OA-1, since all other terms are negative by assumption, we will focus on the following term:

$$\frac{1}{qG^d}\left[\frac{N'(q)T(q) - N(q)T'(q)}{T(q)^3} - \frac{1}{T(q)}\right] = \frac{1}{qG^dT(q)}\left[\frac{N'(q)T(q) - N(q)T'(q)}{T(q)^2} - 1\right]$$

Note that $N(q) = mr(q) - mc(q)$ and $T(q) = mc'(q)$. Substituting these:

$$\begin{aligned}\left[\frac{N'(q)T(q) - N(q)T'(q)}{T(q)^2} - 1\right] &= \left[\frac{(mr'(q) - mc'(q))mc'(q) - (mr(q) - mc(q))mc''(q)}{mc'(q)^2} - 1\right] \\ &= \left[\frac{mr'(q)}{mc'(q)} - mr(q)\frac{mc''(q)}{mc'(q)^2} + mc(q)\frac{mc''(q)}{mc'(q)^2} - 2\right]\end{aligned}$$

We need to show that this expression is negative because $G^d > 0$ and $T(q) > 0$. The first term is negative since $mr'(q) < 0$ and $mc'(q) > 0$. The second term is also negative because $mr(q) > 0$ and $mc''(q) > 0$. Therefore, in order for this term to be negative, we need that $mc(q)mc''(q)/mc'(q)^2 \leq 2$. Note that this condition is equivalent to $(1/mc(q))'' \geq 0$, which concludes the proof. □

Lemma OA-3. *The second-order conditions of the sequential monopolistic bargaining model are satisfied for all β if $mr(q)mr''(q)/mr'(q)^2 \geq -2$ and $mr''(q) \leq 0$.*

Proof. As shown in the proof of Lemma OA-1, since all other terms are negative by assumption, we will focus on the following term in the second-order condition:

$$\frac{1}{q\tilde{G}^u}\left[\frac{\tilde{N}'(q)\tilde{T}(q) - \tilde{N}(q)\tilde{T}'(q)}{\tilde{T}(q)^3} + \frac{1}{\tilde{T}(q)}\right] = \frac{1}{q\tilde{G}^u\tilde{T}(q)}\left[\frac{\tilde{N}'(q)\tilde{T}(q) - \tilde{N}(q)\tilde{T}'(q)}{\tilde{T}(q)^2} + 1\right]$$

Note that $\tilde{N}(q) = mr(q) - mc(q)$ and $\tilde{T}(q) = mr'(q)$. Substituting these:

$$\begin{aligned}\left[\frac{\tilde{N}'(q)\tilde{T}(q) - \tilde{N}(q)\tilde{T}'(q)}{\tilde{T}(q)^2} + 1\right] &= \left[\frac{(mr'(q) - mc'(q))mr'(q) - (mr(q) - mc(q))mr''(q)}{mr'(q)^2} + 1\right] \\ &= \left[-\frac{mc'(q)}{mr'(q)} - mc(q)\frac{mr''(q)}{mr'(q)^2} + mr(q)\frac{mr''(q)}{mr'(q)^2} + 2\right]\end{aligned}$$

We need to show that this expression is positive because $\tilde{G}^u > 0$ and $\tilde{T}(q) < 0$. The first

[40]Even though this condition seems strong, we note that this is a sufficient condition. This condition is satisfied by some common functional form classes, such as exponential, polynomials, and linear functions.

term is positive since $mr'(q) < 0$ and $mc'(q) > 0$. The second term is also positive because $mc(q) > 0$ and $mr''(q) < 0$. Therefore, in order for this term to be positive, we need that $mr(q)mr''(q)/mr'(q)^2 \geq -2$. $\square$

C Proofs of Main Results

This section provides the proofs of all results stated in the main text, in the order they appear.

C.1 Proof of Proposition 1

Proof. We will first show part (i) and part (ii). Since second-order conditions do not hold when $mr'(q) = 0$ and $mc'(q) = 0$, we cannot use first-order conditions to find an equilibrium. Therefore, we will directly work with the maximization programs under each bargaining model. We split each problem into multiple cases.

<u>*Monopsonistic Bargaining:*</u>

The equilibrium (w^e, q^e) maximizes the objectives in the monopsonistic model:

$$\begin{cases} \max_q \pi^u(w^e, q) & (U) \\ \max_w (\pi^d(w, q^e) - \pi_0^d)^\beta \, (\pi^u(w, q^e) - \pi_0^u)^{1-\beta} & (B) \end{cases} \quad \text{s.t } \pi^u(w, q) \geq \pi_0^u \qquad \pi^d(w, q) \geq \pi_0^d.$$

The subgame-perfect equilibrium $(w^e, q^{BR}(w^e))$ is determined by backward induction. For a given w, the upstream solves (U) in the second stage, giving the best-response function $q^{BR}(w)$. In the first stage, the parties solve (B) given $q^{BR}(w)$. Thus, $(w^e, q^{BR}(w^e))$ is an equilibrium if there is no other w with $\pi^b(w, q^{BR}(w), \beta) > \pi^b(w^e, q^{BR}(w^e), \beta)$, and $q^{BR}(w)$ maximizes (U). We analyze this under different β values.

Case I: $\beta \in (0, 1)$

If the marginal cost is constant, the best-response function of upstream in the second stage is:

$$q^{BR}(w) = \begin{cases} 0 & w < c \\ p^{-1}(w) & c < w \\ [0, p^{-1}(w)] & c = w, \end{cases}$$

since upstream profit $(w - c)q$ is increasing in q when $w > c$ (bounded by voluntary exchange at $q = p^{-1}(w)$), zero for all q when $w = c$, and negative for $q > 0$ when $w < c$. We now show that no interior equilibrium exists, regardless of the disagreement payoffs. The key observation is that constant marginal cost forces at least one party's profit to zero:

- When $w > c$: $q^{BR}(w) = p^{-1}(w)$, so $p(q) = w$ and $\pi^d = (p(q) - w)q = 0$.
- When $w = c$: $\pi^u = (w - c)q = 0$ for any q.

- When $w < c$: $q^{BR} = 0$, so both profits are zero.

We consider the implications for the Nash product $(\pi^d - \pi_0^d)^\beta(\pi^u - \pi_0^u)^{1-\beta}$ under each configuration of disagreement payoffs:

(a) $\pi_0^d > 0$ and $\pi_0^u > 0$: When $w > c$, $\pi^d = 0 < \pi_0^d$, violating downstream's participation constraint. When $w = c$, $\pi^u = 0 < \pi_0^u$, violating upstream's. No w satisfies both participation constraints simultaneously, so no equilibrium with positive trade exists.

(b) $\pi_0^d = 0$, $\pi_0^u > 0$: When $w > c$, $\pi^d = 0 = \pi_0^d$ (participation holds weakly) and $\pi^u = (w-c)p^{-1}(w) > 0$, which can satisfy $\pi^u \geq \pi_0^u$ for sufficiently large w. However, the Nash product is $(0)^\beta(\pi^u - \pi_0^u)^{1-\beta} = 0$. When $w = c$, $\pi^u = 0 < \pi_0^u$. The Nash product is zero for all feasible w, so no interior equilibrium exists.

(c) $\pi_0^d > 0$, $\pi_0^u = 0$: When $w > c$, $\pi^d = 0 < \pi_0^d$, violating participation. When $w = c$, $\pi^u = 0 = \pi_0^u$ (participation holds weakly) and q can be chosen so that $\pi^d = (p(q)-c)q \geq \pi_0^d$, but the Nash product is $(\pi^d - \pi_0^d)^\beta(0)^{1-\beta} = 0$. No interior equilibrium exists.

(d) $\pi_0^d = 0$, $\pi_0^u = 0$: When $w > c$, $\pi^d = 0$ and the Nash product is $(0)^\beta(\pi^u)^{1-\beta} = 0$. When $w = c$, $\pi^u = 0$ and the Nash product is $(\pi^d)^\beta(0)^{1-\beta} = 0$. Any $w \geq c$ gives a Nash product of zero, so any such w is an equilibrium, but none is interior.

Summary by disagreement payoff configuration:

- $\pi_0^d = 0$, $\pi_0^u = 0$: Any $w \geq c$ is an equilibrium with Nash product equal to zero. Not interior.

- $\pi_0^d = 0$, $\pi_0^u > 0$: Some $w > c$ are feasible (upstream participation holds), but Nash product is zero ($\pi^d = 0$). Not interior.

- $\pi_0^d > 0$, $\pi_0^u = 0$: At $w = c$, Nash product is zero ($\pi^u = 0$). At $w > c$, downstream participation fails. Not interior.

- $\pi_0^d > 0$, $\pi_0^u > 0$: No w satisfies both participation constraints. No equilibrium with positive trade exists.

In no case does an interior equilibrium (with both $\pi^d > \pi_0^d$ and $\pi^u > \pi_0^u$) exist.

Case II: $\beta = 1$

At $\beta = 1$, the Nash product is $(\pi^d - \pi_0^d)^1 \cdot (\pi^u - \pi_0^u)^0 = \pi^d - \pi_0^d$, so bargaining maximizes downstream profit subject to upstream's participation constraint $\pi^u \geq \pi_0^u$. If marginal cost is constant, there are two candidate equilibria for $\beta = 1$ in sequential monopsonistic bargaining:

$$q_1^{BR}(w) = \begin{cases} 0, & w < c, \\ p^{-1}(w), & c < w, \\ (0, p^{-1}(w)), & c = w. \end{cases} \qquad w_1^e = c \qquad \text{and} \qquad q_2^{BR}(w) = \begin{cases} 0, & w < c, \\ p^{-1}(w), & c < w, \\ p^{-1}(w), & c = w. \end{cases} \qquad w_2^e \geq c.$$

We analyze the feasibility of each under different disagreement payoffs.

Candidate Equilibrium 1: $w_1^e = c$. At $w = c$, upstream is indifferent over q (since $\pi^u = 0$ for all q), so any $q \in (0, p^{-1}(c))$ is a best response. Downstream earns $\pi^d = (p(q) - c)q > 0$. Any deviation to $\tilde{w} > c$ gives $q^{BR} = p^{-1}(\tilde{w})$ and $\pi^d = 0$, which is worse. So $w_1^e = c$ maximizes $\pi^d - \pi_0^d$ over all feasible w.

However, at $w = c$: $\pi^u = 0$. If $\pi_0^u > 0$, upstream's participation constraint is violated, so this equilibrium does not exist. If $\pi_0^u = 0$, the equilibrium exists and downstream earns $\pi^d > 0 \geq \pi_0^d$.

Candidate Equilibrium 2: $q_2^{BR}(c) = p^{-1}(c)$, $w_2^e \geq c$. Here upstream selects $q = p^{-1}(w)$ for all $w \geq c$ (including $w = c$), driving downstream profit to zero: $\pi^d = (p(q) - w)q = 0$. The objective $\pi^d - \pi_0^d = -\pi_0^d \leq 0$, so this is a degenerate equilibrium. If $\pi_0^d > 0$, downstream would prefer disagreement, so this equilibrium does not exist. If $\pi_0^d = 0$, the objective is zero for all $w \geq c$, so any such w is trivially an equilibrium. Additionally, upstream participation requires $(w - c)p^{-1}(w) \geq \pi_0^u$, which restricts the set of feasible w when $\pi_0^u > 0$.

Summary by disagreement payoff configuration:

- $\pi_0^d = 0$, $\pi_0^u = 0$: Both equilibria exist. In equilibrium 1, $\pi^d > 0$; in equilibrium 2, $\pi^d = 0$. Neither is interior (upstream earns zero in both).
- $\pi_0^d = 0$, $\pi_0^u > 0$: Equilibrium 1 fails (upstream participation). Equilibrium 2 exists for w large enough that $\pi^u \geq \pi_0^u$, but $\pi^d = 0$. Not interior.
- $\pi_0^d > 0$, $\pi_0^u = 0$: Equilibrium 1 exists ($\pi^d > 0 > \pi_0^d$ if π_0^d is small enough). Equilibrium 2 fails ($\pi^d = 0 < \pi_0^d$). But upstream earns zero, so not interior.
- $\pi_0^d > 0$, $\pi_0^u > 0$: Both equilibria fail. No equilibrium with positive trade exists.

In no case does an interior equilibrium (with both $\pi^d > \pi_0^d$ and $\pi^u > \pi_0^u$) exist.

Case III: $\beta = 0$

At $\beta = 0$, the Nash product is $(\pi^d - \pi_0^d)^0 \cdot (\pi^u - \pi_0^u)^1 = \pi^u - \pi_0^u$, so bargaining maximizes upstream profit subject to downstream's participation constraint $\pi^d \geq \pi_0^d$. If marginal cost is constant for $\beta = 0$, the candidate equilibrium is given by:

$$q_1^{BR}(w) = \begin{cases} 0, & w < c, \\ p^{-1}(w), & c < w, \\ (0, p^{-1}(w)), & c = w. \end{cases} \qquad w_1^e = \text{argmax}_w \ (w-c)p^{-1}(w).$$

When $w = c$, upstream profit is zero, which cannot be an equilibrium since $w > c$ leads to positive profit for the upstream. For $w > c$, the best response in the second stage is given by $p^{-1}(w)$, which leads to the profit function $(w-c)p^{-1}(w)$. The equilibrium w maximizes this profit function.

We now check feasibility under different disagreement payoffs. At the equilibrium $w_1^e > c$, the best response gives $q = p^{-1}(w_1^e)$, so $\pi^d = (p(q) - w_1^e)q = 0$ and $\pi^u = (w_1^e - c)p^{-1}(w_1^e) > 0$.

(a) $\pi_0^d > 0$: Downstream's participation constraint requires $\pi^d \geq \pi_0^d > 0$, but $\pi^d = 0$ at the equilibrium. No feasible $w > c$ satisfies downstream participation, so no equilibrium with positive trade exists.

(b) $\pi_0^d = 0$, $\pi_0^u \geq 0$: Downstream participation holds weakly ($\pi^d = 0 = \pi_0^d$). Upstream participation requires $(w_1^e - c)p^{-1}(w_1^e) \geq \pi_0^u$, which holds for π_0^u sufficiently small (since $\pi^u > 0$ at the maximizer). However, $\pi^d = 0 = \pi_0^d$, so the equilibrium is not interior.

Summary by disagreement payoff configuration:

- $\pi_0^d = 0$, $\pi_0^u = 0$: The equilibrium exists with $\pi^u > 0$ and $\pi^d = 0$. Not interior (downstream earns zero).
- $\pi_0^d = 0$, $\pi_0^u > 0$: The equilibrium exists provided $\pi^u(w_1^e) \geq \pi_0^u$ (upstream participation). Still $\pi^d = 0$. Not interior.
- $\pi_0^d > 0$, $\pi_0^u = 0$: No equilibrium with positive trade exists ($\pi^d = 0 < \pi_0^d$ for all $w > c$).
- $\pi_0^d > 0$, $\pi_0^u > 0$: No equilibrium with positive trade exists (downstream participation always fails).

In no case does an interior equilibrium (with both $\pi^d > \pi_0^d$ and $\pi^u > \pi_0^u$) exist.

Monopolistic Bargaining:

Since this proof closely follows the proofs of monopsonistic bargaining, the detailed analysis is omitted. We list the equilibria for different values of β and summarize the implications of disagreement payoffs. In monopolistic bargaining, the roles are reversed: downstream sets quantity in stage 2, and with constant marginal revenue ($p'(q) = 0$, i.e.,

p is constant), downstream's best response drives upstream profit to zero when $w < p$, or earns zero itself when $w = p$.

Case I: $\beta \in (0, 1)$

$$q^{BR}(w) = \begin{cases} 0 & p < w \\ c^{-1}(w) & p > w \\ [0, c^{-1}(w)] & p = w \end{cases} \qquad w^e < p.$$

By the same logic as monopsonistic Case I (with the roles of upstream and downstream reversed): when $w < p$, $q^{BR} = c^{-1}(w)$ gives $\pi^u = (w - c(q))q = 0$; when $w = p$, $\pi^d = (p - w)q = 0$. In all configurations of (π_0^d, π_0^u), at least one gain from trade is non-positive, precluding an interior equilibrium.

Case II: $\beta = 1$

$$q_1^{BR}(w) = \begin{cases} 0 & p < w \\ c^{-1}(w) & p > w \\ (0, c^{-1}(w)) & p = w \end{cases} \qquad w_1^e = p, \qquad \text{and} \qquad q_2^{BR}(w) = \begin{cases} 0 & p < w \\ c^{-1}(w) & p > w \\ c^{-1}(w) & p = w \end{cases} \qquad w_2^e \leq p.$$

At $\beta = 1$, bargaining maximizes $\pi^d - \pi_0^d$. In equilibrium 1 ($w_1^e = p$), $\pi^d = 0$ and $\pi^u = (p - c(q))q > 0$. In equilibrium 2 ($w_2^e \leq p$), downstream can earn positive profit but upstream earns $\pi^u = 0$ when $w = c(q)$.

Summary:

- $\pi_0^d = 0, \pi_0^u = 0$: Both equilibria exist. Neither is interior.
- $\pi_0^d > 0, \pi_0^u = 0$: Equilibrium 1 fails ($\pi^d = 0 < \pi_0^d$). Equilibrium 2 exists if upstream earns zero. Not interior.
- $\pi_0^d = 0, \pi_0^u > 0$: Equilibrium 1 exists if $\pi^u \geq \pi_0^u$. Equilibrium 2 fails ($\pi^u = 0 < \pi_0^u$). Not interior ($\pi^d = 0$).
- $\pi_0^d > 0, \pi_0^u > 0$: Both fail. No equilibrium with positive trade.

Case III: $\beta = 0$

$$q_1^{BR}(w) = \begin{cases} 0 & p < w \\ c^{-1}(w) & p > w \\ (0, c^{-1}(w)) & p = w \end{cases} \qquad w_1^e = \text{argmax}_w \ (p - w)c^{-1}(w).$$

At $\beta = 0$, bargaining maximizes $\pi^u - \pi^u_0$. At $w^e_1 < p$, $q^{BR} = c^{-1}(w^e_1)$ gives $\pi^u = (w^e_1 - c(q))q = 0$ and $\pi^d = (p - w^e_1)q > 0$.

Summary:

- $\pi^d_0 = 0, \pi^u_0 = 0$: Equilibrium exists with $\pi^d > 0, \pi^u = 0$. Not interior.
- $\pi^d_0 = 0, \pi^u_0 > 0$: Upstream participation fails ($\pi^u = 0 < \pi^u_0$). No equilibrium.
- $\pi^d_0 > 0, \pi^u_0 = 0$: Equilibrium exists if $\pi^d \geq \pi^d_0$. Still $\pi^u = 0$. Not interior.
- $\pi^d_0 > 0, \pi^u_0 > 0$: Upstream participation fails. No equilibrium with positive trade.

Note that for all β values in both monopsonistic and monopolistic bargaining, constant marginal cost (or constant marginal revenue) forces at least one party's profit to zero. Since an interior solution requires both $\pi^d > \pi^d_0$ and $\pi^u > \pi^u_0$, no interior equilibrium exists when marginal cost or marginal revenue is constant, regardless of the disagreement payoffs.

What's left to show is part (iii). That is, when $mc'(q) > 0$ and $mr'(q) < 0$, equilibrium exists for an interior solution within the β ranges specified in the proposition in both monopsonistic and monopolistic bargaining.

For monopsonistic bargaining, Lemma OA-4 shows that the FOC solution converges as $\beta \to 1$ to the constrained-optimization solution in Section D.1.1, and Lemma OA-6 shows convergence as $\beta \to 0$ to the solution in Section D.1.2. Therefore, the FOC solution is well-defined in a neighborhood of both corners. The continuity of the FOCs implies that the solution exists for any $\beta \in (0, 1)$.

The solution is interior in the sense that both gains from trade are strictly positive: $\pi^d(q) - \pi^d_0 > 0$ and $\pi^u(q) - \pi^u_0 > 0$. This follows from the structure of the FOC (Section B.1), which contains $(\pi^d - \pi^d_0)$ and $(\pi^u - \pi^u_0)$ in the denominators. At any FOC solution with $\beta \in (0, 1)$, both denominators must be nonzero. They must be strictly positive (rather than negative) because Assumption OA-1 guarantees the existence of a quantity at which both gains are positive, and the Nash product equals zero when either gain is zero but is strictly positive at such a feasible quantity. Since the FOC characterizes a maximum of the Nash product, the solution must have both gains strictly positive.

For monopolistic bargaining, Lemma OA-7 shows that the FOC solution converges as $\beta \to 0$ to the constrained-optimization solution in Section D.1.4, and Lemma OA-5 shows convergence as $\beta \to 1$ to the solution in Section D.1.3. The same continuity argument establishes existence for all $\beta \in (0, 1)$, and the same reasoning shows the solution is interior. □

C.2 Proof of Theorem 1

Proof. In both parts, we write the B-FOC as $f(\beta, \pi_0^d, \pi_0^u, w) = \beta \cdot A(w) + (1-\beta) \cdot B(w) = 0$ and apply the Implicit Function Theorem. The second-order condition ensures $\partial f/\partial w < 0$ throughout.

Part (i): Monopsonistic bargaining. We prove $dq^{ms}/d\beta < 0$, $dq^{ms}/d\pi_0^d < 0$, and $dq^{ms}/d\pi_0^u > 0$.

Under monopsonistic conduct, the U-B-FOC gives:

$$A(w) = \frac{d\pi^d/dw}{\pi^d - \pi_0^d} = \frac{-q + (p'(q)q + [p(q) - w])\,(dq^u/dw)}{[p(q) - w] \cdot q - \pi_0^d},$$

$$B(w) = \frac{d\pi^u/dw}{\pi^u - \pi_0^u} = \frac{q}{c'(q)q^2 - \pi_0^u}.$$

We note that $B > 0$ (since the FOC characterizes an interior equilibrium where $\pi^u > \pi_0^u$, as established in Section B.1). Since $\beta A + (1-\beta)B = 0$ with $B > 0$ and $\beta \in (0,1)$, it follows that $A < 0$. Note also that $dq/dw = 1/(2c'(q) + c''(q)q) > 0$ under monopsonistic bargaining.

$\underline{dq^{ms}/d\beta < 0}$: We have $\partial f/\partial\beta = A - B < 0$ (since $A < 0$ and $B > 0$). By the IFT:

$$\frac{dw}{d\beta} = -\frac{\partial f/\partial\beta}{\partial f/\partial w} = -\frac{\overbrace{(A-B)}^{<0}}{\underbrace{\partial f/\partial w}_{<0}} < 0.$$

Since $dq/dw > 0$ under monopsonistic bargaining: $dq^{ms}/d\beta = \underbrace{(dq/dw)}_{>0} \cdot \underbrace{(dw/d\beta)}_{<0} < 0$.

$\underline{dq^{ms}/d\pi_0^d < 0}$: Only A depends on π_0^d, through its denominator $G^d \equiv [p(q) - w]q - \pi_0^d$. Write $A = N_A/G^d$ where $N_A \equiv d\pi^d/dw < 0$. Differentiating: $\partial A/\partial\pi_0^d = N_A/(G^d)^2 < 0$. Therefore $\partial f/\partial\pi_0^d = \beta \cdot (\partial A/\partial\pi_0^d) < 0$. By the IFT:

$$\frac{dw}{d\pi_0^d} = -\frac{\overbrace{\partial f/\partial\pi_0^d}^{<0}}{\underbrace{\partial f/\partial w}_{<0}} < 0.$$

Hence $dq^{ms}/d\pi_0^d = \underbrace{(dq/dw)}_{>0} \cdot \underbrace{(dw/d\pi_0^d)}_{<0} < 0$.

$\underline{dq^{ms}/d\pi_0^u > 0}$: Only B depends on π_0^u, through its denominator $G^u \equiv c'(q)q^2 - \pi_0^u$. Differentiating: $\partial B/\partial \pi_0^u = q/(G^u)^2 > 0$. Therefore $\partial f/\partial \pi_0^u = (1-\beta)\cdot(\partial B/\partial \pi_0^u) > 0$. By the IFT:

$$\frac{dw}{d\pi_0^u} = -\frac{\overbrace{\partial f/\partial \pi_0^u}^{>0}}{\underbrace{\partial f/\partial w}_{<0}} > 0.$$

Hence $dq^{ms}/d\pi_0^u = \underbrace{(dq/dw)}_{>0}\cdot\underbrace{(dw/d\pi_0^u)}_{>0} > 0$.

Part (ii): Monopolistic bargaining. We prove $dq^{mp}/d\beta > 0$, $dq^{mp}/d\pi_0^d > 0$, and $dq^{mp}/d\pi_0^u < 0$.

Under monopolistic conduct, the D-B-FOC gives:

$$A(w) = \frac{d\pi^d/dw}{\pi^d - \pi_0^d} = \frac{-q}{-p'(q)q^2 - \pi_0^d},$$

$$B(w) = \frac{d\pi^u/dw}{\pi^u - \pi_0^u} = \frac{q + ([w - c(q)] - c'(q)q)\,(dq^d/dw)}{[w - c(q)]\cdot q - \pi_0^u}.$$

We note that $A < 0$ (since $-p'(q)q^2 > 0$ as $p'(q) < 0$ by assumption). Since $\beta A + (1-\beta)B = 0$ with $A < 0$ and $\beta \in (0,1)$, it follows that $B > 0$. Note also that $dq/dw = 1/(2p'(q) + p''(q)q) < 0$ under monopolistic bargaining (since $mr'(q) < 0$).

$\underline{dq^{mp}/d\beta > 0}$: $\partial f/\partial \beta = A - B < 0$. By the IFT:

$$\frac{dw}{d\beta} = -\frac{\overbrace{(A-B)}^{<0}}{\underbrace{\partial f/\partial w}_{<0}} < 0.$$

Since $dq/dw < 0$ under monopolistic bargaining: $dq^{mp}/d\beta = \underbrace{(dq/dw)}_{<0}\cdot\underbrace{(dw/d\beta)}_{<0} > 0$.

$\underline{dq^{mp}/d\pi_0^d > 0}$: Only A depends on π_0^d, through $\tilde{G}^d \equiv -p'(q)q^2 - \pi_0^d$. Differentiating:

$\partial A/\partial\pi_0^d = -q/(\tilde{G}^d)^2 < 0$. Therefore $\partial f/\partial\pi_0^d = \beta \cdot (\partial A/\partial\pi_0^d) < 0$. By the IFT:

$$\frac{dw}{d\pi_0^d} = -\frac{\overbrace{\partial f/\partial\pi_0^d}^{<0}}{\underbrace{\partial f/\partial w}_{<0}} < 0.$$

Hence $dq^{mp}/d\pi_0^d = \underbrace{(dq/dw)}_{<0} \cdot \underbrace{(dw/d\pi_0^d)}_{<0} > 0$.

$\underline{dq^{mp}/d\pi_0^u < 0}$: Only B depends on π_0^u, through $\tilde{G}^u \equiv [w - c(q)]q - \pi_0^u$. Write $B = N_B/\tilde{G}^u$ where $N_B > 0$. Differentiating: $\partial B/\partial\pi_0^u = N_B/(\tilde{G}^u)^2 > 0$. Therefore $\partial f/\partial\pi_0^u = (1-\beta) \cdot (\partial B/\partial\pi_0^u) > 0$. By the IFT:

$$\frac{dw}{d\pi_0^u} = -\frac{\overbrace{\partial f/\partial\pi_0^u}^{>0}}{\underbrace{\partial f/\partial w}_{<0}} > 0.$$

Hence $dq^{mp}/d\pi_0^u = \underbrace{(dq/dw)}_{<0} \cdot \underbrace{(dw/d\pi_0^u)}_{>0} < 0$.

□

C.3 Proof of Corollary 1

Proof. Part (i): Under monopolistic bargaining, the D-FOC implies $w = mr(q)$, so the downstream markdown $\Delta^d(q) = (mr(q) - w)/mr(q) = 0$. The upstream markup is $\mu^u = (w - mc(q))/mc(q) = (mr(q) - mc(q))/mc(q)$. Since $w = mr(q)$ and q increases with buyer power (Theorem 1), while $mr(q)$ is decreasing and $mc(q)$ is increasing, the markup μ^u decreases with buyer power.

Part (ii): Under monopsonistic bargaining, the U-FOC implies $w = mc(q)$, so the upstream markup $\mu^u(q) = (w - mc(q))/mc(q) = 0$. The downstream markdown is $\Delta^d = (mr(q) - w)/mr(q) = (mr(q) - mc(q))/mr(q)$. Since $w = mc(q)$ and q decreases with buyer power (Theorem 1), while $mc(q)$ is increasing and $mr(q)$ is decreasing, a decrease in q raises $mr(q)$ and lowers $mc(q)$, increasing the markdown Δ^d with buyer power. □

C.4 Proof of Proposition 2

We prove that there exists a unique $\beta^* \in (0, 1)$ at which both the monopsonistic and monopolistic bargaining models yield the joint-profit maximizing quantity q^*, with the formula:

$$\beta^* = \frac{-p'(q^*)(q^*)^2 - \pi_0^d}{\big(c'(q^*) - p'(q^*)\big)(q^*)^2 - (\pi_0^u + \pi_0^d)}.$$

Proof. By Lemma OA-10, define $w^* \equiv mc(q^*) = mr(q^*)$ and the profit levels $\pi^{d*} \equiv -p'(q^*)(q^*)^2$ and $\pi^{u*} \equiv c'(q^*)(q^*)^2$, which are the same under both conduct models.

Vanishing indirect effects at q^*. Under sequential timing, the total derivative of each party's profit with respect to w has a direct and an indirect component:

$$\frac{d\pi^i}{dw} = \underbrace{\frac{\partial \pi^i}{\partial w}}_{\text{direct}} + \underbrace{\frac{\partial \pi^i}{\partial q} \cdot \frac{dq}{dw}}_{\text{indirect (through } q(w))}.$$

We show that at $q = q^*$, the indirect effects vanish for both parties under both conduct models. The argument relies on two observations:

(i) Envelope theorem: The quantity-setting party chooses q to maximize its own profit. By the envelope theorem, $\partial \pi^{QS}/\partial q = 0$ at the optimum, so the indirect effect on the quantity-setting party's profit is zero:

$$\frac{d\pi^{QS}}{dw} = \frac{\partial \pi^{QS}}{\partial w} + \underbrace{\frac{\partial \pi^{QS}}{\partial q}}_{=0} \cdot \frac{dq}{dw} = \frac{\partial \pi^{QS}}{\partial w}.$$

This holds at any q, not just at q^*. Under monopsonistic conduct, $\partial \pi^u/\partial w = q$, so $d\pi^u/dw = q$. Under monopolistic conduct, $\partial \pi^d/\partial w = -q$, so $d\pi^d/dw = -q$.

(ii) Joint surplus is stationary at q^*: Define joint surplus $S(q) \equiv \pi^d + \pi^u = (p(q) - c(q))q$. Since the input price w cancels between buyer and seller, $\partial S/\partial w = 0$. Moreover, $\partial S/\partial q = mr(q) - mc(q) = 0$ at q^*. Therefore, the total derivative of joint surplus with respect to w vanishes at q^*:

$$\left.\frac{dS}{dw}\right|_{q^*} = \underbrace{\frac{\partial S}{\partial w}}_{=0} + \underbrace{\left.\frac{\partial S}{\partial q}\right|_{q^*}}_{=0} \cdot \frac{dq}{dw} = 0.$$

Combining (i) and (ii): Since $dS/dw = d\pi^{QS}/dw + d\pi^{NQS}/dw = 0$ at q^*, and $d\pi^{QS}/dw =$

$\partial\pi^{QS}/\partial w$ by the envelope theorem, it follows that:

$$\left.\frac{d\pi^{NQS}}{dw}\right|_{q^*} = -\frac{\partial\pi^{QS}}{\partial w}.$$

Under monopsonistic conduct: $d\pi^d/dw\big|_{q^*} = -\partial\pi^u/\partial w = -q^*$. Under monopolistic conduct: $d\pi^u/dw\big|_{q^*} = -\partial\pi^d/\partial w = q^*$.

Verification by direct calculation:

- Under monopsonistic bargaining (U-FOC: $w = mc(q)$), we have $d\pi^u/dw = q + ([w - c(q)] - c'(q)q) \cdot dq/dw = q$ at any q (since $w - c(q) = c'(q)q$). For the downstream term: $d\pi^d/dw = -q + (p'(q)q + [p(q) - w]) \cdot dq/dw$. At q^*, $p(q^*) - w^* = -p'(q^*)q^*$, so $p'(q^*)q^* + [p(q^*) - w^*] = p'(q^*)q^* - p'(q^*)q^* = 0$. Hence $d\pi^d/dw\big|_{q^*} = -q^*$.

- Under monopolistic bargaining (D-FOC: $w = mr(q)$), we have $d\pi^d/dw = -q$ at any q (since $p(q) - w = -p'(q)q$). For the upstream term: $d\pi^u/dw = q + ([w - c(q)] - c'(q)q) \cdot dq/dw$. At q^*, $w^* - c(q^*) = c'(q^*)q^*$, so $[w^* - c(q^*)] - c'(q^*)q^* = 0$. Hence $d\pi^u/dw\big|_{q^*} = q^*$.

Therefore, at q^*, the B-FOC under *both* conduct models reduces to:

$$\frac{\beta \cdot (-q^*)}{\pi^{d*} - \pi_0^d} + \frac{(1-\beta) \cdot q^*}{\pi^{u*} - \pi_0^u} = 0.$$

Dividing by $q^* > 0$ and rearranging:

$$\frac{\beta}{\pi^{d*} - \pi_0^d} = \frac{1-\beta}{\pi^{u*} - \pi_0^u}$$

$$\beta(\pi^{u*} - \pi_0^u) = (1-\beta)(\pi^{d*} - \pi_0^d).$$

Solving for β:

$$\beta^* = \frac{\pi^{d*} - \pi_0^d}{\pi^{d*} + \pi^{u*} - \pi_0^d - \pi_0^u} = \frac{-p'(q^*)(q^*)^2 - \pi_0^d}{\big(c'(q^*) - p'(q^*)\big)(q^*)^2 - (\pi_0^u + \pi_0^d)}.$$

Conditions for $\beta^* \in (0,1)$. By Proposition OA-1, β^* exists if and only if $\pi_0^d < \pi^{d*}$ and $\pi_0^u < \pi^{u*}$. For $\beta^* > 0$: $\pi_0^d < \pi^{d*}$ ensures the numerator is positive, and $\pi_0^d + \pi_0^u < \pi^{d*} + \pi^{u*}$ ensures the denominator is positive. For $\beta^* < 1$: we need numerator < denominator, which requires $\pi_0^u < \pi^{u*}$. When either condition fails, q^* is not achievable and the analysis of Section A (Propositions OA-2–OA-3) applies.

Uniqueness. Since q^* is the unique solution to $mc(q^*) = mr(q^*)$ (by Lemma OA-12), and the formula maps uniquely to β^*, the bilaterally-efficient bargaining weight is unique.

□

C.5 Proof of Corollary 2

Proof. Part (i): We examine changes in $c'(q^*)$ and $|p'(q^*)|$ with respect to β^*, holding all else equal. In particular, changes in $|p'(q)|$ and $c'(q)$ also affect q^* that is present in the formula of β^*. Therefore, we condition on q^* and analyze the changes of $|p'(q)|$ and $c'(q)$ at q^*. Using the formula $\beta^* = \dfrac{-p'(q^*)(q^*)^2 - \pi_0^d}{\big(c'(q^*) - p'(q^*)\big)(q^*)^2 - (\pi_0^u + \pi_0^d)}$:

An increase in $c'(q^*)$ increases the denominator while leaving the numerator unchanged (since $p'(q^*)$ is fixed), so β^* weakly decreases.

For the effect of $|p'(q^*)|$: write $1/\beta^* = 1 + (c'(q^*)(q^*)^2 - \pi_0^u)/(-p'(q^*)(q^*)^2 - \pi_0^d)$. An increase in $|p'(q^*)| = -p'(q^*)$ increases the denominator of the second term, decreasing $1/\beta^*$, hence increasing β^*.

Part (ii): We differentiate β^* with respect to π_0^d and π_0^u.

Write $\beta^* = N/D$ where $N = \pi^{d*} - \pi_0^d$ and $D = \pi^{d*} + \pi^{u*} - \pi_0^d - \pi_0^u$.

Effect of π_0^d:

$$\frac{\partial \beta^*}{\partial \pi_0^d} = \frac{(-1) \cdot D - N \cdot (-1)}{D^2} = \frac{N - D}{D^2} = \frac{-(\pi^{u*} - \pi_0^u)}{D^2} < 0,$$

since $\pi^{u*} > \pi_0^u$ by Proposition OA-1.

Effect of π_0^u:

$$\frac{\partial \beta^*}{\partial \pi_0^u} = \frac{0 \cdot D - N \cdot (-1)}{D^2} = \frac{N}{D^2} = \frac{\pi^{d*} - \pi_0^d}{D^2} > 0,$$

since $\pi^{d*} > \pi_0^d$ by Proposition OA-1. □

C.6 Proof of Proposition 3

Proof. Consumer surplus $CS(\beta) = \int_0^{q(\beta)} (p(h) - p(q(\beta)))\, dh$ is monotonically increasing in output q (since $p'(q) < 0$). Therefore, CS is maximized at the β that maximizes output.

Under monopolistic bargaining, $dq^{mp}/d\beta > 0$ by Theorem 1. Hence $CS(\beta)$ is increasing in β, and consumer surplus is maximized at $\beta = 1$.

Under monopsonistic bargaining, $dq^{ms}/d\beta < 0$ by Theorem 1. Hence $CS(\beta)$ is decreasing in β, and consumer surplus is maximized at $\beta = 0$. □

C.7 Proof of Proposition 4

Proof. By Lemma OA-13, total welfare is maximized at $q^\dagger$ defined by $p(q^\dagger) = mc(q^\dagger)$, with $q^\dagger > q^*$.

First, consider monopsonistic bargaining. As shown in Section D.1.2, $\beta = 0$ results in the condition $p(q) = mc(q)$. This is the first-best any planner could achieve, so total welfare is maximized at this point.

Second, consider monopolistic bargaining. At $\beta = \beta^*$, $mr(q^*) = mc(q^*)$. Given that prices are set by downstream at a markup above marginal costs, this implies that $p(q) > mc(q)$ at the joint-profit-maximization level of buyer power β^*, which is lower than the first-best total welfare maximizing quantity.

As is proven in Section D.1.3, at $\beta = 1$ we have that $mr(q) = c(q)$. Hence, prices are above average costs:

$$p(q(\beta = 1)) = c(q(\beta = 1)) + \mu(q(\beta = 1)).$$

Let $b(q(\beta = 1)) = mc(q(\beta = 1)) - c(q(\beta = 1))$. It follows that there are three possibilities:

$$\begin{cases} b(q(\beta = 1)) = \mu(q(\beta = 1)) & \Rightarrow \beta^\dagger = 1 \\ b(q(\beta = 1)) < \mu(q(\beta = 1)) & \Rightarrow \beta^\dagger = 1 \\ b(q(\beta = 1)) > \mu(q(\beta = 1)) & \Rightarrow \beta^\dagger \in (\beta^*, 1) \end{cases}$$

First, if $b(q(\beta = 1)) = \mu(q(\beta = 1))$, $\beta = 1$ maximizes total welfare and leads to the first-best solution $p(q(\beta = 1)) = mc(q(\beta = 1))$. Second, if $b(q(\beta = 1)) < \mu(q(\beta = 1))$, prices are still too high at $\beta = 1$, as $p(q(\beta = 1)) > mc(q(\beta = 1))$. However, given that $\beta = 1$ is the highest possible value of β, welfare is maximized at this value. Third, if $b(q(\beta = 1)) > \mu(q(\beta = 1))$, the price at $\beta = 1$ is below marginal costs, meaning that there is overproduction. Given that $\frac{\partial q}{\partial \beta} > 0$ under monopolistic conduct and β^* leads to a total quantity lower than first-best, this implies that total welfare is maximized at $\beta^* < \beta < 1$. □

C.8 Proof of Theorem 2

Proof. We show that under Constraint 4.1, at any $\beta \in [0, 1]$, the vertical distortion exists only under monopolistic conduct if $\beta < \beta^*$ and under monopsonistic conduct if $\beta > \beta^*$. We consider three cases depending on the disagreement payoffs.

Case 1: $\pi_0^d \leq \pi^{d*}$ and $\pi_0^u \leq \pi^{u*}$ ($\beta^* \in [0, 1]$)

Monopolistic conduct:

Under monopolistic bargaining, the D-FOC gives $w = mr(q)$, so the markdown $\Delta^d = 0$ automatically. The constraint that must be satisfied is the nonnegative markup: $w \geq mc(q)$, i.e., $mr(q) \geq mc(q)$, i.e., $q \leq q^*$.

- If $\beta \leq \beta^*$: From Theorem 1, q^{mp} is increasing in β, and $q^{mp}(\beta^*) = q^*$. Therefore $q^{mp}(\beta) \leq q^*$ for $\beta \leq \beta^*$, so $w = mr(q^{mp}) \geq mr(q^*) = mc(q^*) \geq mc(q^{mp})$. The constraint $w \geq mc(q)$ is satisfied (slack). The standard unconstrained equilibrium prevails with a vertical distortion ($\mu^u > 0$ when $\beta < \beta^*$).

- If $\beta > \beta^*$: The unconstrained solution would give $q^{mp} > q^*$, implying $w = mr(q^{mp}) < mr(q^*) = mc(q^*) < mc(q^{mp})$, which violates $w \geq mc(q)$. The constraint binds: $w = mc(q)$ together with $w = mr(q)$ forces $mc(q) = mr(q)$, hence $q = q^*$ and $w = w^*$. No distortion.

Monopsonistic conduct:

Under monopsonistic bargaining, the U-FOC gives $w = mc(q)$, so the markup $\mu^u = 0$ automatically. The constraint that must be satisfied is the nonnegative markdown: $w \leq mr(q)$, i.e., $mc(q) \leq mr(q)$, i.e., $q \leq q^*$.

- If $\beta \geq \beta^*$: From Theorem 1, q^{ms} is decreasing in β, and $q^{ms}(\beta^*) = q^*$. Therefore $q^{ms}(\beta) \leq q^*$ for $\beta \geq \beta^*$, so $w = mc(q^{ms}) \leq mc(q^*) = mr(q^*) \leq mr(q^{ms})$. The constraint $w \leq mr(q)$ is satisfied (slack). The standard unconstrained equilibrium prevails with a vertical distortion ($\Delta^d > 0$ when $\beta > \beta^*$).

- If $\beta < \beta^*$: The unconstrained solution would give $q^{ms} > q^*$, implying $w = mc(q^{ms}) > mc(q^*) = mr(q^*) > mr(q^{ms})$, which violates $w \leq mr(q)$. The constraint binds: $w = mr(q)$ together with $w = mc(q)$ forces $mc(q) = mr(q)$, hence $q = q^*$ and $w = w^*$. No distortion.

Case 2: $\pi_0^d > \pi^{d*}$ ($\beta^* = 0$)

By Proposition OA-2, $q^{ms}(\beta) < q^*$ for all β under monopsonistic conduct, and $q^{mp}(\beta) > q^*$ for all β under monopolistic conduct.

- *Monopsonistic conduct:* Since $q^{ms}(\beta) < q^*$ for all β, we have $w = mc(q^{ms}) < mc(q^*) = mr(q^*) < mr(q^{ms})$, so $w \leq mr(q)$ is satisfied. The constraint is slack and the monopsonistic equilibrium has a distortion ($\Delta^d > 0$) for all $\beta > 0$.

- *Monopolistic conduct:* Since $q^{mp}(\beta) > q^*$ for all β, we have $w = mr(q^{mp}) < mr(q^*) = mc(q^*) < mc(q^{mp})$, so $w \geq mc(q)$ is violated. The constraint requires $q \leq q^*$, and the constrained optimum would be at $q = q^*$ with $w = w^*$. However, at q^* downstream earns $\pi^d(q^*) = \pi^{d*} < \pi^d_0$, violating downstream's participation constraint. Therefore, the monopolistic equilibrium does not exist under Constraint 4.1.

Since $\beta^* = 0$, the condition "monopolistic if $\beta < \beta^*$" is never satisfied, and "monopsonistic if $\beta > \beta^*$" holds for all $\beta \in (0, 1]$, consistent with the theorem.

Case 3: $\pi^u_0 > \pi^{u*}$ ($\beta^* = 1$)

By Proposition OA-3, $q^{ms}(\beta) > q^*$ for all β under monopsonistic conduct, and $q^{mp}(\beta) < q^*$ for all β under monopolistic conduct.

- *Monopolistic conduct:* Since $q^{mp}(\beta) < q^*$ for all β, we have $w = mr(q^{mp}) > mr(q^*) = mc(q^*) > mc(q^{mp})$, so $w \geq mc(q)$ is satisfied. The constraint is slack and the monopolistic equilibrium has a distortion ($\mu^u > 0$) for all $\beta < 1$.

- *Monopsonistic conduct:* Since $q^{ms}(\beta) > q^*$ for all β, we have $w = mc(q^{ms}) > mc(q^*) = mr(q^*) > mr(q^{ms})$, so $w \leq mr(q)$ is violated. The constraint requires $q \leq q^*$, and the constrained optimum would be at $q = q^*$ with $w = w^*$. However, at q^* upstream earns $\pi^u(q^*) = \pi^{u*} < \pi^u_0$, violating upstream's participation constraint. Therefore, the monopsonistic equilibrium does not exist under Constraint 4.1.

Since $\beta^* = 1$, the condition "monopolistic if $\beta < \beta^*$" holds for all $\beta \in [0, 1)$, and "monopsonistic if $\beta > \beta^*$" is never satisfied, consistent with the theorem.

In all three cases, for any $\beta < \beta^*$, monopolistic conduct operates unconstrained (with distortion $\mu^u > 0$) while monopsonistic conduct is constrained to q^* (no distortion). For $\beta > \beta^*$, monopsonistic conduct operates unconstrained (with distortion $\Delta^d > 0$) while monopolistic conduct is constrained to q^* (no distortion). At $\beta = \beta^*$, both models yield q^* with no distortion. □

C.9 Proof of Proposition 5

Proof. Under Constraint 4.1, output satisfies $q \leq q^*$ under both conduct models (as shown in the proof of Theorem 2).

Part (i): Monopolistic bargaining $\Rightarrow w \geq w^*$.

Under monopolistic bargaining, $w = mr(q^{mp})$. Since $q^{mp} \leq q^*$ and $mr'(q) < 0$:

$$w = mr(q^{mp}) \geq mr(q^*) = w^*.$$

Part (ii): Monopsonistic bargaining $\Rightarrow w \leq w^*$.

Under monopsonistic bargaining, $w = mc(q^{ms})$. Since $q^{ms} \leq q^*$ and $mc'(q) > 0$:

$$w = mc(q^{ms}) \leq mc(q^*) = w^*.$$

Therefore, $w > w^*$ identifies monopolistic conduct and $w < w^*$ identifies monopsonistic conduct. $\square$

D Auxiliary Lemmas and Results

D.1 Equilibrium Under Limit Cases for β

We solve the bargaining model (monopolistic and monopsonistic) as a constrained profit-maximization model in the limiting cases of $\beta = 1$ and $\beta = 0$ and compare these corner solutions to the solutions obtained from the first-order conditions derived in Section B.1.

D.1.1 Monopsonistic Bargaining, $\beta = 1$

At $\beta = 1$, downstream has full bargaining power. The stage-2 quantity is determined by upstream's FOC ($w = mc(q)$). The stage-1 problem is:

$$\max_q (p(q) - mc(q))q \quad \text{s.t.} \quad c'(q)q^2 \geq \pi_0^u.$$

The unconstrained FOC gives the classical monopsony condition:

$$mr(q) = mc'(q)q + mc(q), \quad w = mc(q).$$

If the participation constraint is slack at this solution (i.e., $c'(q)q^2 \geq \pi_0^u$), this is the equilibrium. If the constraint binds, the equilibrium quantity is instead determined by $c'(q)q^2 = \pi_0^u$. To see this, note that the bargaining SOC (Section B.2) evaluated at $\beta = 1$ implies that $(p(q) - mc(q))q$ is concave in q (via the monotone change of variables $w = mc(q)$). Since the unconstrained maximizer lies below the feasible set $\{q : c'(q)q^2 \geq \pi_0^u\}$, concavity implies the objective is decreasing on the feasible set, so the constrained maximum is at the boundary.

D.1.2 Monopsonistic Bargaining, $\beta = 0$

At $\beta = 0$, upstream has full bargaining power. The stage-2 quantity is determined by upstream's FOC ($w = mc(q)$). The stage-1 problem is:

$$\max_q c'(q)q^2 \quad \text{s.t.} \quad (p(q) - mc(q))q \geq \pi_0^d.$$

Upstream maximizes its own profit $\pi^u = c'(q)q^2$, which is strictly increasing in q. Therefore, upstream wants q as large as possible. The binding constraint is downstream's participation: $(p(q) - mc(q))q = \pi_0^d$, which determines the equilibrium quantity. When $\pi_0^d = 0$, this gives $p(q) = mc(q)$, i.e., price equals marginal cost (the first-best outcome). When $\pi_0^d > 0$, the equilibrium quantity solves $(p(q) - mc(q))q = \pi_0^d$.

D.1.3 Monopolistic Bargaining, $\beta = 1$

At $\beta = 1$, downstream has full bargaining power. The stage-2 quantity is determined by downstream's FOC ($w = mr(q)$). The stage-1 problem is:

$$\max_q \; -p'(q)q^2 \quad \text{s.t.} \quad (mr(q) - c(q))q \geq \pi_0^u.$$

Downstream maximizes its own profit $\pi^d = -p'(q)q^2$, which is strictly increasing in q. Therefore, downstream wants q as large as possible. The binding constraint is upstream's participation: $(mr(q) - c(q))q = \pi_0^u$, which determines the equilibrium quantity. When $\pi_0^u = 0$, this gives $mr(q) = c(q)$, i.e., $w = c(q)$. When $\pi_0^u > 0$, the equilibrium quantity solves $(mr(q) - c(q))q = \pi_0^u$.

D.1.4 Monopolistic Bargaining, $\beta = 0$

At $\beta = 0$, upstream has full bargaining power. The stage-2 quantity is determined by downstream's FOC ($w = mr(q)$). The stage-1 problem is:

$$\max_q \; (mr(q) - c(q))q \quad \text{s.t.} \quad -p'(q)q^2 \geq \pi_0^d.$$

The unconstrained FOC gives the full double marginalization condition:

$$\text{mr}(q) = w, \quad mc(q) = mr'(q)q + mr(q).$$

If the participation constraint is slack at this solution (i.e., $-p'(q)q^2 \geq \pi_0^d$), this is the equilibrium. If the constraint binds, the equilibrium quantity is instead determined by $-p'(q)q^2 = \pi_0^d$. To see this, note that the bargaining SOC (Section B.2) evaluated at $\beta = 0$ implies that $(mr(q) - c(q))q$ is concave in q (via the monotone change of variables $w = mr(q)$). Since the unconstrained maximizer lies above the feasible set $\{q : -p'(q)q^2 \geq \pi_0^d\}$, concavity implies the objective is increasing on the feasible set, so the constrained maximum is at the boundary.

D.2 Auxiliary Lemmas on Equilibrium Existence

In this appendix, we discuss the existence and unicity of the monopolistic and monopsonistic equilibrium in the bargaining model.

For each conduct model, we show that the FOC solution for interior β converges to the direct constrained-optimization solution at the corner values $\beta = 0$ and $\beta = 1$. We then rely on continuity to establish existence for all $\beta \in [0, 1]$.

Lemma OA-4. *The solution to the monopsonistic bargaining with disagreement payoffs π_0^d and π_0^u, characterized by its FOCs given in Appendix B.1, approaches as $\beta \to 1$ to the solution of the constraint-optimization problem at $\beta = 1$ provided in Appendix D.1.1.*

Proof. The B-FOC under monopsonistic conduct is $\beta\, A(q) + (1-\beta)\, B(q) = 0$ where:

$$A(q) = \frac{d\pi^d/dw}{\pi^d - \pi_0^d},$$

$$B(q) = \frac{q}{c'(q)q^2 - \pi_0^u}.$$

Write $\beta = 1 - \varepsilon$. The equation becomes $(1-\varepsilon)A + \varepsilon B = 0$, i.e., $A = -\frac{\varepsilon}{1-\varepsilon}B$.

As $\varepsilon \to 0$, the right side tends to zero. Two cases arise:

Case (a): If B remains bounded (i.e., $c'(q)q^2 - \pi_0^u$ stays bounded away from zero), then $A \to 0$, which gives $d\pi^d/dw = 0$. This is the classical monopsony condition $mr(q) = mc'(q)q + mc(q)$ from Appendix D.1.1, with the participation constraint $c'(q)q^2 \geq \pi_0^u$ slack.

Case (b): If $B \to \pm\infty$ (i.e., $c'(q)q^2 - \pi_0^u \to 0$), then the limit gives $c'(q)q^2 = \pi_0^u$, which is the binding participation constraint from Appendix D.1.1. This case arises when the classical monopsony quantity q_{cm} (defined by $mr(q_{cm}) = mc'(q_{cm})q_{cm} + mc(q_{cm})$) satisfies $c'(q_{cm})q_{cm}^2 < \pi_0^u$, so that upstream would not participate at q_{cm}. For any $\beta < 1$, the interior FOC solution satisfies $\pi^u(q(\beta)) > \pi_0^u$ (both participation constraints hold with strict inequality). Since $dq^{ms}/d\beta < 0$ (Theorem 1), output $q(\beta)$ decreases as β increases, and upstream profit $\pi^u(q) = c'(q)q^2$ decreases with q. As $\beta \to 1$, $\pi^u(q(\beta))$ is squeezed toward π_0^u from above, and the equilibrium converges to $\bar{q}$ where $c'(\bar{q})\bar{q}^2 = \pi_0^u$. In the B-FOC, since $(1-\varepsilon)A + \varepsilon B = 0$ holds at each $\varepsilon > 0$, we have $\varepsilon B = -(1-\varepsilon)A$. As $\varepsilon \to 0$, $(1-\varepsilon) \to 1$ and $A(q(\varepsilon)) \to A(\bar{q})$, which is finite (since $\pi^d(\bar{q}) > \pi_0^d$ at the limit). Therefore $\varepsilon B \to -A(\bar{q})$, confirming that εB remains finite.

In both cases, the FOC solution converges to the direct constrained-optimization solution at $\beta = 1$. □

Lemma OA-5. *The solution to the monopolistic bargaining with disagreement payoffs π_0^d and π_0^u, characterized by its FOCs given in Appendix B.1, approaches as $\beta \to 1$ to the solution of the constraint-optimization problem at $\beta = 1$ provided in Appendix D.1.3.*

Proof. Define

$$A(q) \equiv \frac{d\pi^d/dw}{\pi^d - \pi_0^d} = \frac{-q}{-p'(q)q^2 - \pi_0^d},$$

$$B(q) \equiv \frac{d\pi^u/dw}{\pi^u - \pi_0^u} = \frac{q + [p(q) - c(q) + p'(q)q - c'(q)q]\frac{1}{2p'(q)+p''(q)q}}{[p(q) - c(q) + p'(q)q]q - \pi_0^u}$$

The FOC that characterizes equilibrium q is $\beta\, A(q) \;+\; (1-\beta)\, B(q) \;=\; 0$. Rewrite β as $1-\varepsilon$. Then, the equation becomes

$$(1-\varepsilon)\, A(q) \;+\; \varepsilon\, B(q) \;=\; 0 \quad\Longrightarrow\quad \frac{1-\varepsilon}{\varepsilon}\, A(q) \;=\; -\, B(q).$$

As $\varepsilon \to 0$, the left side tends to $\pm\infty$ (since $A(q) = -q/(-p'(q)q^2 - \pi_0^d) \neq 0$). Thus, $B(q)$ must also become unbounded in magnitude. The numerator of $B(q)$ is bounded, so the denominator must vanish:

$$[p(q) - c(q) + p'(q)q]q - \pi_0^u \;\to\; 0.$$

When $\pi_0^u = 0$, this gives $p(q) - c(q) + qp'(q) = 0$, i.e., $mr(q) = c(q)$. When $\pi_0^u > 0$, the condition becomes $[p(q)+p'(q)q-c(q)]q = \pi_0^u$, which pins down the equilibrium quantity at $\beta = 1$. In both cases, this corresponds to the solution of the constraint-optimization problem at $\beta = 1$ given in Appendix D.1.3. □

Lemma OA-6. *The solution to monopsonistic bargaining with disagreement payoffs π_0^d and π_0^u, characterized by its FOC given in Appendix B.1, approaches as $\beta \to 0$ to the solution of the constraint-optimization problem at $\beta = 0$ given in Appendix D.1.2.*

Proof. Define

$$A(q) \equiv \frac{d\pi^u/dw}{\pi^u - \pi_0^u} = \frac{q}{c'(q)q^2 - \pi_0^u},$$

$$B(q) \equiv \frac{d\pi^d/dw}{\pi^d - \pi_0^d} = \frac{-q + \left([p(q) - c(q)] + [p'(q)q - c'(q)q]\frac{1}{2c'(q)+c''(q)q}\right)}{[p(q) - c(q) - c'(q)q]q - \pi_0^d}$$

The FOC that characterizes equilibrium q is $(1-\beta)\, A(q) \;+\; \beta\, B(q) \;=\; 0$. Rewrite β as ε. Then, the equation becomes

$$(1-\varepsilon)A(q) + \varepsilon B(q) = 0 \quad\Longrightarrow\quad \frac{\varepsilon}{1-\varepsilon} = -\frac{A(q)}{B(q)}$$

As $\varepsilon \to 0$ (i.e., $\beta \to 0$), the multiplier $\frac{\varepsilon}{1-\varepsilon}$ tends to 0. If $A(q) \neq 0$ is finite, we must have $B(q)$

become unbounded (go to $\pm\infty$). The numerator of $B(q)$ is bounded, so the denominator must vanish:

$$[p(q) - c(q) - c'(q)q]q - \pi_0^d \;\to\; 0.$$

When $\pi_0^d = 0$, this gives $p(q) - c(q) - c'(q)q = 0$, i.e., $p(q) = mc(q)$. When $\pi_0^d > 0$, the condition becomes $[p(q) - c(q) - c'(q)q]q = \pi_0^d$, which pins down the equilibrium quantity at $\beta = 0$. In both cases, this corresponds to the solution of the constraint-optimization problem in Appendix D.1.2. □

Lemma OA-7. *The solution to the monopolistic bargaining with disagreement payoffs π_0^d and π_0^u, characterized by its FOCs given in Appendix B.1, approaches as $\beta \to 0$ to the solution of the constraint-optimization problem at $\beta = 0$ provided in Appendix D.1.4.*

Proof. The B-FOC under monopolistic conduct is $\beta\, A(q) + (1 - \beta)\, B(q) = 0$ where:

$$A(q) = \frac{-q}{-p'(q)q^2 - \pi_0^d},$$
$$B(q) = \frac{d\pi^u/dw}{\pi^u - \pi_0^u}.$$

Write $\beta = \varepsilon$. The equation becomes $\varepsilon A + (1 - \varepsilon)B = 0$, i.e., $B = -\frac{\varepsilon}{1-\varepsilon}A$.

As $\varepsilon \to 0$, the right side tends to zero. Two cases arise:

Case (a): If A remains bounded (i.e., $-p'(q)q^2 - \pi_0^d$ stays bounded away from zero), then $B \to 0$, which gives $d\pi^u/dw = 0$. This is the full double marginalization condition $mc(q) = mr'(q)q + mr(q)$ from Appendix D.1.4, with the participation constraint $-p'(q)q^2 \geq \pi_0^d$ slack.

Case (b): If $A \to \pm\infty$ (i.e., $-p'(q)q^2 - \pi_0^d \to 0$), then the limit gives $-p'(q)q^2 = \pi_0^d$, which is the binding participation constraint from Appendix D.1.4. This case arises when the double marginalization quantity would violate $\pi^d \geq \pi_0^d$.

In both cases, the FOC solution converges to the direct constrained-optimization solution at $\beta = 0$. □

Lemma OA-8. *Total surplus $TS(q) \equiv \int_0^q [p(h) - mc(h)]\, dh$ is maximized at $q^\dagger$ defined by $p(q^\dagger) = mc(q^\dagger)$, and $q^\dagger > q^*$.*

Proof. The derivative of total surplus is $\partial TS/\partial q = p(q) - mc(q)$. Since $p'(q) < 0$ and $mc'(q) > 0$, $p(q) - mc(q)$ is strictly decreasing, so TS is strictly concave. The unique maximum is at $q^\dagger$ where $p(q^\dagger) = mc(q^\dagger)$.

To show $q^\dagger > q^*$: since $mr(q) = p(q) + p'(q)q < p(q)$ for all $q > 0$ (because $p'(q) < 0$), at q^* we have $p(q^*) > mr(q^*) = mc(q^*)$. Since $p(q) - mc(q)$ is strictly decreasing and equals zero at $q^\dagger$, it follows that $q^\dagger > q^*$. □

Lemma OA-9. *In the bargaining model, the following inequalities about profits apply:*

$$\begin{cases} \pi_d^{ms} > \pi_d^* \\ \pi_u^{ms} < \pi_u^* \end{cases} (\beta < \beta^*) \quad \begin{cases} \pi_d^{mp} > \pi_d^* \\ \pi_u^{mp} < \pi_u^* \end{cases} (\beta < \beta^*) \quad \begin{cases} \pi_d^{ms} < \pi_d^* \\ \pi_u^{ms} > \pi_u^* \end{cases} (\beta > \beta^*) \quad \begin{cases} \pi_d^{mp} < \pi_d^* \\ \pi_u^{mp} > \pi_u^* \end{cases} (\beta > \beta^*)$$

Proof. We prove the result for monopsonistic bargaining; the monopolistic case follows the same structure with the roles of upstream and downstream interchanged. Define $S(q) \equiv (p(q) - c(q))q$ as total joint surplus. Since $dS/dq = mr(q) - mc(q)$ and mr is strictly decreasing while mc is strictly increasing, q^* is the unique maximizer of S.

Step 1: Profits agree at β^*. By Proposition 2, at $\beta = \beta^*$ both monopsonistic bargaining and joint-profit maximization yield q^*. By Lemma OA-10, the profits at q^* are the same under both conduct models, so $\pi_u^{ms}(\beta^*) = \pi_u^*(\beta^*)$ and $\pi_d^{ms}(\beta^*) = \pi_d^*(\beta^*)$.

Step 2: Corner verification at $\beta = 1$. Under joint-profit maximization at $\beta = 1$, downstream has full power and drives upstream to its disagreement payoff: $\pi_u^*(\beta = 1) = \pi_0^u$. Under monopsonistic bargaining at $\beta = 1$ (Section D.1.1), the equilibrium quantity q_1 satisfies either $mr(q_1) = mc'(q_1)q_1 + mc(q_1)$ (classical monopsony, if the participation constraint $c'(q)q^2 \geq \pi_0^u$ is slack) or $c'(q_1)q_1^2 = \pi_0^u$ (if the constraint binds). In either case, $\pi_u^{ms} = c'(q_1)q_1^2 \geq \pi_0^u = \pi_u^*$.

Step 3: Corner verification at $\beta = 0$. Under monopsonistic bargaining at $\beta = 0$ (Section D.1.2), upstream makes a TIOLI offer, driving downstream to its participation constraint: $\pi_d^{ms} = \pi_0^d$. Upstream earns $\pi_u^{ms} = S(q_0) - \pi_0^d$ where q_0 satisfies $(p(q_0) - mc(q_0))q_0 = \pi_0^d$. Under joint-profit maximization at $\beta = 0$, upstream captures the entire surplus above disagreement payoffs: $\pi_u^*(\beta = 0) = S(q^*) - \pi_0^d$. Since $S(q_0) < S(q^*)$ (because $q_0 \neq q^*$ and S is maximized at q^*), we have $\pi_u^{ms} = S(q_0) - \pi_0^d < S(q^*) - \pi_0^d = \pi_u^*$.

Step 4: Continuity. From Steps 1–3: $\pi_u^{ms} = \pi_u^*$ at β^*, $\pi_u^{ms} \geq \pi_u^*$ at $\beta = 1$, and $\pi_u^{ms} < \pi_u^*$ at $\beta = 0$. By Proposition 2, β^* is the unique value at which the profits agree. For any $\beta \in (\beta^*, 1)$, both participation constraints are strictly slack, so $\pi_u^{ms} > \pi_0^u = \pi_u^*(\beta = 1)$ and the profit functions are continuous. Since π_u^{ms} and π_u^* agree only at β^*, continuity implies $\pi_u^{ms} > \pi_u^*$ for all $\beta \in (\beta^*, 1]$ and $\pi_u^{ms} < \pi_u^*$ for all $\beta \in [0, \beta^*)$. The downstream inequalities follow from the total surplus comparison: $\pi_d^{ms} = S(q^{ms}) - \pi_u^{ms}$ and $\pi_d^* = S(q^*) - \pi_u^*$. Since $S(q^{ms}) \leq S(q^*)$ and $\pi_u^{ms} > \pi_u^*$ (for $\beta > \beta^*$), we have $\pi_d^{ms} < \pi_d^*$. The case $\beta < \beta^*$ follows analogously. $\square$

D.3 Other Auxiliary Lemmas

Lemma OA-10. *At the joint-profit maximizing quantity q^* (where $mc(q^*) = mr(q^*)$), the input price $w^* \equiv mc(q^*) = mr(q^*)$ is the same under both monopsonistic and monopolistic conduct, and the upstream and downstream profits are:*

$$\pi^{u*} \equiv (w^* - c(q^*))q^* = c'(q^*)(q^*)^2,$$
$$\pi^{d*} \equiv (p(q^*) - w^*)q^* = -p'(q^*)(q^*)^2.$$

These expressions are independent of the conduct model.

Proof. Under monopsonistic conduct, $w = mc(q)$. At q^*: $w^* = mc(q^*) = mr(q^*)$. The upstream profit is $\pi^u = (w^* - c(q^*))q^* = (mc(q^*) - c(q^*))q^* = c'(q^*)(q^*)^2$. The downstream profit is $\pi^d = (p(q^*) - w^*)q^* = (p(q^*) - mr(q^*))q^* = -p'(q^*)(q^*)^2$.

Under monopolistic conduct, $w = mr(q)$. At q^*: $w^* = mr(q^*) = mc(q^*)$, which is the same w^*. The profit expressions are therefore identical. $\square$

Lemma OA-11. *Under Assumption OA-1 and the second-order condition (Lemma OA-1), the equilibrium quantity $q(\beta)$ is continuously differentiable in β for $\beta \in (0, 1)$ under both monopsonistic and monopolistic conduct.*

Proof. The equilibrium is characterized by the B-FOC $f(\beta, w) = 0$. The second-order condition ensures $\partial f/\partial w < 0$ at the equilibrium. By the Implicit Function Theorem, $w(\beta)$ is continuously differentiable in β in a neighborhood of any $\beta \in (0, 1)$. Under monopsonistic conduct, $q = q^u(w)$ where $dq^u/dw = 1/(2c'(q)+c''(q)q)$ is continuous (since $mc'(q) > 0$). Under monopolistic conduct, $q = q^d(w)$ where $dq^d/dw = 1/(2p'(q) + p''(q)q)$ is continuous (since $mr'(q) < 0$). Therefore $q(\beta) = q(w(\beta))$ is continuously differentiable as the composition of continuously differentiable functions. $\square$

Lemma OA-12. *The joint profit-maximizing quantity q^* is unique.*

Proof. q^* is characterized by $mr(q^*) - mc(q^*) = 0$. Since $mr'(q) < 0$ and $mc'(q) > 0$, there is a unique solution to this equation. $\square$

Lemma OA-13. *Total surplus $TS(q) \equiv \int_0^q [p(h) - mc(h)]\, dh$ is maximized at $q^\dagger$ defined by $p(q^\dagger) = mc(q^\dagger)$, and $q^\dagger > q^*$.*

Proof. The derivative of total surplus is $\partial TS/\partial q = p(q) - mc(q)$. Since $p'(q) < 0$ and $mc'(q) > 0$, $p(q) - mc(q)$ is strictly decreasing, so TS is strictly concave. The unique maximum is at $q^\dagger$ where $p(q^\dagger) = mc(q^\dagger)$.

To show $q^\dagger > q^*$: since $mr(q) = p(q) + p'(q)q < p(q)$ for all $q > 0$ (because $p'(q) < 0$), at q^* we have $p(q^*) > mr(q^*) = mc(q^*)$. Since $p(q) - mc(q)$ is strictly decreasing and equals zero at $q^\dagger$, it follows that $q^\dagger > q^*$. □

Lemma OA-14. *In the bargaining model, the following inequalities about profits apply:*

$$\begin{cases} \pi_d^{ms} > \pi_d^* \\ \pi_u^{ms} < \pi_u^* \end{cases} (\beta < \beta^*) \quad \begin{cases} \pi_d^{mp} > \pi_d^* \\ \pi_u^{mp} < \pi_u^* \end{cases} (\beta < \beta^*) \quad \begin{cases} \pi_d^{ms} < \pi_d^* \\ \pi_u^{ms} > \pi_u^* \end{cases} (\beta > \beta^*) \quad \begin{cases} \pi_d^{mp} < \pi_d^* \\ \pi_u^{mp} > \pi_u^* \end{cases} (\beta > \beta^*)$$

Proof. We prove the result for monopsonistic bargaining; the monopolistic case follows the same structure with the roles of upstream and downstream interchanged. Define $S(q) \equiv (p(q) - c(q))q$ as total joint surplus. Since $dS/dq = mr(q) - mc(q)$ and mr is strictly decreasing while mc is strictly increasing, q^* is the unique maximizer of S.

Step 1: Profits agree at β^*. By Proposition 2, at $\beta = \beta^*$ both monopsonistic bargaining and joint-profit maximization yield q^*. By Lemma OA-10, the profits at q^* are the same under both conduct models, so $\pi_u^{ms}(\beta^*) = \pi_u^*(\beta^*)$ and $\pi_d^{ms}(\beta^*) = \pi_d^*(\beta^*)$.

Step 2: Corner verification at $\beta = 1$. Under joint-profit maximization at $\beta = 1$, downstream has full power and drives upstream to its disagreement payoff: $\pi_u^*(\beta = 1) = \pi_0^u$. Under monopsonistic bargaining at $\beta = 1$ (Section D.1.1), the equilibrium quantity q_1 satisfies either $mr(q_1) = mc'(q_1)q_1 + mc(q_1)$ (classical monopsony, if the participation constraint $c'(q)q^2 \geq \pi_0^u$ is slack) or $c'(q_1)q_1^2 = \pi_0^u$ (if the constraint binds). In either case, $\pi_u^{ms} = c'(q_1)q_1^2 \geq \pi_0^u = \pi_u^*$.

Step 3: Corner verification at $\beta = 0$. Under monopsonistic bargaining at $\beta = 0$ (Section D.1.2), upstream makes a TIOLI offer, driving downstream to its participation constraint: $\pi_d^{ms} = \pi_0^d$. Upstream earns $\pi_u^{ms} = S(q_0) - \pi_0^d$ where q_0 satisfies $(p(q_0) - mc(q_0))q_0 = \pi_0^d$. Under joint-profit maximization at $\beta = 0$, upstream captures the entire surplus above disagreement payoffs: $\pi_u^*(\beta = 0) = S(q^*) - \pi_0^d$. Since $S(q_0) < S(q^*)$ (because $q_0 \neq q^*$ and S is maximized at q^*), we have $\pi_u^{ms} = S(q_0) - \pi_0^d < S(q^*) - \pi_0^d = \pi_u^*$.

Step 4: Continuity. From Steps 1–3: $\pi_u^{ms} = \pi_u^*$ at β^*, $\pi_u^{ms} \geq \pi_u^*$ at $\beta = 1$, and $\pi_u^{ms} < \pi_u^*$ at $\beta = 0$. By Proposition 2, β^* is the unique value at which the profits agree. For any $\beta \in (\beta^*, 1)$, both participation constraints are strictly slack, so $\pi_u^{ms} > \pi_0^u = \pi_u^*(\beta = 1)$ and the profit functions are continuous. Since π_u^{ms} and π_u^* agree only at β^*, continuity implies $\pi_u^{ms} > \pi_u^*$ for all $\beta \in (\beta^*, 1]$ and $\pi_u^{ms} < \pi_u^*$ for all $\beta \in [0, \beta^*)$. The downstream inequalities follow from the total surplus comparison: $\pi_d^{ms} = S(q^{ms}) - \pi_u^{ms}$ and $\pi_d^* = S(q^*) - \pi_u^*$. Since $S(q^{ms}) \leq S(q^*)$ and $\pi_u^{ms} > \pi_u^*$ (for $\beta > \beta^*$), we have $\pi_d^{ms} < \pi_d^*$. The case $\beta < \beta^*$ follows analogously. $\square$

D.4 Loglinear Version of the Model

We solve the bargaining model with log-linear costs and demand:

$$c(q) = \frac{1}{1+\psi} q^{\psi} \quad \text{and} \quad p(q) = q^{\frac{1}{\eta}}.$$

Solving the first-order condition for output in the monopsonistic conduct case (U-FOC) results in the factor supply curve $w = q^{\psi}$. Solving the first-order condition for output in the monopolistic conduct case (D-FOC) results in the factor demand curve $w = q^{\frac{1}{\eta}}(\frac{\eta+1}{\eta})$. The joint-profit-maximizing output level is found by equating marginal costs to marginal revenue, which results in $q^* = (\frac{1+\eta}{\eta})^{\frac{1}{\psi - \frac{1}{\eta}}}$.

Solving the bargaining problem (B-FOC) and setting it equal to the monopsonistic and monopolistic cases to find the intersection of the two output-buyer power curves results in the output-maximizing bargaining parameter

$$\beta^* = \left(\frac{1+\eta}{1+\psi} - \eta \right)^{-1}.$$

D.4.1 Corollary 2 in terms of elasticities

In the log-linear version of the model, we can write Corollary 2 as a function of supply and demand elasticities rather than the first derivatives of costs and demand.

Corollary OA-1. *The bilaterally-efficient level of buyer power β^* weakly decreases with the elasticity of downstream demand and weakly increases with the elasticity of upstream supply.*

Proof. The β^* expression in the log-linear model that was derived above:

$$\beta^* = \left(\frac{1+\eta}{1+\psi} - \eta \right)^{-1}.$$

The elasticity of downstream demand is η is negative, we conduct comparative statics in terms of $(-\eta)$ in order to have a higher value of this parameter indicate more elastic demand. Taking the first derivative of β^* to $(-\eta)$ results in:

$$\frac{\partial \beta^*}{\partial(-\eta)} = -\frac{\psi(1+\psi)}{(1-\eta\psi)^2} \leq 0$$

Hence, more elastic downstream demand implies a lower bilaterally-efficient level of buyer power β^*.

The elasticity of upstream supply is $\frac{1}{\psi}$. Taking the first derivative of β^* to ψ results in:

$$\frac{\partial \beta^*}{\partial \psi} = -\left(\frac{1+\eta}{1+\psi} - \eta\right)(-(1+\eta)) \leq 0$$

This expression is weakly positive because $\eta \leq -1$ is required for profit maximization, and because $\frac{1+\eta}{1+\psi} - \eta = \frac{1}{\beta^*} \geq 0$. It follows that the more inelastic the upstream supply is, the lower β^*. Hence, the more elastic the upstream supply, the higher β^*. □

E Extensions

In this Appendix, we extend our model by incorporating (i) multi-input downstream production and (ii) an alternative distortion classification criterion based on incentive compatibility of linear pricing. We show that our main results are robust to these extensions.

E.1 Multi-Input Downstream Production

Our baseline model assumes a single-input production function where the downstream firm simply resells the input in the downstream market. In this Appendix, we extend this model to incorporate multi-input downstream production. We show that while the bargaining problem remains largely unchanged, it requires some modifications. Specifically, under monopsonistic bargaining, the upstream firm's output choice influences downstream output through a monotone function rather than by determining it directly, as the downstream firm can substitute to other inputs in its production process. Similarly, under monopolistic bargaining, the downstream firm's output choice no longer directly dictates the upstream quantity; rather, it influences it via a monotone input demand function.

The multi-input production introduces a key parameter affecting the model's comparative statics: the elasticity of substitution between inputs. When this elasticity approaches zero, the production function converges to a Leontief form, creating a one-to-one mapping between upstream and downstream output, mirroring our baseline model. Conversely, as the elasticity of substitution grows, the relationship between upstream and downstream outputs weakens, reducing the scope of buyer and seller power in the vertical chain.

E.1.1 Monopolistic Bargaining

Assume that the downstream firm produces according to the following CES production function with two inputs :

$$q = (\alpha_1 x_1^{\rho} + \alpha_2 x_2^{\rho})^{1/\rho}.$$

For simplicity, we assume that there is no productivity term. For input x_2, the downstream firm negotiates through monopolistic bargaining while taking the price of input x_1 as given in the market. Under monopolistic bargaining, the downstream firm takes the input price w_1 as given and negotiates over w_2. Given the negotiation outcome of w_2 and market price w_1, it then determines its profit-maximizing quantity. From this, we can express the firm's demand for x_2 as a function of its target output quantity q:

$$x_2(q) = \left(\frac{\alpha_2}{w_2}\right)^{\frac{1}{1-\rho}} q \left((\alpha_1/w_1)^{1/(1-\rho)} + (\alpha_2/w_2)^{1/(1-\rho)}\right)^{-1/(1-\rho)}.$$

The CES function leads to the following cost function:

$$C_d(q) = q \left((\alpha_1/w_1)^{1/(1-\rho)} + (\alpha_2/w_2)^{1/(1-\rho)}\right)^{1-1/\rho}.$$

Taking the derivative to find the marginal cost,

$$mc_d(w_2) = \left((\alpha_1/w_1)^{1/(1-\rho)} + (\alpha_2/w_2)^{1/(1-\rho)}\right)^{1-1/\rho},$$

which is the same as the average cost $c_d(w_2)$ due to constant returns to scale. With these objects, we can write the firm's maximization problems as

$$\pi^d(w_2, q) = \big(p(q) - mc_d(w_2)\big)q$$
$$\pi^u(w_2, q) = \big(w_2 - c_u(x_2(q))\big)x_2(q).$$

These problems resemble the problem in the paper with the following exceptions: w_2 in the upstream firm's maximization problem shows up in $c_d(w_1, w_2)$ instead of as a simple linear function in w_2. Similarly, q in the downstream firm's problem shows up as $x_2(q)$ instead of as a simple linear function. Since $c_d(w_1, w_2)$ is increasing in w_2 and $x_2(q)$ is increasing in q, having a multi-input downstream firm does not change the main economics of the problem.

We evaluate this model for two extreme cases for elasticity of substitution ρ. First, consider $\rho \to -\infty$ where the production function takes the Leontief form:

$$q = \min\{\alpha_1 x_1, \alpha_1 x_2\}$$

In this case, we are back to the model in the main text, but with an additional marginal cost term w_1, which is non-negotiated.

Second, consider the case $\rho = 1$, which corresponds to perfect substitutes:

$$q = \alpha_1 x_1 + \alpha_2 x_2$$

Under this production function, the firm will only use x_2 if $\frac{a_2}{w_2} \geq \frac{a_1}{w_1}$. Hence, the bargaining problem over w_2 is only relevant to the extent that this condition holds.

Finally, for any $-\infty < \rho < 1$, firms bargain over w_2 while internalizing that x_1 and x_2 are either gross complements (if $\rho < 0$) or gross substitutes (if $\rho > 0$). We refer to Rubens

(2025) for an empirical implementation of the monopsonistic bargaining model with a CES production function under gross complements.

E.1.2 Monopsonistic Bargaining

Now we will consider monopsonistic bargaining. In monopsonistic bargaining, the production function remains the same, but since the upstream firm determines the input x_2, the downstream firm will take x_2 as given. Therefore, the firm will solve the constrained cost minimization problem conditional on x_2

$$\min_{x_1} \; w_1 x_1 \quad \text{s.t} \quad q \leq (\alpha_1 x_1^\rho + \alpha_2 x_2^\rho)^{1/\rho}.$$

This leads to a conditional factor demand conditional on x_2:

$$x_1(q, x_2) = \left(\frac{\alpha_1}{w_1}\right)^{\frac{1}{1-\rho}} \left(\frac{q^\rho - \alpha_2 x_2^\rho}{(\alpha_1/w_1^\rho)^{1/(1-\rho)}}\right)^{1/\rho}.$$

Similarly, we obtain a conditional cost function:

$$C_d(q, x_2) = (q - \alpha_2 x_2^\rho)^{1/\rho} \left((\alpha_1/w_1^\rho)^{1/(1-\rho)}\right)^{(\rho-1)/\rho} + w_2 x_2.$$

Denote the average cost as $c_d(q, w_2) = C_d(q, w_2)/q$. Taking the derivative to find the marginal cost,

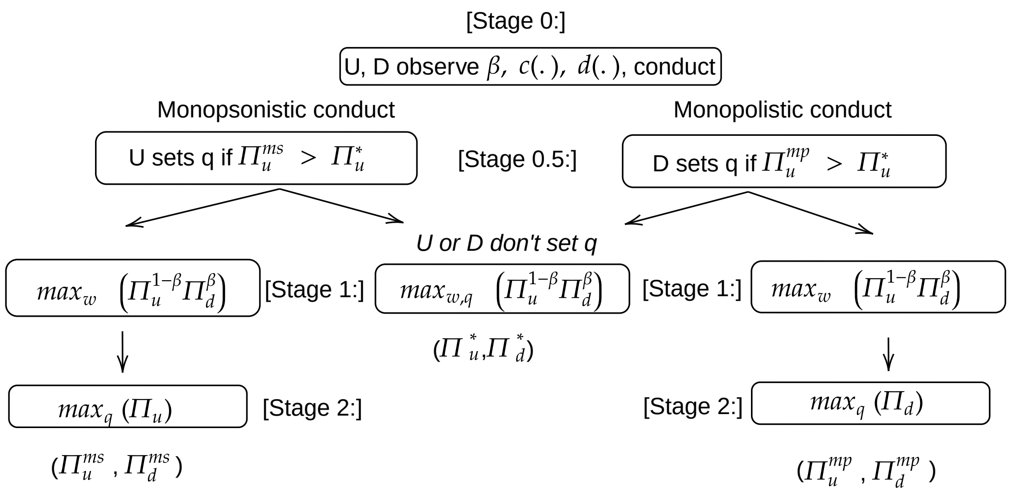
$$mc_d(q, x_2) = \frac{1}{\rho} \left(\frac{w_1^\rho}{\alpha_1}\right)^{1/\rho} (q - \alpha_2 x_2^\rho)^{\frac{1}{\rho}-1}$$

Given x_2, the firm will set marginal cost to marginal revenue: $mc_d(q, x_2) = p'(q)q + p(q)$. Let the solution to this problem be $q_d(x_2)$. Now, we can write the firms' maximization problems as

$$\begin{aligned}
\pi^d(w_2, x_2) &= \big(p(q) - c_d(q_d(x_2), w_2)\big) q_d(x_2) \\
\pi^u(w_2, x_2) &= \big(w_2 - c_u(x_2)\big) x_2.
\end{aligned}$$

where $c_u(x_2)$ is the average cost function of the upstream firm. These problems resemble the problem in the paper with the following exceptions: w_2 in the downstream firm's maximization problem shows up as $c_d(q, w_2)$ instead of as a simple linear function w_2. Similarly, q in the downstream problem appears as $q_d(x_2)$ instead of as itself. In this case, we do not see any change in the upstream firm's cost function. Since both $c_d(w_2)$ and $q_d(x_2)$ are monotone functions, they do not change the basic mechanisms of the model.

Figure OA-2: Decision Tree: Incentive Compatibility of Linear Pricing



Notes: This decision tree illustrates the augmented bargaining game under the distortion selection rule defined in Appendix E.2. The addition is Stage 0.5, where the upstream or downstream firm decides to set q based on Constraint E.2. π^{ms}, π^{mp}, and π^* correspond to profit under monopsonistic bargaining, monopolistic bargaining, and joint-profit maximization, respectively. The game is formally defined in Online Appendix E.2.1.

E.2 Selecting Conduct: Incentive Compatibility of Linear Pricing

The sequential bargaining model provides another criterion to distinguish the source of the vertical distortions without directly imposing nonnegative markups and markdowns. We augment our bargaining model by introducing the possibility that firms bargain over a two-part tariff instead of over a linear contract if it is incentive-compatible.

Nonnegative Markup/Markdown Constraint. *Under monopsonistic conduct, the upstream firm has the right to choose q, and it does so if it cannot earn higher profits by bargaining over (q, w). Under monopolistic conduct, the downstream firm has the right to choose q, and it does so if it cannot earn higher profits by bargaining over (q, w). If neither party is willing to choose q, the parties bargain over a two-part tariff (q, w) instead of over a linear contract.*

Constraint E.2 states that U or D is willing to make an output choice and bargain over a linear price only if the resulting profit surpasses the profit it would earn under joint-profit maximization (i.e., the profit it would earn when *not* setting output unilaterally, but by jointly bargaining over input prices and quantities). Figure OA-2 illustrates the resulting decision tree by adding a stage before bargaining where either the upstream or the downstream firm chooses whether to set q unilaterally.

Theorem 3. *Under Constraint E.2, for any $\beta \neq \beta^*$, either (q^{ms}, w^{ms}) or (q^{mp}, w^{mp}) is an equilibrium with a linear price contract, but not both. Specifically, this equilibrium occurs under monopsonistic conduct if $\beta \geq \beta^*$, and under monopolistic conduct if $\beta \leq \beta^*$.*

Theorem 3 implies, by revealed preference, that if we observe a linear price contract, the equilibrium is monopsonistic when $\beta > \beta^*$ or monopolistic when $\beta < \beta^*$. If $\beta = \beta^*$, neither party has the incentive to determine output, and firms simply maximize joint profits.

This theorem rests on the insight that parties may earn greater profits under monopsonistic or monopolistic vertical conduct than under joint-profit maximization.[41] Under monopsony, the upstream firm's profit from linear pricing exceeds the two-part tariff profit if $\beta > \beta^*$, whereas under monopolistic bargaining, the downstream firm's profit from linear pricing exceeds the two-part tariff profit if $\beta < \beta^*$. Thus, this distortion selection rule ensures that the party choosing the output is never worse off than it would be under joint-profit maximization.

The finding that firms do not always choose two-part tariffs has broader implications for full-information vertical bargaining models with linear contracts. A potential criticism of this class of models is that firms could achieve Pareto improvements by switching to nonlinear pricing (Lee, Whinston, and Yurukoglu, 2021). In our model, the possibility of linear contracting arises from a holdup problem due to a lack of commitment. Under commitment, a two-part tariff would always be reached because either party can convince the other not to set quantities in Stage 0.5 by committing to bargaining less aggressively in Stage 1. However, since the bargaining weights are predetermined and fixed, such a commitment would not be credible. This holdup rationalization of linear price contracts has similarities to prior work on vertical contracts, such as Iyer and Villas-Boas (2003), where shocks between contract signing and delivery induce firms not to negotiate over two-part tariffs.

E.2.1 Proof of Theorem 3

Proof. We formally define the bargaining game as follows:

- **Players:** i = {U, D}
- **Actions:** $a_i = \{L, NL\}$ (Linear pricing (set q), Nonlinear pricing (bargain over q))
- **States:** $s = \{MS, MP\}$ (Monopsonistic Conduct, Monopolistic Conduct)
- **Payoffs:**

[41] To see this, assume $\beta = 1$. Under monopsonistic bargaining, which corresponds to classical monopsony, the upstream profit is $c'(q)q^2 > \pi_0^u$, whereas under a two-part tariff, the upstream profit equals its disagreement payoff π_0^u. The possibility of higher profit from linear prices than from a two-part tariff holds only under sequential timing, because the two-part tariff profit weakly dominates the linear price profit for any β under simultaneous timing.

$$\begin{aligned}
\pi_i(a_i = NL, s) &= \Pi_i^* && \forall i \\
\pi_u(a_u = L, s = MS) &= \Pi_u^{ms} \\
\pi_d(a_u = L, s = MS) &= \Pi_d^{ms} \\
\pi_u(a_d = L, s = MP) &= \Pi_u^{mp} \\
\pi_d(a_d = L, s = MP) &= \Pi_d^{mp}
\end{aligned}$$

We assume that the side that can pick q (upstream under monopsony, downstream under monopoly) sets the action a_i: it decides whether to set q_i unilaterally (in which case $a_i = L$) or to bargain over q_i (in which case $a_i = NL$). Hence, there are four possible subgame equilibria that we need to examine, two for each conduct state: (i) $s = MS, a_u = L$, (ii) $s = MS, a_u = NL$, (iii) $s = MP, a_d = L$, (iv) $s = MP, a_d = NL$.

First, consider monopsonistic conduct. Both players decide on whether to set quantities or bargain over quantities in stage 0.5 of the game by comparing their respective expected profits. Suppose $\beta < \beta^*$. From Lemma OA-14, it follows that $\pi_u^{ms} < \pi_u^*$. Hence, $a_u = L$ is not a subgame perfect equilibrium in this case, whereas $a_u = NL$ is: the game ends at stage 1 when upstream chooses nonlinear pricing, by bargaining over both w and q.

In contrast, if $\beta > \beta^*$, Lemma OA-14 implies that $\pi_u^{ms} > \pi_u^*$. Hence, $a_u = NL$ is not a subgame perfect equilibrium in this case, whereas $a_u = L$ is: the game proceeds to stage 1 in which U and D bargain over input prices, which is followed by stage 2, in which U sets a quantity.

Second, consider monopolistic conduct. Suppose $\beta < \beta^*$. From Lemma OA-14, it follows that $\pi_d^{mp} > \pi_d^*$: downstream expects that if it sets quantities in stage 1, this will result in higher profits than under nonlinear pricing. Hence, $a_d = NL$ is not a subgame perfect equilibrium in this case, whereas $a_d = L$ is: firms bargain over input prices in stage 1, which is followed by downstream setting quantities in stage 2.

In contrast, if $\beta > \beta^*$, Lemma OA-14 implies that $\pi_d^{mp} < \pi_d^*$. Hence, $a_d = L$ is not a subgame perfect equilibrium in this case, whereas $a_d = NL$ is: the game ends at stage 1 when downstream chooses to bargain over both input prices and quantities.

In summary, if a linear price contract is observed (either $a_d = L$ or $a_u = L$), this only happens under monopsonistic bargaining if $\beta > \beta^*$ and under monopolistic bargaining if $\beta < \beta^*$. □

F Empirical Application Appendix: Unions and Cooperatives

F.1 Labor Unions Application

F.1.1 U.S. Construction Workers

In our labor unions application, we rely on the estimates for labor supply and demand for U.S. construction workers from Kroft et al. (2023). Given that their labor supply model is log-linear, it has the same functional form as the numerical example used in the calibrations in the main text. Using their notation of firms being indexed as j, wages W_j and number of workers L_j, their inverse labor supply curve at the firm level is

$$W_{jt} = L_{jt}^{\theta} U_{jt}.$$

Denoting output as Q_{jt}, the goods price as P_{jt}, and an aggregate price index as $\bar{P}_t$, their downstream residual demand curve is

$$Q_{jt} = \left(\frac{P_{jt}}{\bar{P}_t}\right)^{\frac{-1}{\epsilon}}.$$

Hence, their inverse elasticity of labor supply is θ and their inverse elasticity of goods demand is ϵ.

The production function is Leontief in materials and a composite term of labor and capital. Given that we study wage bargaining on the short term, we treat capital as fixed, which results in output being proportional to the labor input. Translating their notation into ours, we have that $\eta = -\frac{1}{\epsilon}$ and $\psi = \theta$.

We use the β^* formula applied to the loglinear example, as worked out in Supplemental Material D.4. Using the notation from Kroft et al. (2023), this gives

$$\beta^* = \left(\frac{1 - \frac{1}{\epsilon}}{1 + \theta} + \frac{1}{\epsilon}\right)^{-1}.$$

We use the estimated inverse demand elasticity $\epsilon = 0.137$ from Table 2, Panel B, and the RDD estimate for the inverse labor supply elasticity $\theta = 0.286$ from Table 2, Panel A. Plugging these into the β^* formula above results in $\beta^* = 0.417$.

F.1.2 Brazilian Labor Markets

Similarly to above, we use the estimated labor supply and product demand elasticities from Lobel (2024). We transform the reported residual labor supply elasticities for small and large firms of to the inverse labor supply elasticity as

$$\psi^{small} = \frac{1}{4.15} = 0.24$$

Second, we use their estimated demand elasticities of $\eta = -1.43$. We treat capital as a fixed input, meaning that the residual demand elasticity is equal to the labor demand elasticity. Also, we set disagreement payoffs to zero. Using the elasticity-based formula for β^* above results in the estimates reported in Table 1.

F.1.3 Chinese Tobacco Farmers

In our farmer cooperatives application, we focus on the setting of tobacco farmers in China, as analyzed in Rubens (2023). Although Rubens (2023) presents a discrete-choice oligopsony model in Appendix A1, we take a first-order approximation of this model by modeling loglinear leaf supply and assuming monopsonistic competition instead. Denoting total leaf production at manufacturer f as M_f, the leaf price at firm f as P_f^m, an aggregate leaf price as $\bar{P}^m$, and a demand residual as A_f, leaf supply is given by

$$M_f = \left(\frac{P_f^m}{\bar{P}^m}\right)^{\frac{1}{\psi}} A_f.$$

In equilibrium, the ratio of the marginal revenue product of tobacco leaf $MRPM$ over the leaf price is equal to one plus the inverse leaf-supply elasticity:

$$\frac{MRPM}{P^M} = 1 + \psi.$$

Using the preferred GMM specification in the third column of Table 1, Panel B, the MRPM/Leaf price ratio is estimated at 2.904, which implies an inverse leaf-supply elasticity of $\psi = 1.904$.

Given that the production function is Leontief in tobacco leaf, cigarette production is proportional to tobacco-leaf usage. We approximate cigarette demand by the same loglinear demand function used above, denoting cigarette production at firm f as Q_f, the cigarette price as P_f, a price aggregator as $\bar{P}_t$, and the inverse demand elasticity as ϵ:

$$Q_f = \left(\frac{P_f}{\bar{P}_t}\right)^{\frac{-1}{\epsilon}}.$$

As an estimate of the cigarette demand elasticity $-1/\epsilon$, we rely on the estimates of Ciliberto and Kuminoff (2010). Given that only median own-price elasticities (rather than average elasticities) are reported, we rely on these median elasticities. We use the estimate of $\frac{1}{\epsilon} = 1.14$ from Table 4, column 6, since this is one of the two preferred specifications that relies on GMM. Using the formula above results in $\beta^* = 0.916$. The key driver of the large bilaterally-efficient-level of buyer power is the inelastic demand for cigarettes due to its addictive nature. Alternatively, using the other GMM specification (in column 7) of $\frac{1}{\epsilon} = 1.11$ results in a similar $\beta^* = 0.93$.

G Empirical Application Appendix: Coal Procurement

G.1 Data Sources

Coal-Mine Characteristics and Production Data. For coal mines, we use two datasets: one from the Mine Safety and Health Administration (MSHA) (Mine Safety and Health Administration, 2024) and one from Velocity Suite. The MSHA data provides information on production, employment, and technical mine characteristics. Quarterly production and employment (in hours worked and total employment) come from MSHA Form 8. Mine characteristics, such as the number and type of openings and the vein thickness, are obtained from MSHA Form 10. We merge these MSHA datasets with each other by matching on the MSHA mine identifier. We use the Velocity data to obtain ownership information. While ownership details are also available in the MSHA data, we found it unreliable due to the lack of unique owner IDs, inconsistent spellings, and unaccounted ownership changes.

Coal-Mine Cost Data. We purchased cost information for coal mines from the 2019 Coal Cost Guide published by Costmine Intelligence. This data source provides detailed data on operating costs, capital costs, labor requirements, and equipment expenses for different mining technologies used in the United States and Canada. The Coal Cost Guide provides data on five types of operating expenses (in USD per short tons) and nine types of capital expenses (in USD). The data is provided at the level of mine characteristics, which consist of a combination of: (i) the mine type (Surface, Continuous Underground with Ramp Access, Continuous Underground with Shaft Access, Longwall Underground with Ramp Access, Longwall Underground with Shaft Access, Room & Pillar Underground with Ramp Access, Room & Pillar Underground with Shaft Access), (ii) the mine's daily capacity (in short kilotons: 1, 2, 4, 8, 24, 72, 216 short kilotons), (iii) the average vein thickness (in meters: 1.5, 2.5, 3.5 meters for underground mines, 1, 3, 12, 18, 24, and 27 meters for surface mines). We obtain this data from pages 5-74 in Chapter 3 of the 2019 Coal Cost Guide (InfoMine USA, 2019).

Power-Plant Characteristics, Cost and Generation Data. For data on power plants, we rely on data from Velocity Suite, which compiled data from EIA 860, EIA 906, EIA 923, NERC 411 forms, EPA, as well as from their own proprietary research. We use five different datasets from Velocity. The first dataset is at the month-generator level and includes the universe of all generators in the U.S., capturing characteristics such as age, fuel type, boiler type, capacity, location, ISO region, installation date, operating status, ownership and regulation status of the owner. Velocity collects this data from various public sources and their proprietary research. The second dataset provides hourly generation data for fossil fuel generation units, sourced from the EPA's CEMS database, which includes details on generation, fuel usage, heat rate, and emissions. The third dataset contains monthly plant-level data, offering information on plant characteristics and monthly generation by fuel type, compiled from the EIA-923 form. The fourth dataset is hourly load data for ERCOT, sourced directly from ERCOT's website. Finally, the fifth dataset consists of hourly generation data for generation units in ERCOT, obtained from ERCOT's 60-Day SCED Disclosure Report.

BLS Wage Data. We obtain weekly earnings of coal miners from the Quarterly Census of Employment of Wages of the U.S. Bureau of Labor Statistics (U.S. Bureau of Labor Statistics, 2024). We download the data at the 4-digit industry level for industry '2121 Coal Mining'.[42] We keep weekly earnings at the state-quarter level and average across quarters to obtain annual averages of weekly earnings per state. For some states in some years, earnings are not reported. In these cases, we impute wages by averaging over broader regions that correspond to the coal basins: Northwest, Southwest, Midwest, Appalachia, and Southeast. We recompute weekly earnings into hourly wages by assuming a 45-hour work week, following average data reported by the CDC.[43]

Coal Transaction Data [2005-2014]. Velocity Suite provides two datasets related to coal transactions between power plants and coal mines. The first dataset is transaction-level, where each record includes coal mine and plant IDs, quantity shipped, FOB price, transportation price, contract information (ID and duration), and coal characteristics (ash, sulfur, and type). Most of the information in this dataset comes from the EIA-923 form, and Velocity augments this data with FOB prices obtained from railroad waybills and their own internal model. The second dataset focuses on coal routes and includes leg-level transportation information, such as the mode of transport (railroad, truck, vessel), carrier details for railroads, costs, and routing points. This data is sourced from waybills and Velocity's proprietary research.

Coal Transaction Data [1979-2000]. This dataset provides historical information on coal transactions and contracts from 1979 to 2000, sourced from the EIA's Coal Transportation Rate Database. It includes details on transportation rates, contract terms, and other relevant information about coal shipments during this period. We use this dataset to obtain historical information on contract types and duration.

[42] The data is available at more disaggregated industry levels (e.g., surface vs. underground mining) and geographical levels (e.g., county), but both of these more detailed data sources have the disadvantage of being much sparser, both along the time and geographical dimension.

[43] https://www.cdc.gov/niosh/mining/content/economics/safetypayscostesttechguide.html

G.2 Hourly Electricity Generation Construction for Power Plants

Since we estimate Cournot competition for every hour, we must observe hourly generation data of all generators operating in ERCOT. This data is sourced from three main datasets: (i) the CEMS database of hourly generation from the EPA, (ii) the ERCOT 60-day hourly generation report, and (iii) EIA monthly generation data at the plant-fuel level. The CEMS data cover all fossil-fuel generation units subject to environmental regulations but exclude renewables and other plants not regulated under these standards. For renewables, we rely on unit-level data from the ERCOT 60-day generation report. For a small subset of units without hourly generation data from either source, we use EIA Form 923 to obtain monthly generation information and assume that monthly generation is uniformly distributed across hours within the given month.

G.3 Capacity Estimation of Coal Mines and Power Plants

Power Plants

We calculate capacities separately for fossil-fuel power plants and other generation sources. For fossil fuel power plants, we obtain capacity factor information by fuel type from the GADS database, calculated based on the maintenance frequency of power plants using different fuel types. In our analysis, these capacity factors are applied uniformly across all hours; we do not account for strategic maintenance timing, as this involves a complex, dynamic problem that is beyond the scope of this study. The effective capacity of each unit is thus determined by multiplying its capacity factor by its nameplate capacity.

For solar, wind, hydroelectric, geothermal, other renewables, and nuclear power plants, we calculate capacity factors based on their generation, as these are zero-marginal-cost generators, and their actual generation should reflect their availability to produce electricity. For these generators, we compute a unit-level capacity factor by averaging their generation within a given month-hour-weekend/weekday bin and dividing it by their nameplate capacity. Multiplying this capacity factor with the nameplate capacity provides the effective capacity of the generator by hour type.

Coal Mines

The EIA collects data on coal mine capacity, which have been used in prior research (Johnsen, LaRiviere, and Wolff, 2019). However, the EIA no longer makes this data available to researchers. Consequently, we infer mine capacity from production data. For each year, we define a mine's capacity as the maximum historical production observed at that mine up to that year. This approach makes mine capacity time-varying, as it reflects changes in production over time.

G.4 Heat Rate Calculations and Coal Weight to Heat Content Conversion

To determine the cost of fossil-fuel generators, we calculate their heat rate annually by dividing their total MMBtu fuel consumption by their total electricity generation. This heat rate is assumed to remain constant throughout the year. To estimate the cost per MMBtu, we multiply the inverse of the heat rate by the per-MMBtu cost of coal.

To convert coal quantities from short tons to MMBtu, we calculate an annual conversion factor by dividing the total coal production (in short tons) by the total heat content of coal produced during the same year. This conversion factor is then assumed to remain constant throughout the year.

G.5 Marginal Cost Estimation Details for Coal Mines

Matching the Coal Cost Guide with the MSHA data

We match the Coal Cost Guide dataset to the MSHA data as follows. First, we rely on the 'technology' variable in the MSHA data to match the mine types. We group 'surface', 'strip/open pit', and 'mountain top' mines in the MSHA data into the 'surface mine' category of the Coal Cost Guide, and the 'continuous', 'longwall', and 'room-pillar' technologies from the MSHA into the corresponding technologies in the Coal Cost Guide. If the technology variable is unobserved in the MSHA dataset, we categorize the mine type as 'other'. We keep only mines of the types 'Auger', 'Bank', 'Surface', 'Underground', and 'Surface at Underground' from the MSHA data, in order to exclude non-production units such as administrative offices.

We categorize mines for which there are one or more shafts reported in the MSHA dataset as 'shaft access' in the Coal Cost Guide, and the remaining mines as 'ramp access'. We use the 'maximum vein thickness' variable in the MSHA data to classify the mines into the corresponding vein thickness category in the Coal Cost Guide. If the vein thickness is unobserved in the MSHA data, we assign the smallest vein thickness type (which corresponds to the highest marginal cost). We assign each mine to the Coal Cost Guide capacity categories based on its capacity as calculated in Section G.3.

Computation of Labor-to-Material-Cost Ratios

We use the 'hourly labor cost' subdivision of the operating costs variable in the Coal Cost Guide as our definition of variable labor costs, and the remaining operating costs reported as intermediate input costs. We also add the 'equipment costs' reported under capital expenditure into intermediate input costs because extracting more coal requires more equipment. Given that equipment costs are unlikely to fully depreciate within a year, unlike the other operating costs listed, we let it depreciate linearly over a period of 5 years. We consider the remaining capital expenses that are listed in the Coal Cost Guide, 'Preproduction Development', 'Surface Facilities', 'Working Capital', 'Engineering & Construction Management', and 'Contingency' to be fixed costs, so we do not include these in our marginal costs measures.

Taking the ratio of these two operating costs, variable labor and intermediate input expenditure, results in the variable $\bar{\gamma}_{\theta(iu)} \frac{p^m}{w^l}$. We assume that the ratio of intermediate input prices over wages is the same across mines in a given year, which allows us to recover $\bar{\gamma}_{\theta(iu)}$ up to a constant. Combining this information with the mine-specific hourly wage that we described below allows us to recover $\gamma_{\theta(iu)}$.

Computation of Mining Marginal Costs

The unit labor and intermediate input costs in the Coal Cost Guide are based on the average input requirements within a mine type. However, the mines in our dataset can diverge from the average daily labor requirements in the Coal Cost Guide because mines are heterogeneous in terms of their productivity. This prevents us from directly using the cost information in the Coal Cost Guide. To address this, we compute observed labor productivity for mine i as hours worked per ton of coal extracted l_{iu}/q^c_{iu} from the MSHA data. This labor productivity corresponds to total factor productivity under the Leontieff production function assumption as shown in Equation (5).

One complication is that the MSHA data reports the total hours worked per year for all labor, including production and non-production workers. Since non-production workers should be considered as fixed costs, we isolate the production worker hours using the 'hourly labor' information in the Coal Cost Guide. In particular, we convert the total hours reported in the MSHA data to the total hours worked by hourly workers by taking the ratio of the hourly worker requirement to the total worker requirement reported in the Coal Cost Guide. We compute this ratio as an average at the level of the mine type θ, as surface and underground mines and mines with different capacities differ in their labor-to-material ratios. We multiply this ratio by the total hours reported in the MSHA data to obtain the total hours worked by hourly workers in mine i in a given year.

We compute hourly wages w^l_{iu} from the BLS data as explained above. We multiply hourly wages by labor productivity to compute the labor cost per ton in each mine, and combine this with the ratio of intermediate to labor costs to compute marginal costs for each mine, as shown in Equation (5).

We then aggregate these marginal costs to firm-level as described in Section 6.3. For firms with a small number of mines, this leads to a discrete cost function, which makes it difficult to obtain the solution of the Nash bargaining. Therefore, we smooth the cost function at the firm level using a polynomial approximation.

G.6 Cournot Demand Estimation Details

As described in the main text, we assume a Cournot competition model with strategic and fringe firms. We estimate a separate and independent Cournot competition model in each hour type, which is a month-hour-weekday/weekend combination. We assume that all regulated firms and firms whose total capacity is below 5% are fringe firms in a given year. With these assumptions, modeling downstream competition requires the consumer demand and supply of fringe firms every hour type.

We assume that total demand is fully inelastic in the short run and calculate the short-run demand by averaging the observed demand for each hour type. We assume that this average is the expected demand during the bargaining between upstream and downstream firms. For fringe supply, we first calculate the cost curve of each fringe firm and aggregate them to the industry level. We assume that fringe firms supply a quantity in a given hour such that the price equals the marginal cost.

Subtracting this fringe supply curve from the inelastic demand yields the industry demand curve faced by strategic firms. The analysis then follows standard Cournot competition modeling, in which each strategic firm faces a residual demand curve given by industry demand minus the observed generation from other strategic firms.

G.7 Disagreement Payoff Estimation

Coal Mines

We assume that the disagreement payoff for mining firms equals the profit from sales to all other firms, implying that if a negotiation fails, the coal mine will not produce the negotiated quantity. We think this assumption is reasonable because for mining firms, each transaction is small relative to total capacity as mining firms transact with many partners.

Power Plants

For power firms, the assumption of no production in the event of a disagreement is unrealistic, as coal power plants contribute significantly to the total capacity of power firms and require substantial upfront capital investments. Moreover, power plants are subject to reliability requirements. Thus, it is more reasonable to assume that coal power generators would continue operating even if negotiations fail. In such cases, we assume that the power firm would procure coal from the spot market. However, the spot market is volatile, and delivery is not guaranteed. If power firms are risk-averse, as supported by Jha (2022) in the context of regulated power plants, it is necessary to account for disutility from uncertainty. To address this, we calculate the yearly mean spot price of coal sold from the same basin and coal type, along with its standard deviation. Using the estimates from Jha (2022), who finds that "plants are willing to trade a $1.62 increase in their expected costs for a $1 decrease in their standard deviation of costs", we assume that the power plants perceive the cost of coal in the spot market as the mean spot price plus 1.62 times its standard deviation.

G.8 Bargaining Model Estimation Algorithm

Consider a grid of potential coal prices, denoted by $[0, \bar{w}]$. The following steps outline the procedure to compare the resulting equilibria across these different prices:

Monopsonistic Bargaining

1. First, calculate how much quantity will be supplied by the upstream firm at any price w. Denote this as $q^{ms}(w)$.
2. Calculate upstream profits as a function of w and $q^{ms}(w)$.
3. Calculate downstream profit as a function of w and $q^{ms}(w)$. To find the downstream profit, we need to construct the cost curve of the downstream firm for a given $q^{ms}(w)$. Since in monopsonistic bargaining, the upstream firm chooses the quantity, we assume that the downstream firm will use that quantity in production. The way we operationalize this is as follows:

    - We construct the supply curve of all other power plants in the power company's portfolio. We take that as given, and it is not affected by the negotiation between coal mines and the power company.
    - We also take other prices as given, such as the prices of natural gas and coal from other mining companies. This, together with the assumption above, constructs the supply curve from all inputs other than the one negotiated with the mining company.
    - We assume that the quantity supplied by U, $q^{ms}(w)$, is allocated to each coal generator in the portfolio of the power company proportionally to their capacity. For example, if the downstream quantity is 100 tons, and we have two coal power plants, A and B, whose capacities are 50 tons and 200 tons, respectively,

we assume that 20 tons (100 tons x (50/250)) will go to plant A and 80 tons will go to plant B. This will matter to the extent that plant A's heat rate differs from plant B's. If their heat rates are the same, this is without loss of generality.

- We further assume that the coal quantity is distributed uniformly throughout each hour of the day. For example, there are 8,760 hours in a given year, so plant A will have 20/8,760 tons of coal to use in a given hour. This assumption ignores the optimal intertemporal allocation of a limited coal quantity over the course of the year. For example, if coal is limited, Plant A might want to use all of it when the price is high. However, we think the potential role of this dynamic channel is limited as coal generators operated the majority of the time during our sample period.
- With these steps, we now have the tons of coal allocated to each plant in a given hour. The downstream firm takes as given that the allocated coal is used for electricity generation under any market conditions. Then, it optimizes the production of the rest of its portfolio following static profit maximization under Cournot competition.

4. Now we have the profits as a function of coal prices for both parties. Construct the Nash product.

5. For each β, find w that maximizes the Nash product. This gives us $q^{ms}(\beta)$ and $w^{ms}(\beta)$.

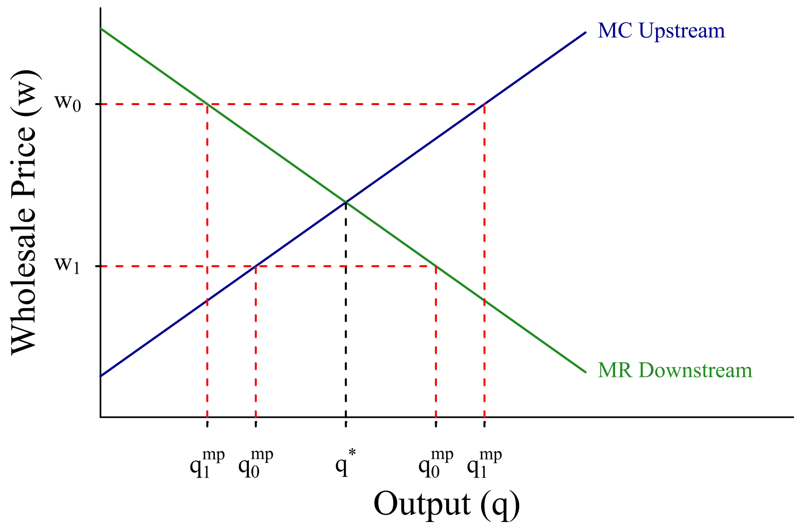
Monopolistic Bargaining

1. In the monopoly setting, we calculate how much quantity will be demanded by the downstream firm. To find this quantity, take w as given and construct the downstream firm's supply curve as a function of w. When doing this, we condition on all prices except w_{ud} so the part of the supply curve that comes from non-coal generators or parts of the coal generators that come from other coal mines $(-u)$ remains fixed. In other words, if d receives coal from multiple suppliers, then w_{ud} affects only the remaining capacity of coal generators in the firm's cost curve. After obtaining the cost curve as a function of w_{ud}, we solve the Cournot model to calculate the quantity produced by firm d. The firm produces this quantity from its lowest-cost generators. Using this, we calculate the corresponding coal input demand of d from u, $q^{mp}(w_{ud})$.

2. Given $q^{mp}(w_{ud})$ and w_{ud}, find the upstream profit for each w_{ud}.

3. We fix a β and construct the Nash product.

Figure OA-3: Conduct Identification: Intuition



Notes: This figure illustrates how we identify vertical conduct for two different observed values of the coal price: w_0 and w_1.

4. For each β, find w that maximizes the Nash product, which gives us $q^{mp}(\beta)$ and $w^{mp}(\beta)$ for $\beta \in (0, 1)$.

Vertical Conduct Identification

To identify the vertical conduct, we use $w^{ms}(\beta)$ and $w^{mp}(\beta)$ calculated in the estimation. Then we find β^{ms} and β^{mp} such that $w^{ms}(\beta^{ms}) = w^{obs}$ and $w^{mp}(\beta^{mp}) = w^{obs}$. With these β values, we apply Theorem 2 to select the conduct, which implies that the monopsonistic equilibrium exists if $\beta \geq \beta^*$, while the monopolistic equilibrium exists if $\beta \leq \beta^*$. Lemmas 1-1 guarantee that either $\beta^{ms} < \beta^*$ and $\beta^{mp} < \beta^*$ or $\beta^{ms} > \beta^*$ and $\beta^{mp} > \beta^*$. To see this, note that $dq^{ms}/d\beta < 0$ by Lemma 1 and $dq^{mp}/d\beta > 0$ by Lemma 1. By the FOCs, we also have that $dq/dw^{ms} > 0$ and $dq/dw^{mp} < 0$. The chain rule implies that $dw^{ms}/d\beta < 0$ and $dw^{mp}/d\beta < 0$. Therefore, w is decreasing with β under both types of conduct. Since $w^{ms}(\beta^*) = w^{mp}(\beta^*)$, it follows that either $\beta^{mp} < \beta^*$ or $\beta^{mp} > \beta^*$, implying a unique conduct.

Figure OA-3 provides a graphical intuition of our procedure for identifying vertical conduct. Assume a certain coal price w_0 is observed. The corresponding monopsonistic and monopolistic quantities q_0^{ms} and q_0^{mp} are computed using the two estimation algorithms that were explained above. We determine vertical conduct by comparing these quantity values to the joint-profit-maximizing output value q^*, following Proposition 4.2. In this particular example, $q_0^{ms} < q^*$ and $q_0^{mp} > q^*$, so vertical conduct is monopsonistic. For a different coal price w_1, we obtain the opposite result: vertical conduct is monopolistic at w_1. The bargaining parameter β is estimated as the value that rationalizes the observed coal price under the vertical conduct model that was found to apply.

Standard Error Calculations

The only source of uncertainty in our model is the inelastic market demand for a given hour type. We calculate the inelastic demand in a given hour type t as the average demand value across a finite sample of hours of that type in our estimation. To account for this uncertainty in the estimates, we implement a bootstrap procedure with 100 iterations, in which we calculate the average demand for hour type t by resampling hours within that hour type with replacement. We then repeat the entire estimation procedure to obtain a bootstrap distribution of our estimates.

H Additional Tables

Table OA-1: Summary of limit cases for β

Case	Equilibrium Condition		Explanation	
	FOC	Cons. Max	FOC	Cons. Max
Sim. MP, $\beta = 1$	$mr(q) = c(q)$	$mr(q) = c(q)$	(D-TIOLI)	(D-TIOLI)
Sim. MP, $\beta = 0$	–	–	–	–
Sim. MS, $\beta = 1$	–	–	–	–
Sim. MS, $\beta = 0$	$mc(q) = p(q)$	$mc(q) = p(q)$	(U-TIOLI)	(U-TIOLI)
Seq. MP, $\beta = 1$	–	$mr(q) = c(q)$	–	(D-TIOLI)
Seq. MP, $\beta = 0$	$mr(q) = w$	$mr(q) = w$	(D.M.)	(D.M.)
Seq. MS, $\beta = 1$	$mc(q) = w$	$mc(q) = w$	(C.M.)	(C.M.)
Seq. MS, $\beta = 0$	–	$mc(q) = p(q)$	–	(U-TIOLI)

Notes: This table summarizes the equilibrium of monopsonistic and monopolistic bargaining in the limit cases separately using FOCs and from the constraint maximization problems. We use the following abbreviations: "D-TIOLI" (downstream take-it-or-leave-it) and "U-TIOLI" (upstream take-it-or-leave-it), "D.M." (double marginalization), "C.M." (classical monopsony). "–" denotes that equilibrium does not exist.

Table OA-2: Key Notation Used in the Model

Mining (Upstream)		Power (Downstream)	
q^c_{iu}	Coal output of mine i (short tons)	Q^D_t	Electricity demand in hour t
l_{iu}	Labor hours used at mine i	Q^{fr}_t	Fringe supply in hour t
m_{iu}	Intermediate inputs at mine i	Q^{st}_t	Strategic supply in hour t
$\theta(iu)$	Mine type for mine i	P_t	Electricity price in hour t
$\gamma_{\theta(iu)}$	Labor–material ratio by mine type	c_{jd}	Marginal cost of generator j in firm d
ω_{iu}	Productivity shifter at mine i	k_{jdt}	Capacity of generator j in hour t
w^l_{iu}	Hourly wage at mine i	$C_{dt}(Q_{dt})$	Downstream cost function in hour t
p^m_{iu}	Material unit cost at mine i	Q_{-dt}	Output of other downstream firms in hour t
λ_{iu}	Conversion factor (short ton to MMBtu)		
c_{iu}	Marginal cost at mine i		**Bargaining**
c_u	Vector of mine marginal costs for firm u	D_u	Set of downstream partners for firm u
$\bar{q}_u$	Vector of mine capacities for firm u	q_{ud}	Quantity traded between u and d
$C_u(Q_u)$	Upstream cost curve for firm u	w_{ud}	Transacted coal price between u and d
Q_u	Total coal output of firm u	π_u	Profit function of firm u
n_u	Number of mines owned by u	π_{dt}	Period-t profit of downstream firm d
$\bar{q}_{iu}$	Capacity of mine i	Q^{ms}_{dt}	Monopsonistic downstream quantity in hour t
		Q^{mp}_{dt}	Monopolistic downstream quantity in hour t
		β_{ud}	Bargaining power of buyer in pair (u, d)
		Q_{-d}	Total coal output supplied to downstream partners other than d
		Q^{-u}_{dt}	Disagreement output (without upstream u) for firm d in hour t

Notes: Subscripts i, j, u, d, and t denote mine index, generator index, upstream firm, downstream firm, and hour type, respectively.

Table OA-3: Buyer Power from the Empirical Bargaining Literature

Sources	Industry	Range	Mean	Median	Std. Dev.
			Panel A. Firm-to-Firm		
Crawford and Yurukoglu (2012)	Television	[0.170, 0.770]	0.559	0.595	0.159
Gowrisankaran et al. (2015)	Healthcare	{0.500}	0.500	0.500	0
Crawford et al. (2018)	Television	[0.280, 0.370]	0.325	0.325	0.064
Ho and Lee (2017, 2019)	Healthcare	[0.310, 0.470]	0.387	0.38	0.080
Hosken et al. (2024)	Healthcare	[0.370, 0.990]	-	0.93	-
Alviarez et al. (2025)	Intl. Trade	-	0.812	-	0.101
Cuesta et al. (2025)	Healthcare	[0.167, 0.680]	0.476	0.518	0.187
			Panel B. Union-to-Firm		
Svejnar (1986)	Multiple	[0.140, 0.890]	0.513	0.555	0.281
Doiron (1992)	Woodworking	[0.499, 0.791]	0.648	0.678	0.116
Abowd and Lemieux (1993)	Multiple	[0.608, 0.850]	0.727	0.738	0.078
Mumford and Dowrick (1994)	Coal	{0.141}	-	-	-
Cahuc, Postel-Vinay, and Robin (2006)	Multiple	[0.020, 1.000]	0.804	0.855	0.247
Coles and Hildreth (2000)	Manufacturing	-	0.144	-	0.088
Breda (2014)	Multiple	[0.636, 0.805]	0.713	0.709	0.061
Fuess (2001)	Multiple	[0.176, 0.592]	0.357	0.250	0.099
			Panel C. Cooperative-to-Firm		
Prasertsri and Kilmer (2008)	Dairy	[0.217, 0.334]	0.267	0.267	0.034
Ahn and Sumner (2012)	Dairy	[0.861, 0.912]	0.889	0.896	0.020
Ge, Flores-Lagunes, and Kilmer (2015)	Dairy	[0.880, 0.881]	0.8804	0.8803	0.000
Hayashida (2018)	Dairy	[0.209, 1.179]	0.8739	0.9765	0.2470
Shokoohi, Chizari, and Asgari (2019)	Dairy	{0.690}	-	-	-
Sano, Sato, Kawasaki, Suzuki, and Kaiser (2022)	Vegetables	[0.481, 0.844]	0.707	0.707	0.089

Notes: This table summarizes bargaining parameters from empirical papers implementing models with a *Nash-in-Nash* bargaining protocol. All weights denote buyer power. We summarize the estimates from the authors' preferred specifications when available. Some papers (e.g., Crawford and Yurukoglu (2012)) provide weights from bargaining pairs (some only report the distribution, e.g., Coles and Hildreth (2000)), and others (e.g., Abowd and Lemieux (1993)) provide weights under different model specifications. Cahuc et al. (2006) do not specifically study union-based wage negotiation, but we include them in this panel for their focus on wage bargaining.

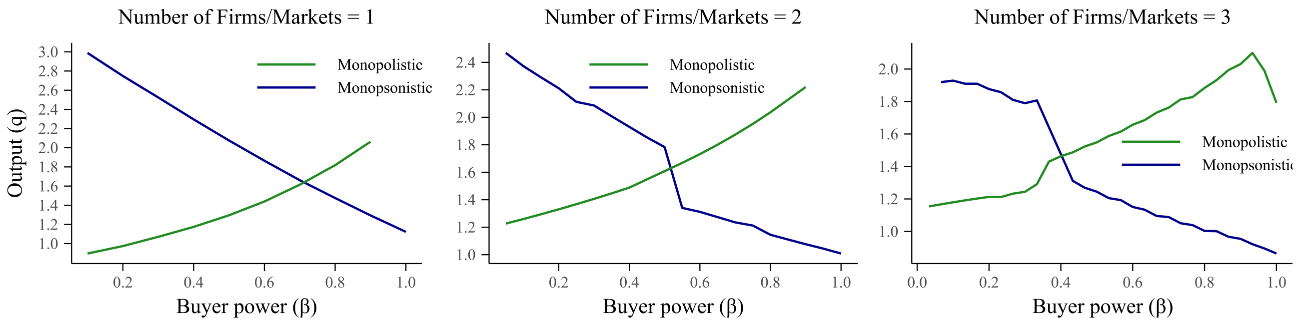
Table OA-4: List of Firms

Upstream Firms	**Downstream Firms**	
	A. With Coal Plants	*B. Without Coal Plants*
Peabody Energy Corp	NRG Energy Inc	Energy Capital Partners
Rio Tinto Energy America	Vistra Energy	Energy Future Holdings Corp
Westmoreland Coal Co		NextEra Energy Inc
Cloud Peak Energy		
Arch Resources Inc		
Foundation Coal Corp		
Peter Kiewit Sons Inc		
Alpha Natural Resources LLC		
Vistra Energy		

Notes: This table lists the upstream and downstream firms in Section 6. The list of upstream firms and downstream firms with coal power plants is included in the bargaining model. Downstream firms without coal power plants are strategic firms in the demand model with more than 5% market share.

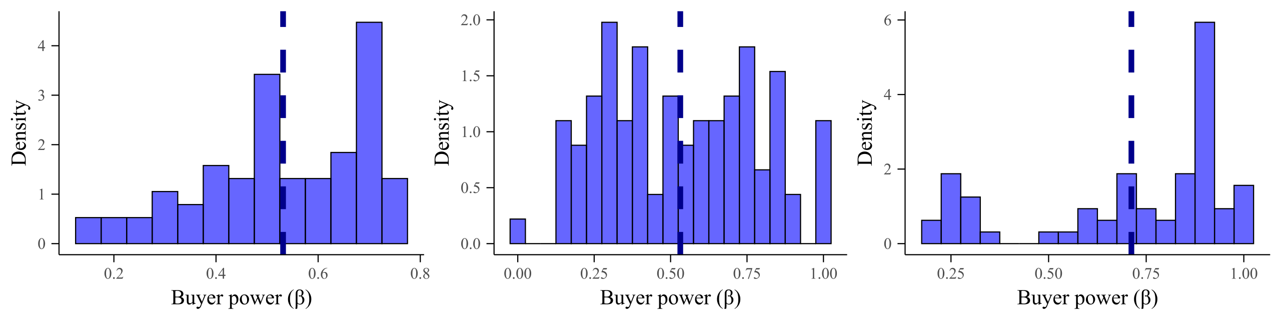
I Additional Figures

Figure OA-4: Change in β^* with the Number of Downstream Firms



Notes: This figure presents numerical simulation results showing the relationship between output (q) and buyer power (β) when there are one to three firms in each downstream market.

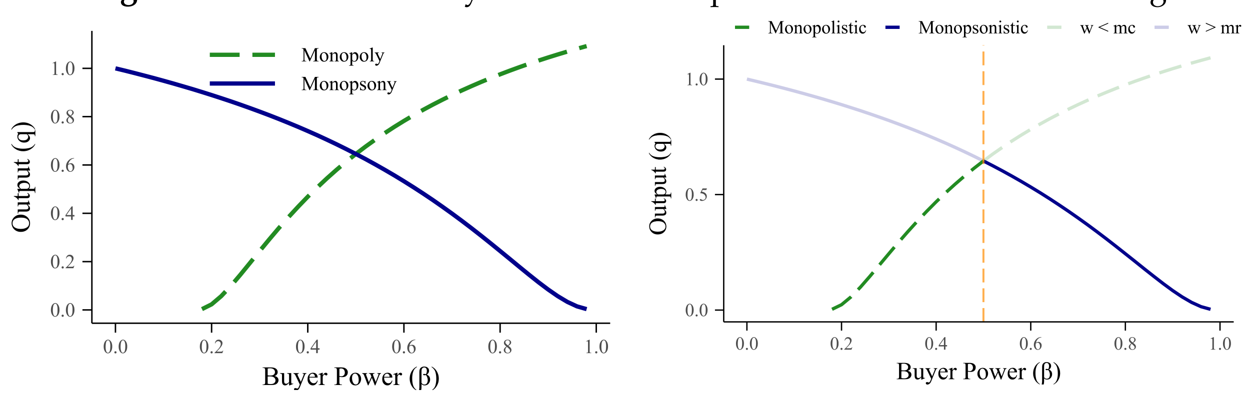
Figure OA-5: Buyer Power from the Empirical Bargaining Literature



(a) Firm-to-Firm **(b)** Union-to-Firm **(c)** Cooperative-to-Firm

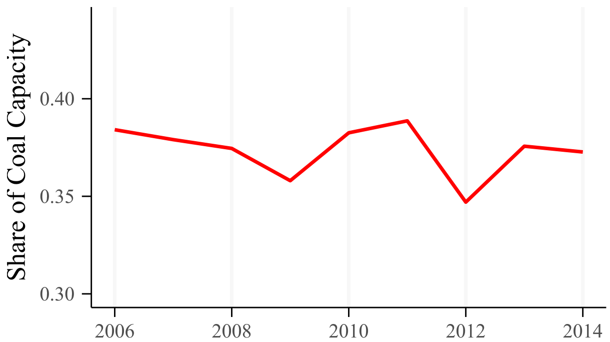
Notes: This figure presents the distribution of bargaining weights from Table OA-3.

Figure OA-6: Effects of Buyer Power on Output Under Simultaneous Timing



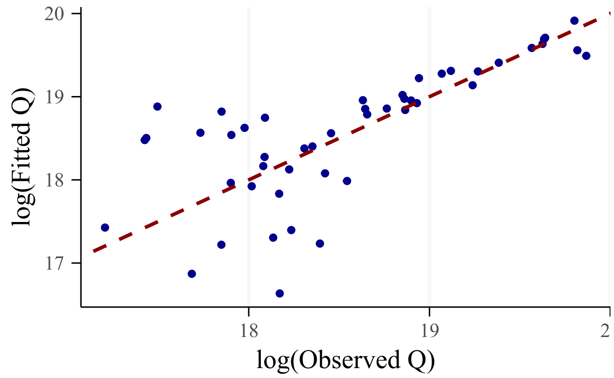
Notes: Panel (a) simultaneous case of Figure 1(c). Panel (b) is the simultaneous case of Figure **??**. The simultaneous monopolistic bargaining model does not have a solution for $\beta < 1/6$ as we show in Appendix J.1.

Figure OA-7: ERCOT Share of Coal Generation



Notes: This figure reports the share of electricity generation by coal-fired generators in the ERCOT market during the sample period between 2006 and 2014.

Figure OA-8: Observed vs Model-predicted Quantities



Notes: This figure shows the scatter plot of observed vs. model-predicted coal transactions in MMBtu. An observation is a combination of firm pair and year.

J Additional Results

This section contains auxiliary results that are not directly used in the proofs above but may be of independent interest.

J.1 Equilibrium Existence Under Simultaneous Bargaining (Loglinear Case)

The following results characterize equilibrium existence under *simultaneous* timing for the loglinear model with $c(q) = q^{\psi}/(1+\psi)$ and $p(q) = q^{1/\eta}$.

Simultaneous monopolistic bargaining

Solving the first-order conditions for the simultaneous monopolistic bargaining problem, $q(\beta)$, is given by

$$q^{mpl} = \left(\frac{\psi+1}{\beta\eta} + 1 + \psi\right)^{\frac{1}{\psi - \frac{1}{\eta}}}.$$

Given that $\psi - \frac{1}{\eta} = \frac{5}{12} < 1$ in our numerical example, equilibrium existence requires

$$\left(\frac{\psi+1}{\beta\eta}\right) + 1 + \psi > 0.$$

Hence, it must hold that $\beta > -1/\eta$. In our numerical example, this condition is satisfied for $\beta > 1/6$, so the simultaneous monopolistic equilibrium is defined only for this range of bargaining parameters.

Simultaneous monopsonistic bargaining

Solving the FOCs of the simultaneous monopsonistic bargaining model delivers the following $\beta(q)$ relationship:

$$\beta = \frac{q^{\psi} - q^{\frac{1}{\eta}}}{q^{\frac{1}{\eta}} + \frac{q^{\psi}}{1+\psi}}.$$

Given that $\psi > 0$ and $\eta < 0$, output is well-defined for any $\beta > 0$. Hence, the simultaneous monopsonistic equilibrium always exists for the range of bargaining parameters we consider.

Limits in the simultaneous numerical example

When we apply the bounds for existence under simultaneous timing from Proposition 1, the limit of monopolistic bargaining corresponds to

$$\lim_{q\to 0^+} -\frac{p'(q)\,q}{p(q)-c(q)} = \lim_{q\to 0} \frac{q^{1/\eta}}{q^{1/\eta}-(1+\psi)^{-1}q^{\psi}}.$$

Using l'Hôpital's rule, this limit can be found as

$$\lim_{q\to 0^+} -\frac{(1/\eta)q^{1/\eta}}{q^{1/\eta}-(1+\psi)^{-1}q^{\psi}} = -\frac{1}{\eta}$$

Since we set $\eta = -6$, the limit is $1/6$.

In monopsonistic bargaining, the upper bound is given by the limit:

$$\lim_{q\to 0^+} \frac{c'(q)\,q}{p(q)-c(q)} = \lim_{q\to 0^+} \frac{(\psi/(1+\psi))q^{\psi}}{q^{1/\eta}-(1+\psi)^{-1}q^{\psi}}$$

Using l'Hôpital's rule, this limit can be found as

$$\lim_{q\to 0^+} \frac{(\psi/(1+\psi))q^{\psi}}{q^{1/\eta}-(1+\psi)^{-1}q^{\psi}} = 0.$$

J.2 Lemmas on Markups and Markdowns

Lemma OA-15. *The results on equilibrium conditions, quantity, markdown, markup, upstream profit, and downstream profits in Table OA-5 hold.*

Proof. We will prove the results column by column.

(i) Equilibrium conditions: These are derived in Section D.1.

(ii) Quantities: We will only show this result for sequential monopsonistic bargaining because the results for other cases are identical and similar to derive. The equilibrium quantities are characterized by

$$\begin{aligned} \text{For } \beta = 1: &\quad mr(q_1) = mc'(q_1)q_1 + mc(q_1) \\ \text{For } \beta = \beta^*: &\quad mr(q^*) = mc(q^*) \\ \text{For } \beta = 0: &\quad mc(q_3) = p(q_3) \end{aligned}$$

where q_1 and q_3 are equilibrium quantities when $\beta = 1$ and $\beta = 0$. Note that since $mc'(q) > 0$, we have that $mc'(q)q + mc(q) > mc(q)$. Since $mr(q)$ is a decreasing function, it follows that $q^* > q_1$. For the comparison of q_3 and q^* note that $p(q) > mr(q)$ because

$p'(q) < 0$, Since $mc(q)$ is an increasing function, it follows that $q_3 > q^*$.

(iii) Markups: Observe that in monopsonistic bargaining $w = mr(q)$, so the markup is given by $(mc(q) - mr(q))/mc(q)$ as defined in Section 2.5. It immediately follows from the relationship between $mc(q)$ and $mr(q)$ for different β values that $mc(q_1) - mr(q_1) < 0$, $mc(q) = mr(q)$ and $mc(q_3) - mr(q_3) > 0$.

(iv) Markdowns: Markdown results can be developed analogously to markup results and therefore are omitted.

(v) Upstream Profit: The equivalence of profit at β^* to joint-profit maximization profit follows from Proposition 2. The cases where $\pi_u = 0$ are also trivial because either the quantity approaches zero or the downstream firm makes a take-it-or-leave-it offer. Therefore, the only nontrivial cases are (i) Seq.MS with $\beta = 0$, (ii) Seq.MS with $\beta = 1$, and (iii) Seq.MP with $\beta = 0$. We will analyze these cases one by one.

Consider first sequential monopsony (Seq.MS) with $\beta = 0$. When $\beta = 0$, under joint profit maximization, the upstream firm captures the entire profit so that $\pi_u(\beta = 0) = \pi_u^*$. Note that in this case, $mc(q) = p(q)$, meaning that the equilibrium quantity is less than the optimal quantity. This implies that the total profit, which is captured entirely by the upstream firm since $\beta = 0$, is below π_u^*.

Now consider the sequential monopsony case (Seq.MS) with $\beta = 1$, which corresponds to the classical monopsony model. Under joint profit maximization, $\pi_u = 0$ because the downstream firm captures the entire profit. However, in the case of sequential monopsony, $w = mc(q) > c(q)$ and the upstream profit is $\pi_u = (w - c(q))q > 0$. This implies that $\pi_u^{ms}(\beta = 1) > \pi_u^*(\beta = 1)$.

Finally, consider the case of the sequential monopoly (Seq.MP) with $\beta = 0$. The joint-profit maximization gives the entire joint profit, which also equals the highest profit the upstream firm can reach. However, when Seq.MP with $\beta = 0$, the downstream profit is positive because $\pi_d = (p(q) - w)q$ and $w = mr(q) < p(q)$. Since the total joint profit cannot be greater than $\pi_u^* + \pi_d^*$, it follows that $\pi_u^{mp}(\beta = 0) < \pi_u^* + \pi_d^*$.

(vi) Downstream Profit: The proof for downstream profit follows very similarly to that of upstream profit and is therefore omitted. □

Lemma OA-16. *In the monopsonistic bargaining model, $\Delta^d = 0$ if and only if q equals q^*.*

Proof. Since, in monopsonistic bargaining (U-FOC) implies that $\mu^u(q) = 0$, $\Delta^d(q) = 0$ if and only if $\mu^u(q) + \Delta^d(q) = 0$. Furthermore $\mu^u(q) + \Delta^d(q) = 0$ implies that $mc(q) = mr(q)$. Note that this equation only holds at q^* by Lemma OA-12, which concludes the proof. □

Lemma OA-17. *In monopolistic bargaining, $\mu^u = 0$ if and only if the equilibrium q equals q^*.*

Table OA-5: Equilibrium Outcomes under Limit Cases for β

Model	Equilibrium Condition	Explanation	q	Δ^d	μ^u	π_u	π_d
Sim. MS, $\beta = 1$	$q \to 0$	–	$q < q^*$	$\Delta^d > 0$	$\mu^u = 0$	$\pi_u = 0$	$\pi_d = 0$
Sim. MS, $\beta = \beta^*$	$mc(q) = mr(q)$	(JPM)	$q = q^*$	$\Delta^d = 0$	$\mu^u = 0$	$\pi_u = \pi_u^*$	$\pi_d = \pi_d^*$
Sim. MS, $\beta = 0$	$mc(q) = p(q)$	(U-TIOLI)	$q > q^*$	$\Delta^d < 0$	$\mu^u = 0$	$\pi_u < \pi_u^*$	$\pi_d = 0$
Sim. MP, $\beta = 1$	$mr(q) = c(q)$	(D-TIOLI)	$q < q^*$	$\Delta^d = 0$	$\mu^u < 0$	$\pi_u = 0$	$\pi_d = 0$
Sim. MP, $\beta = \beta^*$	$mr(q) = mc(q)$	(JPM)	$q = q^*$	$\Delta^d = 0$	$\mu^u = 0$	$\pi_u = \pi_u^*$	$\pi_d = \pi_d^*$
Sim. MP, $\beta = 0$	$q \to 0$	–	$q < q^*$	$\Delta^d = 0$	$\mu^u > 0$	$\pi_u = 0$	$\pi_d > \pi_d^*$
Seq. MS, $\beta = 1$	$mr(q) = mc'(q)q + mc(q)$	(C.M.)	$q < q^*$	$\Delta^d > 0$	$\mu^u = 0$	$\pi_u > \pi_u^*$	$\pi_d = 0$
Seq. MS, $\beta = \beta^*$	$mc(q) = mr(q)$	(JPM)	$q = q^*$	$\Delta^d = 0$	$\mu^u = 0$	$\pi_u = \pi_u^*$	$\pi_d = \pi_d^*$
Seq. MS, $\beta = 0$	$mc(q) = p(q)$	(U-TIOLI)	$q > q^*$	$\Delta^d < 0$	$\mu^u = 0$	$\pi_u < \pi_u^*$	$\pi_d = 0$
Seq. MP, $\beta = 1$	$mr(q) = c(q)$	(D-TIOLI)	$q < q^*$	$\Delta^d = 0$	$\mu^u < 0$	$\pi_u = 0$	$\pi_d < \pi_d^*$
Seq. MP, $\beta = \beta^*$	$mr(q) = mc(q)$	(JPM)	$q = q^*$	$\Delta^d = 0$	$\mu^u = 0$	$\pi_u = \pi_u^*$	$\pi_d = \pi_d^*$
Seq. MP, $\beta = 0$	$mc(q) = mr'(q)q + mr(q)$	(D.M.)	$q < q^*$	$\Delta^d = 0$	$\mu^u > 0$	$\pi_u < \pi_u^*$	$\pi_d > \pi_d^*$

Proof. Since, in monopolistic bargaining (D-FOC) implies that $\Delta^d(q) = 0$, $\mu^u = 0$ if and only if $\mu^u(q) + \Delta^d(q) = 0$. Furthermore $\mu^u(q) + \Delta^d(q) = 0$ implies that $mc(q) = mr(q)$. Note that this equation only holds at q^* by Lemma OA-12, which concludes the proof. □

Lemma OA-18. *In sequential monopsonistic bargaining $\Delta^d < 0$ when $\beta \in (0, \beta^*)$ and $\Delta^d > 0$ when $\beta \in (\beta^*, 1)$.*

Proof. By Lemma OA-15, we have $\Delta^d(\beta = 0) < 0$ and by Lemma OA-16 $\Delta^d(q^*) = 0$. Lemma OA-12 shows that β^* is the unique value of β that gives q^* as the equilibrium quantity. The continuity of Δ^d as a function of β, this proves that $\Delta^d < 0$ when $\beta \in (0, \beta^*)$. Similarly, we showed that at β^*, $\Delta^d = 0$ and at $\beta = 1$, $\Delta^d > 0$. Continuity of Δ^d as a function of β implies that $\Delta^d > 0$ when $\beta \in (\beta^*, 1)$ □

Lemma OA-19. *In sequential monopolistic bargaining $\mu^u > 0$ when $\beta \in (0, \beta^*)$ and $\mu^u < 0$ for $\beta \in (\beta^*, 1)$.*

Proof. This proof is identical to the proof of Lemma OA-18 and therefore omitted. □

J.3 Other Auxiliary Lemmas

Lemma OA-20. *The condition $c''(q)q + c'(q) > 0$ is equivalent to $\frac{\partial(mc(q)-c(q))}{\partial q} > 0$. The condition $p''(q)q + p'(q) < 0$ is equivalent to $\frac{\partial(mr(q)-p(q))}{\partial q} < 0$*

Proof. Since $mc(q) = d(c(q)q)/dq$, the difference between marginal and average cost is given by $mc(q) - c(q) = c'(q)q$ whose derivative is $c''(q)q + c'(q)$. Since $c'(q) > 0$, the condition given in Assumption **??**, $c''(q)q + c'(q) > 0$ implies $mc'(q) > 0$. The proof with respect to marginal revenue is the same after replacing $c(q)$ functions with $p(q)$ functions. □

J.4 Alternative Representation of Profit Functions

Table OA-6 collects the profit functions, stage-2 problems, and Nash bargaining objectives under each conduct model, expressed both in terms of the input price w and in terms of quantity q.

Table OA-6: Summary of Profit Functions, Stage-2 Problems, and Nash Products

	Monopsonistic Conduct	**Monopolistic Conduct**
Stage 2: Quantity Setting		
Quantity setter	Upstream (U)	Downstream (D)
Problem	$\max_{q^u} \, (w - c(q^u))\, q^u$	$\max_{q^d} \, (p(q^d) - w)\, q^d$
FOC	$w = mc(q) = c(q) + c'(q)\, q$	$w = mr(q) = p(q) + p'(q)\, q$
SOC	$mc'(q) > 0$	$mr'(q) < 0$
Profits in Terms of w (both conduct models)		
$\pi^d(w) = \big(p(q(w)) - w\big)\, q(w), \qquad \pi^u(w) = \big(w - c(q(w))\big)\, q(w)$		
Profits in Terms of q (substituting stage-2 FOC)		
$\pi^d(q)$	$\big(p(q) - c(q) - c'(q)\, q\big)\, q$	$-p'(q)\, q^2$
$\pi^u(q)$	$c'(q)\, q^2$	$\big(p(q) + p'(q)\, q - c(q)\big)\, q$
Stage 1: Nash Bargaining over w		
$\max_w \left[\left(\pi^d(w) - \pi_0^d\right)^\beta \left(\pi^u(w) - \pi_0^u\right)^{1-\beta}\right]$		
In terms of q	$\max_w \left(G^d\right)^\beta \left(c'q^2 - \pi_0^u\right)^{1-\beta}$	$\max_w \left(-p'q^2 - \pi_0^d\right)^\beta \left(G^u\right)^{1-\beta}$
	$G^d = (p - c - c'q)\, q - \pi_0^d$	$G^u = (p + p'q - c)\, q - \pi_0^u$
	$q = mc^{-1}(w)$	$q = mr^{-1}(w)$

Log-Concavity of Gains from Trade as a Sufficient Condition

Under each conduct model, define the gains from trade as $G^d(q) \equiv \pi^d(q) - \pi_0^d$ and $G^u(q) \equiv \pi^u(q) - \pi_0^u$, where $\pi^d(q)$ and $\pi^u(q)$ are the profit functions in terms of q given in Table

OA-6. We say that $G^i(q)$ is *log-concave* in q if $\ln G^i(q)$ is concave in q on the domain where $G^i(q) > 0$.

Lemma OA-21. *Under a given conduct model, if both $G^d(q)$ and $G^u(q)$ are log-concave in q, then:*

(i) *The Nash bargaining product is log-concave in q for every $\beta \in [0, 1]$.*

(ii) *The FOC is sufficient for a global maximum: any solution to the FOC is the unique global maximizer.*

(iii) *For $\beta \in (0, 1)$, the equilibrium exists, is interior, and is unique.*

(iv) *For $\beta \in \{0, 1\}$, the constrained optimization problems in Section D.1 are well-posed: if the participation constraint binds, the constrained optimum is at the boundary of the feasible set.*

Proof. We prove each part in turn.

Part (i): Log-concavity of the Nash product. The Nash bargaining objective is $(G^d)^\beta (G^u)^{1-\beta}$. Its logarithm is:

$$f(q) \equiv \ln\left[(G^d)^\beta \, (G^u)^{1-\beta}\right] = \beta \, \ln G^d(q) + (1-\beta) \, \ln G^u(q).$$

By assumption, $\ln G^d(q)$ and $\ln G^u(q)$ are each concave in q. Since $\beta \geq 0$ and $1 - \beta \geq 0$, $f(q)$ is a non-negative weighted sum of concave functions, hence concave in q. This holds for every $\beta \in [0, 1]$.

Part (ii): FOC is sufficient for a global maximum. Since $f(q) = \ln[(G^d)^\beta (G^u)^{1-\beta}]$ is concave in q, any critical point where $f'(q) = 0$ is a global maximizer of f. Because ln is strictly increasing, a maximizer of f is also a maximizer of the Nash product $(G^d)^\beta (G^u)^{1-\beta}$. If f is strictly concave, the critical point is unique.

The bargaining is over w, not q directly. Under each conduct model, q and w are related by a monotone mapping: $w = mc(q)$ (monopsonistic, with $mc'(q) > 0$) or $w = mr(q)$ (monopolistic, with $mr'(q) < 0$). Define $h(w) \equiv f(q(w))$. At any critical point where $h'(w) = 0$:

$$h'(w) = f'(q) \cdot \frac{dq}{dw} = 0 \implies f'(q) = 0,$$

since $dq/dw \neq 0$. For the second-order condition:

$$h''(w) = f''(q) \cdot \left(\frac{dq}{dw}\right)^2 + f'(q) \cdot \frac{d^2 q}{dw^2}.$$

At the critical point where $f'(q) = 0$, the second term vanishes, giving $h''(w) = f''(q) \cdot (dq/dw)^2 < 0$ since $f''(q) \leq 0$ (concavity of f in q) and $(dq/dw)^2 > 0$. Therefore, any FOC solution in w is a local maximum, and by the correspondence with the unique global maximum in q, it is also the global maximum in w.

Part (iii): Existence, interiority, and uniqueness for $\beta \in (0,1)$. For $\beta \in (0,1)$, both terms $\beta \ln G^d$ and $(1-\beta) \ln G^u$ carry positive weight. At the boundary of the feasible set—where either $G^d(q) \to 0$ or $G^u(q) \to 0$—the corresponding logarithm diverges to $-\infty$:

$$G^i(q) \to 0^+ \implies \ln G^i(q) \to -\infty \implies f(q) \to -\infty.$$

By Assumption OA-1 (Gains from Trade), there exists a quantity $\hat{q}$ at which both $G^d(\hat{q}) > 0$ and $G^u(\hat{q}) > 0$, so $f(\hat{q})$ is finite. Since $f(q)$ is concave, finite at an interior point, and diverges to $-\infty$ at the boundaries of its domain, f attains its maximum at an interior point. By strict concavity, this maximizer is unique. By the monotone mapping $w \leftrightarrow q$, the equilibrium input price is also unique.

Part (iv): Corner cases. At $\beta = 1$, the Nash product reduces to $G^d(q)$, and the problem under monopsonistic conduct becomes $\max_q G^d(q)$ subject to $G^u(q) \geq 0$. Log-concavity of G^d implies quasi-concavity (unimodality): G^d increases to a single peak and then decreases. To see that G^d is indeed hump-shaped (rather than monotone), note that $G^d(0) = -\pi_0^d \leq 0$ and $G^d(q) \to -\infty$ as $q \to \infty$ (since eventually $mc(q) > p(q)$), while $G^d(\hat{q}) > 0$ for some interior $\hat{q}$ by the Gains from Trade assumption. Therefore G^d must achieve a positive interior maximum.

If the participation constraint $G^u(q) \geq 0$ is slack at this maximum, the unconstrained solution is the equilibrium. If the constraint binds, the unconstrained maximizer q^{unc} lies outside the feasible set $\mathcal{F} = \{q : G^u(q) \geq 0\}$. Since $G^u(q) = \pi^u(q) - \pi_0^u$ is strictly increasing (see Table OA-6), the feasible set takes the form $\mathcal{F} = [q_{\min}, \infty)$ where $G^u(q_{\min}) = 0$. The unconstrained maximizer satisfies $q^{unc} < q_{\min}$ (otherwise the constraint would be slack). By unimodality, G^d is decreasing on $[q_{\min}, \infty)$, so the constrained maximum is at $q_{\min}$—exactly where $G^u(q) = 0$.

The symmetric argument applies at $\beta = 0$ (where the Nash product reduces to G^u) and under monopolistic conduct. □

Remark. Under each conduct model, one profit function is hump-shaped and the other is strictly increasing in q (see Table OA-6). A strictly increasing function can be log-concave: this requires that its growth rate is decreasing in log terms (i.e., $d^2 \ln G^i / dq^2 \leq 0$). For the loglinear specification used in Section D.4 ($c(q) = q^{1+\psi}/(1+\psi)$, $p(q) = q^{1/\eta}$), both gains from trade are log-concave in q under each conduct model, including for $\pi_0^d, \pi_0^u > 0$.

Comparison with Avignon et al. (2025)

Avignon et al. (2025) adopt an alternative approach to ensuring the SOC holds: rather than assuming the SOC directly or assuming log-concavity, they impose conditions on the *curvature of primitives* that are sufficient for the Nash product to have a unique maximum. We summarize their approach here and relate it to ours.

Define the convexity measure $\sigma_f(q) \equiv qf''(q)/|f'(q)|$ for any function f. **?** impose two assumptions on this measure:

Assumption 1 (on the inverse supply curve $r(q)$ and inverse demand curve $p(q)$):

(i) $r'(q) > 0$ and $\sigma_r(q) > -2$;

(ii) $p'(q) < 0$, $\varepsilon_p(q) > 1$, and $\sigma_p(q) < 2$.

Assumption 2 (on U's marginal cost $MC_U(q) \equiv r'(q)q + r(q)$ and D's marginal revenue $MR_D(q) \equiv p'(q)q + p(q)$):

(i) $\sigma_{MC_U}(q) > -2$;

(ii) $\varepsilon_{MR_D}(q) > 1$ and $\sigma_{MR_D}(q) < 2$.

We now show formally how these conditions map to the first- and second-order conditions of the bargaining model. To facilitate comparison, we use their notation: $r(q)$ denotes the inverse supply (our $c(q)$), $MC_U(q) \equiv r'(q)q + r(q)$ (our $mc(q)$), and $MR_D(q) \equiv p'(q)q + p(q)$ (our $mr(q)$).

Stage-2 quantity-setting SOCs. Assumption 1(i) requires $\sigma_r > -2$, i.e., $qr''(q)/r'(q) > -2$. Since $r'(q) > 0$, this is equivalent to $r''(q)q + 2r'(q) > 0$, which is exactly $MC_U'(q) > 0$ (our U-SOC). To see this:

$$MC_U'(q) = \frac{d}{dq}[r'(q)q + r(q)] = r''(q)q + 2r'(q) = r'(q)\left(\frac{r''(q)q}{r'(q)} + 2\right) = r'(q)(\sigma_r + 2) > 0.$$

Similarly, Assumption 1(ii) requires $\sigma_p < 2$, i.e., $qp''(q)/|p'(q)| < 2$. Since $p'(q) < 0$, $|p'(q)| = -p'(q)$, so this gives $qp''(q)/(-p'(q)) < 2$, or equivalently $p''(q)q + 2p'(q) < 0$, which is exactly $MR_D'(q) < 0$ (our D-SOC):

$$MR_D'(q) = p''(q)q + 2p'(q) = |p'(q)|\left(\frac{p''(q)q}{|p'(q)|} - 2\right) = |p'(q)|(\sigma_p - 2) < 0.$$

Bargaining SOC: monopsonistic side. When U sets the quantity ($w = MC_U(q)$), the Nash product with zero disagreement payoffs is $\pi_U(q)^{\alpha}\,\pi_D(q)^{1-\alpha}$ where $\pi_U(q) = r'(q)q^2$ and $\pi_D(q) = (p(q) - r(q) - r'(q)q)q$. Assumption 2(i) requires $\sigma_{MC_U} > -2$, i.e.:

$$\frac{q\,MC_U''(q)}{MC_U'(q)} > -2 \quad\Longleftrightarrow\quad MC_U''(q)q + 2\,MC_U'(q) > 0.$$

Since $MC_U''(q) = 3r''(q) + r'''(q)q = mc''(q)$ in our notation, the condition $\sigma_{MC_U} > -2$ is equivalent to $mc''(q)q + 2\,mc'(q) > 0$. Our Lemma OA-1 imposes the stronger condition $mc''(q) \geq 0$, which together with $mc'(q) > 0$ implies $mc''(q)q + 2\,mc'(q) > 0$. Thus, $\sigma_{MC_U} > -2$ is a *weaker* condition than $mc''(q) \geq 0$: it allows $mc''(q) < 0$ as long as $|mc''(q)q| < 2\,mc'(q)$.

The role of this condition in the SOC can be seen directly. At $\beta = 1$ (monopsony), the objective is $\pi_D(q) = (p(q) - MC_U(q))q$, whose second derivative is (see Section D.1):

$$\frac{d^2\pi_D}{dq^2} = MR_D'(q) - 2\,MC_U'(q) - MC_U''(q)\,q = \underbrace{MR_D'(q)}_{<0} - \underbrace{\left(MC_U''(q)\,q + 2\,MC_U'(q)\right)}_{>0 \text{ by } \sigma_{MC_U} > -2} < 0.$$

For interior β, the full bargaining SOC involves additional terms (Section B.2), but the same condition $\sigma_{MC_U} > -2$ ensures the critical terms have the correct sign.

Bargaining SOC: monopolistic side. When D sets the quantity ($w = MR_D(q)$), Assumption 2(ii) requires $\sigma_{MR_D} < 2$. Since $MR_D'(q) < 0$, this gives:

$$\frac{q\,MR_D''(q)}{|MR_D'(q)|} < 2 \quad\Longleftrightarrow\quad MR_D''(q)\,q + 2\,MR_D'(q) < 0.$$

Since $MR_D''(q) = 3p''(q) + p'''(q)q = mr''(q)$ in our notation, this is $mr''(q)q + 2\,mr'(q) < 0$. Our Lemma OA-1 imposes the stronger condition $mr''(q) \leq 0$, which together with $mr'(q) < 0$ implies $mr''(q)q + 2\,mr'(q) < 0$. Again, $\sigma_{MR_D} < 2$ is weaker: it allows $mr''(q) > 0$ as long as $mr''(q)q < 2|mr'(q)|$.

At $\beta = 0$ (double marginalization), the objective is $\pi_U(q) = (MR_D(q) - r(q))q$, whose second derivative is:

$$\frac{d^2\pi_U}{dq^2} = 2\,MR_D'(q) + MR_D''(q)\,q - 2\,r'(q) - r''(q)\,q = \underbrace{\left(MR_D''(q)\,q + 2\,MR_D'(q)\right)}_{<0 \text{ by } \sigma_{MR_D} < 2} - \underbrace{\left(r''(q)\,q + 2\,r'(q)\right)}_{>0 \text{ by } \sigma_r > -2} < 0.$$

Key differences. There are three main differences relative to our approach. First, **?** set disagreement payoffs to zero ($\pi_0^d = \pi_0^u = 0$), which simplifies the SOC expressions (the gains from trade equal the profit levels). Second, they express their conditions using curvature measures (σ) and elasticities (ε) rather than sign conditions on derivatives ($mc'' \geq 0, mr'' \leq 0$), which allows them to accommodate a wider range of functional forms (e.g., CES, translog, and AIDS demand satisfy $\sigma_p < 2$). Third, their model determines the quantity-setter endogenously via a "short-side rule" ($q = \min\{q_U, q_D\}$) rather than through an exogenous conduct assumption, so they require the SOC to hold under both regimes simultaneously.

S Supplemental Material: The Model Under Simultaneous Timing

The main text presents the theoretical results of the paper under the sequential timing assumption. This section presents the corresponding analysis under the *simultaneous* timing assumption, allowing for nonzero disagreement payoffs. The section is self-contained: all definitions, assumptions, and intermediate results used below are stated within the section, and results are numbered S.1, S.2, . . . independently of the main text and of the rest of the Online Appendix.

S.1 Setup

A supplier U sells a quantity q to a buyer D at an input price w under a linear pricing contract. The buyer faces a residual inverse demand curve $p(q)$ with $p'(q) \leq 0$, and the supplier produces at a residual average cost curve $c(q)$ with $c'(q) \geq 0$; both functions are three times continuously differentiable. There are gains from trade on an interval $(0, \bar{q})$: $p(q) > c(q)$ for all $q \in (0, \bar{q})$, with $p(\bar{q}) = c(\bar{q})$. The parties' profits are

$$\pi^d(w, q) \equiv \big(p(q) - w\big)q, \qquad \pi^u(w, q) \equiv \big(w - c(q)\big)q,$$

and $\pi_0^d \geq 0$ and $\pi_0^u \geq 0$ denote the (lump-sum) disagreement payoffs of D and U if bargaining breaks down. Marginal revenue and marginal cost are

$$mr(q) \equiv \frac{\partial\big(p(q)q\big)}{\partial q} = p(q) + p'(q)q, \qquad mc(q) \equiv \frac{\partial\big(c(q)q\big)}{\partial q} = c(q) + c'(q)q,$$

and the seller markup and buyer markdown are defined as

$$\mu^u(q) \equiv \frac{w - mc(q)}{mc(q)}, \qquad \Delta^d(q) \equiv \frac{mr(q) - w}{mr(q)}.$$

The joint-profit maximizing quantity q^* solves $mr(q^*) = mc(q^*)$; it is unique whenever $mr(q)$ is strictly decreasing and $mc(q)$ is strictly increasing, which we maintain whenever q^* is used. We write $w^* \equiv mc(q^*) = mr(q^*)$ for the associated input price and

$$\pi^{d*} \equiv \big(p(q^*) - w^*\big)q^* = -p'(q^*)(q^*)^2, \qquad \pi^{u*} \equiv \big(w^* - c(q^*)\big)q^* = c'(q^*)(q^*)^2, \tag{S.1}$$

for the parties' profits at (w^*, q^*), with $\Pi^* \equiv \pi^{d*} + \pi^{u*} = \big(p(q^*) - c(q^*)\big)q^*$ denoting total profit at q^*.

Definition S.1 (Simultaneous Bargaining)**.** *The quantity is determined simultaneously with bargaining over the input price: the parties bargain over w taking the quantity as given, and the quantity-setting party chooses q taking the input price as given. In particular, neither problem* *internalizes the dependence of the other variable on its own choice (dq/dw does not enter the bargaining problem).*

As in the main text, we consider two models of vertical conduct. Under *monopsonistic* bargaining, the transacted quantity lies on the input supply curve, and under *monopolistic* bargaining it lies on the input demand curve:

$$(w^{ms}, q^{ms}) : \begin{cases} \max\limits_{w} \left[(\pi^d(w,q) - \pi_0^d)^\beta \, (\pi^u(w,q) - \pi_0^u)^{1-\beta}\right] \\ w = mc(q) \end{cases}$$

$$(w^{mp}, q^{mp}) : \begin{cases} \max\limits_{w} \left[(\pi^d(w,q) - \pi_0^d)^\beta \, (\pi^u(w,q) - \pi_0^u)^{1-\beta}\right] \\ w = mr(q) \end{cases}$$

where $\beta \in [0,1]$ is the buyer's bargaining weight. Under each conduct model, the input price is pinned down by q, so each party's profit becomes a function of q alone:

$$\text{Monopsonistic } (w = mc(q)) : \quad \pi^u(q) = c'(q)\, q^2, \quad \pi^d(q) = \big(p(q) - c(q) - c'(q)q\big)q, \tag{S.2}$$

$$\text{Monopolistic } (w = mr(q)) : \quad \pi^d(q) = -p'(q)\, q^2, \quad \pi^u(q) = \big(p(q) + p'(q)q - c(q)\big)q. \tag{S.3}$$

We impose the following assumption throughout this section.

Assumption S.1 (Gains from Trade). *Under each conduct model, the set of quantities at which both participation constraints are satisfied is nonempty:*

$$\mathcal{Q}^{ms} \equiv \left\{q > 0 : \big(p(q) - c(q) - c'(q)q\big)q \geq \pi_0^d \text{ and } c'(q)q^2 \geq \pi_0^u\right\} \neq \emptyset,$$
$$\mathcal{Q}^{mp} \equiv \left\{q > 0 : -p'(q)q^2 \geq \pi_0^d \text{ and } \big(p(q) + p'(q)q - c(q)\big)q \geq \pi_0^u\right\} \neq \emptyset.$$

S.2 First-Order Conditions

Under the simultaneous bargaining models with disagreement payoffs $\pi_0^d \geq 0$ and $\pi_0^u \geq 0$, the maximization problems are:

$$\begin{cases} \max\limits_{q} \;\; p(q)q - wq & \text{(Downstream's problem)} \\ \max\limits_{q} \;\; wq - c(q)q & \text{(Upstream's problem)} \\ \max\limits_{w} \;\; \left[(p(q)q - wq - \pi_0^d)^\beta \, (wq - c(q)q - \pi_0^u)^{1-\beta}\right] & \text{(Bargaining problem)} \\ \max\limits_{w,q} \;\; \left[(p(q)q - wq - \pi_0^d)^\beta \, (wq - c(q)q - \pi_0^u)^{1-\beta}\right] & \text{(Efficient bargaining problem)} \end{cases}$$

For $\beta \in (0,1)$ and under Assumption S.1, an interior equilibrium satisfies $\pi^d > \pi_0^d$ and $\pi^u > \pi_0^u$ and is characterized by the following first-order conditions, derived in Section S.4:

$$\begin{cases} w = p'(q)q + p(q) = mr(q) & \text{(D-FOC)} \\ w = c'(q)q + c(q) = mc(q) & \text{(U-FOC)} \\ \beta\left(\pi^u - \pi_0^u\right) = (1-\beta)\left(\pi^d - \pi_0^d\right) & \text{(B-FOC)} \\ q[p'(q) - c'(q)] + [p(q) - c(q)] = 0 & \text{(J-FOC)} \end{cases}$$

Because the bargaining problem takes the quantity as given, the disagreement payoffs enter (B-FOC) only through the net gains from trade. Solving (B-FOC) for the input price yields

$$w = (1-\beta)\,p(q) + \beta\,c(q) + \frac{\beta\,\pi_0^u - (1-\beta)\,\pi_0^d}{q}, \tag{S.4}$$

which reduces to the familiar weighted average $w = (1-\beta)p(q) + \beta c(q)$ when $\pi_0^d = \pi_0^u = 0$. Note also that (J-FOC) does not involve the disagreement payoffs: the efficient bargaining problem selects the joint-profit maximizing quantity q^* regardless of (π_0^d, π_0^u).

Substituting the conduct FOCs into (B-FOC), the equilibrium-quantity conditions are:

$$F(q,\beta) \equiv (1-\beta)\big[p(q) - c(q)\big]q - c'(q)q^2 - (1-\beta)\,\pi_0^d + \beta\,\pi_0^u = 0 \quad \text{(MS-Q-FOC)} \tag{S.5}$$
$$G(q,\beta) \equiv p'(q)q^2 + \beta\big[p(q) - c(q)\big]q + (1-\beta)\,\pi_0^d - \beta\,\pi_0^u = 0 \quad \text{(MP-Q-FOC)} \tag{S.6}$$

With $\pi_0^d = \pi_0^u = 0$, dividing by q recovers the conditions $(1-\beta)[p(q) - c(q)] - c'(q)q = 0$ and $p'(q)q + \beta[p(q) - c(q)] = 0$. The FOCs characterize the solution only for $\beta \in (0,1)$; the corner cases $\beta \in \{0,1\}$ are constrained optimization problems, which we analyze as part of the existence results in Section S.5.

S.3 Second-Order Conditions

The second-order conditions for the quantity-setting problems are:

$$\underbrace{2p'(q) + p''(q)\,q}_{=mr'(q)} < 0 \quad \text{(D-SOC)}, \qquad \underbrace{2c'(q) + c''(q)\,q}_{=mc'(q)} > 0 \quad \text{(U-SOC)}.$$

The second-order condition of the bargaining problem is:

$$-\left[\frac{\beta\, q^2}{\left(\pi^d - \pi_0^d\right)^2} + \frac{(1-\beta)\, q^2}{\left(\pi^u - \pi_0^u\right)^2}\right] < 0 \qquad \text{(B-SOC)}.$$

(B-SOC) holds at every point where the net gains from trade are positive, for any disagreement payoffs. Thus, under simultaneous timing the bargaining second-order condition is *always* satisfied. This contrasts with sequential timing, where the bargaining problem internalizes dq/dw and the second-order condition involves third derivatives of the cost and demand functions.

S.4 Derivations of the First- and Second-Order Conditions

(D-FOC) and (U-FOC) are straightforward and therefore omitted. For (B-FOC), take the natural logarithm of the bargaining objective:

$$\mathcal{L}(w) \equiv \beta \ln\left(p(q)q - wq - \pi_0^d\right) + (1-\beta)\ln\left(wq - c(q)q - \pi_0^u\right). \tag{S.7}$$

Under simultaneous timing, the quantity is taken as given in the bargaining problem, so $d\pi^d/dw = -q$ and $d\pi^u/dw = q$. Differentiating $\mathcal{L}(w)$ with respect to w and setting it to zero:

$$\frac{-\beta\, q}{\pi^d - \pi_0^d} + \frac{(1-\beta)\, q}{\pi^u - \pi_0^u} = 0, \tag{S.8}$$

which gives (B-FOC): $\beta(\pi^u - \pi_0^u) = (1-\beta)(\pi^d - \pi_0^d)$. Expanding $\pi^d = (p(q) - w)q$ and $\pi^u = (w - c(q))q$ and solving for w yields Equation (S.4). Differentiating (S.8) once more with respect to w gives (B-SOC):

$$\mathcal{L}''(w) = -\frac{\beta\, q^2}{\left(\pi^d - \pi_0^d\right)^2} - \frac{(1-\beta)\, q^2}{\left(\pi^u - \pi_0^u\right)^2} < 0.$$

For (J-FOC), take the derivative of $\mathcal{L}$ from Equation (S.7) with respect to q:

$$\beta \cdot \frac{p'(q)q + p(q) - w}{\pi^d - \pi_0^d} + (1-\beta) \cdot \frac{w - c'(q)q - c(q)}{\pi^u - \pi_0^u} = 0.$$

By (B-FOC), $\dfrac{\beta}{\pi^d - \pi_0^d} = \dfrac{1-\beta}{\pi^u - \pi_0^u} \equiv \lambda > 0$. Substituting:

$$\lambda\Big[\left(p'(q)q + p(q) - w\right) + \left(w - c'(q)q - c(q)\right)\Big] = \lambda\Big[q\left(p'(q) - c'(q)\right) + \left(p(q) - c(q)\right)\Big] = 0,$$

which is (J-FOC). Since the disagreement payoffs enter only through λ, they do not affect the efficient-bargaining quantity.

Finally, for the equilibrium-quantity conditions: under monopsonistic bargaining, substituting (U-FOC) $w = c(q) + c'(q)q$ into (B-FOC) gives $\pi^u = c'(q)q^2$ and $\pi^d = (p(q) - c(q) - c'(q)q)q$, so

$$\beta\big[c'(q)q^2 - \pi_0^u\big] = (1-\beta)\big[\big(p(q) - c(q) - c'(q)q\big)q - \pi_0^d\big],$$

which rearranges to (MS-Q-FOC). Under monopolistic bargaining, substituting (D-FOC) $w = p(q) + p'(q)q$ into (B-FOC) gives $\pi^d = -p'(q)q^2$ and $\pi^u = (p(q) + p'(q)q - c(q))q$, so

$$\beta\big[\big(p(q) + p'(q)q - c(q)\big)q - \pi_0^u\big] = (1-\beta)\big[-p'(q)q^2 - \pi_0^d\big],$$

which rearranges to (MP-Q-FOC).

S.5 Existence of Equilibrium

Under monopsonistic conduct, an equilibrium (w^e, q^e) of the simultaneous model solves the following programs:

$$\begin{cases} \max\limits_q \ \pi^u(w^e, q) & (U) \\ \max\limits_w \ \big(\pi^d(w, q^e) - \pi_0^d\big)^\beta \big(\pi^u(w, q^e) - \pi_0^u\big)^{1-\beta} & (B) \end{cases} \quad \text{s.t. } \pi^u(w,q) \geq \pi_0^u, \quad \pi^d(w,q) \geq \pi_0^d.$$

That is, (w^e, q^e) is an equilibrium if there is no deviation in w that increases the Nash product given q^e, and no deviation in q that increases upstream profit given w^e, subject to the participation constraints. Under monopolistic conduct, (U) is replaced by $\max_q \pi^d(w^e, q)$. We call an equilibrium *interior* if both net gains from trade are strictly positive: $\pi^d > \pi_0^d$ and $\pi^u > \pi_0^u$.

Proposition S.1 (Existence). *Under simultaneous timing:*

(i) If the upstream marginal cost is constant, $mc'(q) = 0$, the monopsonistic bargaining problem does not have an interior solution for any β and any disagreement payoffs.

(ii) If the downstream marginal revenue is constant, $mr'(q) = 0$, the monopolistic bargaining problem does not have an interior solution for any β and any disagreement payoffs.

(iii) If $mc'(q) > 0$ and $mr'(q) < 0$, interior solutions exist for the ranges of β characterized in Lemmas S.1 and S.2 below, and fail to exist outside these ranges (Corollary S.1). When $\pi_0^d < \pi^{d*}$ and $\pi_0^u < \pi^{u*}$, these ranges include $(0, \beta^*]$ under monopsonistic bargaining and $[\beta^*, 1)$ under monopolistic bargaining, where β^* is the bilaterally-efficient bargaining weight characterized in Section S.7.

Proof. We first show parts (i) and (ii). Since the second-order conditions of the quantity-setting problems fail when $mc'(q) = 0$ or $mr'(q) = 0$, the first-order conditions do not characterize the equilibrium, and we work directly with the maximization programs.

Monopsonistic bargaining with constant marginal cost ($c(q) = c$):

Case I: $\beta \in (0, 1)$. The key observation is that constant marginal cost forces at least one party's net gain from trade to zero in any candidate equilibrium with positive trade:

- If $w^e > c$: upstream profit $(w^e - c)q$ is strictly increasing in q, so q^e must lie on the downstream participation boundary, where $\pi^d(w^e, q^e) = \pi_0^d$ if $\pi_0^d > 0$ (and $\pi^d = 0$ at $q^e = p^{-1}(w^e)$ if $\pi_0^d = 0$). In either case the downstream net gain is zero.
- If $w^e = c$: $\pi^u = (w^e - c)q = 0$ for any q, so the upstream net gain is $-\pi_0^u \leq 0$.
- If $w^e < c$: upstream profit is negative for any $q > 0$, so $q^e = 0$ and there is no trade.

Consider the implications for the Nash product $(\pi^d - \pi_0^d)^\beta(\pi^u - \pi_0^u)^{1-\beta}$. If $\pi_0^d > 0$ and $\pi_0^u > 0$, no w satisfies both participation constraints with positive net gains: for $w^e > c$ the downstream net gain is zero, and for $w^e = c$ the upstream participation constraint is violated. If exactly one disagreement payoff is zero, the corresponding party's net gain is zero at every candidate equilibrium, so the Nash product is zero and no interior equilibrium exists (degenerate equilibria with a zero Nash product may exist, as in the case without disagreement payoffs). If both disagreement payoffs are zero, every $w^e \geq c$ supports a degenerate equilibrium with a zero Nash product, and none is interior.

Case II: $\beta = 1$. Bargaining maximizes $\pi^d - \pi_0^d$ subject to $\pi^u \geq \pi_0^u$. The downstream-optimal input price is $w^e = c$, at which $\pi^u = 0$: if $\pi_0^u > 0$, the upstream participation constraint fails and no equilibrium with positive trade exists; if $\pi_0^u = 0$, the continuum of equilibria $w^e = c$, $q^e \in [0, p^{-1}(c)]$ from the case without disagreement payoffs survives (subject to $\pi^d(c, q^e) \geq \pi_0^d$), but the upstream net gain is zero in all of them. The alternative candidate family ($w^e \geq c$, $q^e = p^{-1}(w^e)$) has $\pi^d = 0$ and is infeasible when $\pi_0^d > 0$. No interior equilibrium exists.

Case III: $\beta = 0$. Bargaining maximizes $\pi^u - \pi_0^u$ subject to $\pi^d \geq \pi_0^d$. For any $w > c$, upstream profit increases in q up to the downstream participation boundary, so in any candidate equilibrium the downstream net gain is zero ($\pi^d = \pi_0^d$, or $\pi^d = 0$ when $\pi_0^d = 0$ at $q = p^{-1}(w)$). No interior equilibrium exists.

Monopolistic bargaining with constant marginal revenue ($p(q) = p$):

The argument is symmetric, with the roles of the parties reversed: the downstream best response drives the upstream net gain to zero when $w < p$ (output expands to the upstream participation boundary), and the downstream profit is zero when $w = p$. By the same case analysis, at least one party's net gain from trade is zero in every candidate equilibrium for every $\beta \in [0,1]$ and every configuration of disagreement payoffs, so no interior equilibrium exists. We omit the details.

Part (iii): Suppose $mc'(q) > 0$ and $mr'(q) < 0$. Fix $\beta \in (0,1)$ and suppose $1-\beta$ (respectively β) is attained by the share function s^{ms} (respectively s^{mp}) of Lemma S.1 (respectively Lemma S.2) at a quantity q in the interior of the feasible set. Set $w = mc(q)$ (respectively $w = mr(q)$). Then (B-FOC) holds by construction, and (B-SOC) holds globally (Section S.3), so w maximizes the Nash product given q. Moreover, q solves the quantity-setter's problem given w, since (U-FOC) holds with $mc'(q) > 0$ (respectively (D-FOC) with $mr'(q) < 0$). Both participation constraints hold strictly in the interior of the feasible set, so (w,q) is an interior equilibrium. Conversely, any interior equilibrium satisfies the FOCs, hence the share equation. The characterization of the attainable ranges of β, including the claim that they cover $(0,\beta^*]$ and $[\beta^*,1)$ when $\pi_0^d < \pi^{d*}$ and $\pi_0^u < \pi^{u*}$, is given in Lemmas S.1 and S.2. □

Unlike under sequential timing, where an interior equilibrium exists for every $\beta \in (0,1)$, under simultaneous timing the equilibrium may fail to exist for some values of β, depending on the demand and cost curves and on the disagreement payoffs. The following two lemmas characterize the range of β for which an interior equilibrium exists. For each conduct model, define the *share function* implied by (B-FOC): the equilibrium condition $\beta(\pi^u - \pi_0^u) = (1-\beta)(\pi^d - \pi_0^d)$, with profits evaluated along the conduct curve as in Equations (S.2)–(S.3), pins down the bargaining weight as a function of the equilibrium quantity.

Lemma S.1 (Existence Range: Monopsonistic Bargaining)**.** *Assume $mc'(q) > 0$, Assumption S.1, and $\pi_0^d < \pi^{d*}$, $\pi_0^u < \pi^{u*}$. Let $[q_l, q_h]$ denote the connected component of Ω^{ms} containing q^*, and define on it*

$$s^{ms}(q) \equiv \frac{c'(q)\,q^2 - \pi_0^u}{\big(p(q) - c(q)\big)q - \pi_0^d - \pi_0^u} = \frac{\pi^u(q) - \pi_0^u}{\big(\pi^d(q) - \pi_0^d\big) + \big(\pi^u(q) - \pi_0^u\big)}. \tag{S.9}$$

An interior monopsonistic equilibrium with $q \in (q_l, q_h)$ exists at bargaining weight β if and only if $1-\beta$ is in the range of s^{ms} over (q_l, q_h). In particular:

(i) $s^{ms}(q_h) = 1$ and $s^{ms}(q^*) = 1-\beta^*$, so an interior equilibrium exists for every $\beta \in (0,\beta^*]$;

(ii) more generally, an interior equilibrium exists for every $\beta \in \big(0, \bar{\beta}^{ms}\big)$, where $\bar{\beta}^{ms} \equiv 1 - \inf_{q\in[q_l,q_h]} s^{ms}(q) \geq \beta^*$ is the upper existence bound for simultaneous monopsonistic bargaining;

(iii) if $\pi_0^d = \pi_0^u = 0$ *and Assumption S.2 holds, then* $s^{ms}(q) = \dfrac{c'(q)\,q}{p(q) - c(q)}$ *is strictly increasing (its numerator is increasing since* $c''(q)q + c'(q) > 0$*, and its denominator is positive and decreasing), so* $\bar{\beta}^{ms} = 1 - \lim_{q \to 0^+} \dfrac{c'(q)\,q}{p(q) - c(q)}$*, recovering the existence range of the model without disagreement payoffs;*

(iv) if the upstream participation constraint binds at the lower end of the feasible set ($c'(q_l)q_l^2 = \pi_0^u > 0$*) while the downstream constraint is slack there (*$\pi^d(q_l) > \pi_0^d$*), then* $s^{ms}(q_l) = 0$*, so* $\bar{\beta}^{ms} = 1$ *and an interior equilibrium exists for every* $\beta \in (0, 1)$*.*

Proof. Under monopsonistic conduct, $w = mc(q)$ implies $\pi^u(q) = c'(q)q^2$ and $\pi^d(q) = (p(q) - mc(q))q$, with $\pi^d(q) + \pi^u(q) = (p(q) - c(q))q$. Substituting into (B-FOC) and solving for the bargaining weight gives $1 - \beta = s^{ms}(q)$, with s^{ms} as in Equation (S.9). By the argument in part (iii) of Proposition S.1, an interior equilibrium at β exists if and only if $1 - \beta = s^{ms}(q)$ for some q in the interior of the feasible set.

We compute the value of s^{ms} at three points. First, since $mc'(q) > 0$, upstream profit $\pi^u(q) = c'(q)q^2$ is strictly increasing: $\big(c'(q)q^2\big)' = q\big(2c'(q) + c''(q)q\big) = q\,mc'(q) > 0$. Hence the upstream participation constraint cannot bind at the upper end of the feasible set, and at q_h the downstream constraint binds: $\pi^d(q_h) = \pi_0^d$. Substituting into Equation (S.9), the numerator and denominator coincide, so $s^{ms}(q_h) = 1$.

Second, at $q = q^*$: by (J-FOC), $p(q^*) - c(q^*) = q^*\big(c'(q^*) - p'(q^*)\big)$, so $\pi^d(q^*) = \big(p(q^*) - c(q^*) - c'(q^*)q^*\big)q^* = -p'(q^*)(q^*)^2 = \pi^{d*}$ and $\pi^u(q^*) = \pi^{u*}$. Note that $q^* \in (q_l, q_h)$ because $\pi^d(q^*) = \pi^{d*} > \pi_0^d$ and $\pi^u(q^*) = \pi^{u*} > \pi_0^u$. Therefore

$$s^{ms}(q^*) = \frac{\pi^{u*} - \pi_0^u}{\Pi^* - \pi_0^d - \pi_0^u} = 1 - \beta^*,$$

by the formula for β^* in Proposition S.2.

Third, if the upstream constraint binds at q_l with $\pi_0^u > 0$, the numerator of $s^{ms}(q_l)$ is zero while the denominator equals $\pi^d(q_l) - \pi_0^d \geq 0$, so $s^{ms}(q_l) = 0$ whenever $\pi^d(q_l) > \pi_0^d$.

Since s^{ms} is continuous on $[q_l, q_h]$, the Intermediate Value Theorem implies that its range contains every value between any two attained values. Part (i) follows from $s^{ms}(q^*) = 1 - \beta^*$ and $s^{ms}(q_h) = 1$: every $1 - \beta \in [1 - \beta^*, 1)$ is attained in the open interval (q_l, q_h), i.e., every $\beta \in (0, \beta^*]$. Part (ii) follows by the same argument applied to the infimum of s^{ms} (when the infimum is attained in the interior, the endpoint $\beta = \bar{\beta}^{ms}$ is also included). For part (iii), with $\pi_0^d = \pi_0^u = 0$ the share function reduces to $s^{ms}(q) = c'(q)q/(p(q) - c(q))$, the feasible set extends to the left endpoint $q_l = 0$, and the infimum is the stated limit. Part (iv) is immediate from $s^{ms}(q_l) = 0$ and $s^{ms}(q_h) = 1$. □

Lemma S.2 (Existence Range: Monopolistic Bargaining)**.** *Assume $mr'(q) < 0$, Assumption S.1, and $\pi_0^d < \pi^{d*}$, $\pi_0^u < \pi^{u*}$. Let $[q_l, q_h]$ denote the connected component of Ω^{mp} containing q^*, and define on it*

$$s^{mp}(q) \equiv \frac{-p'(q)\,q^2 - \pi_0^d}{\big(p(q) - c(q)\big)q - \pi_0^d - \pi_0^u} \;=\; \frac{\pi^d(q) - \pi_0^d}{\big(\pi^d(q) - \pi_0^d\big) + \big(\pi^u(q) - \pi_0^u\big)}. \tag{S.10}$$

An interior monopolistic equilibrium with $q \in (q_l, q_h)$ exists at bargaining weight β if and only if β is in the range of s^{mp} over (q_l, q_h). In particular:

(i) $s^{mp}(q_h) = 1$ and $s^{mp}(q^*) = \beta^*$, so an interior equilibrium exists for every $\beta \in [\beta^*, 1)$;

(ii) more generally, an interior equilibrium exists for every $\beta \in \big(\underline{\beta}^{mp}, 1\big)$, where $\underline{\beta}^{mp} \equiv \inf_{q\in[q_l,q_h]} s^{mp}(q) \leq \beta^*$ is the lower existence bound for simultaneous monopolistic bargaining;

(iii) if $\pi_0^d = \pi_0^u = 0$ and Assumption S.3 holds, then $s^{mp}(q) = \dfrac{-p'(q)\,q}{p(q) - c(q)}$ is strictly increasing (its numerator is increasing since $p''(q)q + p'(q) < 0$, and its denominator is positive and decreasing), so $\underline{\beta}^{mp} = \lim_{q\to 0^+} \dfrac{-p'(q)\,q}{p(q) - c(q)}$, recovering the existence range of the model without disagreement payoffs;

(iv) if the downstream participation constraint binds at the lower end of the feasible set ($-p'(q_l)q_l^2 = \pi_0^d > 0$) while the upstream constraint is slack there ($\pi^u(q_l) > \pi_0^u$), then $s^{mp}(q_l) = 0$, so $\underline{\beta}^{mp} = 0$ and an interior equilibrium exists for every $\beta \in (0, 1)$.

Proof. The argument mirrors the proof of Lemma S.1. Under monopolistic conduct, $w = mr(q)$ implies $\pi^d(q) = -p'(q)q^2$ and $\pi^u(q) = (mr(q) - c(q))q$, with $\pi^d(q) + \pi^u(q) = (p(q) - c(q))q$. Substituting into (B-FOC) and solving gives $\beta = s^{mp}(q)$.

Since $mr'(q) < 0$, downstream profit is strictly increasing: $\big(-p'(q)q^2\big)' = -q\big(2p'(q) + p''(q)q\big) = -q\,mr'(q) > 0$. Hence the downstream constraint cannot bind at the upper end of the feasible set, and at q_h the upstream constraint binds: $\pi^u(q_h) = \pi_0^u$, which gives $s^{mp}(q_h) = 1$. At $q = q^*$: $\pi^d(q^*) = \pi^{d*}$ directly, and $\pi^u(q^*) = \big(mr(q^*) - c(q^*)\big)q^* = \big(mc(q^*) - c(q^*)\big)q^* = c'(q^*)(q^*)^2 = \pi^{u*}$, so

$$s^{mp}(q^*) = \frac{\pi^{d*} - \pi_0^d}{\Pi^* - \pi_0^d - \pi_0^u} = \beta^*.$$

If the downstream constraint binds at q_l with $\pi_0^d > 0$, then $s^{mp}(q_l) = 0$. Parts (i)–(iv) then follow from the continuity of s^{mp} and the Intermediate Value Theorem, exactly as in Lemma S.1. □

The preceding lemmas characterize where equilibrium exists; the following corollary makes the converse explicit: for sufficiently extreme bargaining weights, the simultaneous bargaining models have *no* interior equilibrium.

Corollary S.1 (Non-Existence of Equilibrium for Extreme Bargaining Weights)**.** *Under the conditions of Lemmas S.1–S.2, the set of bargaining weights at which an interior equilibrium with quantity in $[q_l, q_h]$ exists is an interval. In particular:*

(i) under simultaneous monopsonistic bargaining, no interior equilibrium with $q \in (q_l, q_h)$ exists for any $\beta > \bar{\beta}^{ms}$;

(ii) under simultaneous monopolistic bargaining, no interior equilibrium with $q \in (q_l, q_h)$ exists for any $\beta < \underline{\beta}^{mp}$.

These regions are nonempty whenever $\bar{\beta}^{ms} < 1$ and $\underline{\beta}^{mp} > 0$, respectively; with zero disagreement payoffs, the feasible set is itself an interval, so the qualification on q is vacuous and the bounds equal those in part (iii) of Lemmas S.1–S.2. If the feasible set has further connected components, the same characterization applies to each component, with the bounds given by the range of the share function on that component.

Proof. We prove part (i); part (ii) is symmetric. On the feasible set, both net gains from trade are nonnegative, so $s^{ms}(q) \in [0, 1]$, and s^{ms} is continuous on the interval $[q_l, q_h]$. The range of a continuous function on an interval is itself an interval; since $s^{ms}(q_h) = 1$, the range is exactly $[1 - \bar{\beta}^{ms}, 1]$. Hence, for any $\beta > \bar{\beta}^{ms}$, no quantity in the feasible set solves the share equation $1 - \beta = s^{ms}(q)$. Because every interior equilibrium must satisfy (B-FOC), and (B-FOC) combined with (U-FOC) is equivalent to the share equation, no interior equilibrium exists for $\beta > \bar{\beta}^{ms}$. Since the attainable values of $1 - \beta$ form an interval, the set of bargaining weights supporting an interior equilibrium is an interval as well. □

Remark. The threshold β^* appearing in Lemmas S.1–S.2 is the bilaterally-efficient bargaining weight of Section S.7: it enters the existence characterization because the share function evaluated at q^* equals $1 - \beta^*$ (monopsonistic) and β^* (monopolistic) by construction. Each conduct model is thus guaranteed an interior equilibrium for all bargaining weights between the corner and β^*: $\beta \in (0, \beta^*]$ under monopsonistic and $\beta \in [\beta^*, 1)$ under monopolistic bargaining. As shown in Section S.8, these are exactly the regions where the nonnegative markup/markdown constraint *binds*, so the unconstrained solution whose existence the conduct-selection argument invokes there is always available. On the opposite side of β^*, where the constraint is slack and the unconstrained equilibrium is the equilibrium, existence can fail for extreme β, with the exact bounds given by $\bar{\beta}^{ms}$ and $\underline{\beta}^{mp}$. In the remainder of this section, when we refer to the equilibrium at a bargaining weight

$\beta \in (0,1)$, we implicitly restrict attention to values of β within the existence ranges; for brevity, we do not repeat this qualification.

S.5.1 Illustration: Existence Bounds with Loglinear Demand and Costs

To illustrate that the non-existence regions in Corollary S.1 can be quantitatively meaningful, consider the loglinear specification with zero disagreement payoffs:

$$p(q) = q^{1/\eta}, \qquad c(q) = \frac{q^{\psi}}{1+\psi},$$

where $\eta < -1$ is the downstream demand elasticity and $1/\psi > 0$ is the elasticity of upstream supply, with $\psi - 1/\eta > 0$.

Monopolistic bargaining. Solving (MP-Q-FOC) in closed form yields

$$q^{mp}(\beta) = \left(\frac{1+\psi}{\beta\eta} + 1 + \psi\right)^{\frac{1}{\psi - 1/\eta}},$$

which is well-defined only if the term in parentheses is positive, i.e., if $\beta > -1/\eta$. The same bound follows from Lemma S.2(iii):

$$\underline{\beta}^{mp} = \lim_{q\to 0^+} \frac{-p'(q)\,q}{p(q)-c(q)} = \lim_{q\to 0^+} \frac{-(1/\eta)\,q^{1/\eta}}{q^{1/\eta} - (1+\psi)^{-1}q^{\psi}} = \lim_{q\to 0^+} \frac{-1/\eta}{1-(1+\psi)^{-1}q^{\psi-1/\eta}} = -\frac{1}{\eta},$$

using $q^{\psi-1/\eta} \to 0$ as $q \to 0^+$. Hence the simultaneous monopolistic equilibrium exists only for $\beta \in (-1/\eta, 1)$. For example, under the parametrization $\eta = -6$ and $\psi = 1/4$ used in the numerical illustrations of the paper, the monopolistic bargaining model has no solution for $\beta < 1/6$.

Monopsonistic bargaining. By Lemma S.1(iii):

$$1 - \bar{\beta}^{ms} = \lim_{q\to 0^+} \frac{c'(q)\,q}{p(q)-c(q)} = \lim_{q\to 0^+} \frac{\frac{\psi}{1+\psi}\,q^{\psi}}{q^{1/\eta} - (1+\psi)^{-1}q^{\psi}} = \lim_{q\to 0^+} \frac{\frac{\psi}{1+\psi}\,q^{\psi-1/\eta}}{1-(1+\psi)^{-1}q^{\psi-1/\eta}} = 0.$$

Hence $\bar{\beta}^{ms} = 1$: the simultaneous monopsonistic equilibrium exists for every $\beta \in (0,1)$ in this specification. With positive disagreement payoffs, the bounds shift as described in parts (ii) and (iv) of Lemmas S.1–S.2.

S.6 Output and Buyer Power

We now characterize how the equilibrium output responds to buyer power, measured either by the bargaining weight β or by the disagreement payoffs. Under simultaneous timing, these comparative statics require curvature assumptions on the cost and demand functions, together with a bound on the size of the disagreement payoffs.

Assumption S.2 (Increasing Differences Between Marginal and Average Costs)**.** $c''(q)\,q + c'(q) > 0$.

Assumption S.3 (Decreasing Differences Between Marginal and Average Revenue)**.** $p''(q)\,q + p'(q) < 0$.

Assumption S.2 implies $mc'(q) = c'(q) + \big(c''(q)q + c'(q)\big) > 0$, and Assumption S.3 implies $mr'(q) = p'(q) + \big(p''(q)q + p'(q)\big) < 0$. We need Assumption S.2 only for monopsonistic bargaining and Assumption S.3 only for monopolistic bargaining. Define the curvature terms

$$\Gamma^{ms}(q) \equiv (1-\beta)\big(c'(q) - p'(q)\big) + c''(q)\,q + c'(q),$$
$$\Gamma^{mp}(q) \equiv \beta\big(c'(q) - p'(q)\big) - \big(p''(q)\,q + p'(q)\big),$$

which are strictly positive under Assumptions S.2 and S.3, respectively. The bounds on the disagreement payoffs, evaluated at the equilibrium quantity, are:

$$(1-\beta)\,\pi_0^d - \beta\,\pi_0^u \;<\; q^2\,\Gamma^{ms}(q), \tag{S.11}$$
$$\beta\,\pi_0^u - (1-\beta)\,\pi_0^d \;<\; q^2\,\Gamma^{mp}(q). \tag{S.12}$$

Theorem S.1 (Output and Buyer Power Under Simultaneous Timing)**.**

(i) Under monopsonistic bargaining, if Assumption S.2 and condition (S.11) hold, the output quantity decreases with the buyer's bargaining weight and the buyer's disagreement payoff, and increases with the supplier's disagreement payoff:

$$\frac{dq^{ms}}{d\beta} < 0, \qquad \frac{dq^{ms}}{d\pi_0^d} < 0, \qquad \frac{dq^{ms}}{d\pi_0^u} > 0.$$

(ii) Under monopolistic bargaining, if Assumption S.3 and condition (S.12) hold, the output quantity increases with the buyer's bargaining weight and the buyer's disagreement payoff, and decreases with the supplier's disagreement payoff:

$$\frac{dq^{mp}}{d\beta} > 0, \qquad \frac{dq^{mp}}{d\pi_0^d} > 0, \qquad \frac{dq^{mp}}{d\pi_0^u} < 0.$$

Proof. Part (i): Monopsonistic bargaining. We apply the Implicit Function Theorem to $F(q, \beta) = 0$ from (MS-Q-FOC), treating the disagreement payoffs as parameters.

Step 1: Sign of $\partial F/\partial \beta$. Differentiating Equation (S.5):

$$\frac{\partial F}{\partial \beta} = -\big[p(q) - c(q)\big]q + \pi_0^d + \pi_0^u = -\Big[\big(\pi^d(q) - \pi_0^d\big) + \big(\pi^u(q) - \pi_0^u\big)\Big] < 0,$$

where the inequality holds because both net gains from trade are strictly positive at an interior equilibrium.

Step 2: Sign of $\partial F/\partial q$. Differentiating Equation (S.5):

$$\frac{\partial F}{\partial q} = (1-\beta)\big[p'(q) - c'(q)\big]q + (1-\beta)\big[p(q) - c(q)\big] - c''(q)q^2 - 2c'(q)q.$$

We eliminate $(1-\beta)[p(q) - c(q)]$ using the equilibrium condition (MS-Q-FOC) divided by q:

$$(1-\beta)\big[p(q) - c(q)\big] = c'(q)q - \frac{\beta\,\pi_0^u - (1-\beta)\,\pi_0^d}{q}.$$

Substituting and collecting terms:

$$\frac{\partial F}{\partial q} = -q\,\Gamma^{ms}(q) + \frac{(1-\beta)\,\pi_0^d - \beta\,\pi_0^u}{q}.$$

The first term is strictly negative since $\Gamma^{ms}(q) > 0$ by Assumption S.2 (as $c'(q) - p'(q) \geq 0$ and $c''(q)q + c'(q) > 0$). The second term has ambiguous sign, but under condition (S.11) it is dominated by the first, so $\partial F/\partial q < 0$.

Step 3: Comparative statics. By the Implicit Function Theorem:

$$\frac{dq^{ms}}{d\beta} = -\frac{\partial F/\partial \beta}{\partial F/\partial q} = -\frac{(-)}{(-)} < 0.$$

For the disagreement payoffs, from Equation (S.5), $\partial F/\partial \pi_0^d = -(1-\beta) < 0$ and $\partial F/\partial \pi_0^u = \beta > 0$, so

$$\frac{dq^{ms}}{d\pi_0^d} = -\frac{-(1-\beta)}{\partial F/\partial q} < 0, \qquad \frac{dq^{ms}}{d\pi_0^u} = -\frac{\beta}{\partial F/\partial q} > 0,$$

using $\partial F/\partial q < 0$ from Step 2.

Part (ii): Monopolistic bargaining. We apply the Implicit Function Theorem to $G(q, \beta) = 0$ from (MP-Q-FOC).

Step 1: Sign of $\partial G/\partial\beta$. Differentiating Equation (S.6):

$$\frac{\partial G}{\partial\beta} = \big[p(q) - c(q)\big]q - \pi_0^d - \pi_0^u = \big(\pi^d(q) - \pi_0^d\big) + \big(\pi^u(q) - \pi_0^u\big) > 0,$$

at an interior equilibrium.

Step 2: Sign of $\partial G/\partial q$. Differentiating Equation (S.6):

$$\frac{\partial G}{\partial q} = p''(q)q^2 + 2p'(q)q + \beta\big[p'(q) - c'(q)\big]q + \beta\big[p(q) - c(q)\big].$$

We eliminate $\beta[p(q) - c(q)]$ using the equilibrium condition (MP-Q-FOC) divided by q:

$$\beta\big[p(q) - c(q)\big] = -p'(q)q + \frac{\beta\,\pi_0^u - (1-\beta)\,\pi_0^d}{q}.$$

Substituting and collecting terms:

$$\frac{\partial G}{\partial q} = -q\,\Gamma^{mp}(q) + \frac{\beta\,\pi_0^u - (1-\beta)\,\pi_0^d}{q}.$$

The first term is strictly negative since $\Gamma^{mp}(q) > 0$ by Assumption S.3. Under condition (S.12), the second term is dominated by the first, so $\partial G/\partial q < 0$.

Step 3: Comparative statics. By the Implicit Function Theorem:

$$\frac{dq^{mp}}{d\beta} = -\frac{\partial G/\partial\beta}{\partial G/\partial q} = -\frac{(+)}{(-)} > 0.$$

From Equation (S.6), $\partial G/\partial\pi_0^d = (1-\beta) > 0$ and $\partial G/\partial\pi_0^u = -\beta < 0$, so

$$\frac{dq^{mp}}{d\pi_0^d} = -\frac{(1-\beta)}{\partial G/\partial q} > 0, \qquad \frac{dq^{mp}}{d\pi_0^u} = -\frac{-\beta}{\partial G/\partial q} < 0,$$

using $\partial G/\partial q < 0$ from Step 2. □

Remark (when conditions (S.11)–(S.12) hold). With zero disagreement payoffs, the left-hand sides of conditions (S.11)–(S.12) are zero and both conditions hold automatically under Assumptions S.2–S.3; Theorem S.1 then nests the comparative statics of the model without disagreement payoffs. With nonzero disagreement payoffs, condition (S.11) holds automatically whenever $\beta\,\pi_0^u \geq (1-\beta)\,\pi_0^d$, and condition (S.12) holds automatically whenever $(1-\beta)\,\pi_0^d \geq \beta\,\pi_0^u$; at any given β, at least one of the two conditions is therefore automatic. More generally, both conditions hold whenever the disagreement payoffs are small relative to the curvature of the cost and demand curves. We note that these bounds are a genuine feature of simultaneous timing: under sequential timing, the corresponding comparative statics hold without any restriction on the size of the disagreement payoffs, because the disagreement payoffs enter the sequential bargaining FOC only through the (level) denominators rather than through the equilibrium share equation.

Corollary S.2 (Markups, Markdowns, and the Input Price)**.**

(i) Under monopolistic bargaining, the buyer markdown is always zero, $\Delta^d = 0$*. Under the conditions of Theorem S.1(ii), the seller markup* μ^u *decreases with the buyer's bargaining weight and the buyer's disagreement payoff, and increases with the supplier's disagreement payoff.*

(ii) Under monopsonistic bargaining, the seller markup is always zero, $\mu^u = 0$*. Under the conditions of Theorem S.1(i), the buyer markdown* Δ^d *increases with the buyer's bargaining weight and the buyer's disagreement payoff, and decreases with the supplier's disagreement payoff.*

(iii) Under the conditions of Theorem S.1, the input price decreases with the buyer's bargaining weight under both conduct models: $dw/d\beta < 0$*.*

Proof. Part (i): Under monopolistic bargaining, (D-FOC) implies $w = mr(q)$, so $\Delta^d = (mr(q) - w)/mr(q) = 0$. The markup is $\mu^u = \big(mr(q) - mc(q)\big)/mc(q) = mr(q)/mc(q) - 1$, which is strictly decreasing in q since $mr'(q) < 0$ and $mc'(q) > 0$. By Theorem S.1(ii), q^{mp} increases with β and π_0^d and decreases with π_0^u; composing with the strict monotonicity of μ^u in q yields the stated signs.

Part (ii): Under monopsonistic bargaining, (U-FOC) implies $w = mc(q)$, so $\mu^u = 0$. The markdown is $\Delta^d = 1 - mc(q)/mr(q)$, which is strictly decreasing in q since $mc(q)/mr(q)$ is strictly increasing in q. By Theorem S.1(i), q^{ms} decreases with β and π_0^d and increases with π_0^u; composing yields the stated signs.

Part (iii): Under monopsonistic bargaining, $w = mc(q)$ with $mc'(q) > 0$ and $dq^{ms}/d\beta < 0$, so $dw/d\beta = mc'(q)\, dq^{ms}/d\beta < 0$. Under monopolistic bargaining, $w = mr(q)$ with $mr'(q) < 0$ and $dq^{mp}/d\beta > 0$, so $dw/d\beta = mr'(q)\, dq^{mp}/d\beta < 0$. □

S.7 The Bilaterally-Efficient Bargaining Weight

Proposition S.2 (Bilaterally-Efficient Buyer Power Under Simultaneous Timing)**.** *Suppose $\pi_0^d < \pi^{d*}$ and $\pi_0^u < \pi^{u*}$. Then there exists a unique bargaining weight*

$$\beta^* = \frac{\pi^{d*} - \pi_0^d}{\Pi^* - (\pi_0^d + \pi_0^u)} = \frac{-p'(q^*)(q^*)^2 - \pi_0^d}{\big(c'(q^*) - p'(q^*)\big)(q^*)^2 - (\pi_0^u + \pi_0^d)} \in (0,1) \tag{S.13}$$

at which the equilibrium output of both the monopsonistic and the monopolistic bargaining model equals the joint-profit maximizing output q^*. With zero disagreement payoffs, the formula simplifies to $\beta^* = \dfrac{-p'(q^*)}{c'(q^*) - p'(q^*)}$.

Proof. Monopsonistic bargaining. Under (U-FOC), $w = mc(q)$, so $\pi^u(q) = c'(q)q^2$ and $\pi^d(q) = \big(p(q) - c(q) - c'(q)q\big)q$. At $q = q^*$, (J-FOC) gives $p(q^*) - c(q^*) = q^*\big(c'(q^*) - p'(q^*)\big)$, hence

$$\pi^u(q^*) = c'(q^*)(q^*)^2 = \pi^{u*}, \qquad \pi^d(q^*) = -p'(q^*)(q^*)^2 = \pi^{d*}.$$

Substituting into (B-FOC) at q^*:

$$\beta\left(\pi^{u*} - \pi_0^u\right) = (1-\beta)\left(\pi^{d*} - \pi_0^d\right), \tag{S.14}$$

and solving for β:

$$\beta\big(\pi^{u*} + \pi^{d*} - \pi_0^u - \pi_0^d\big) = \pi^{d*} - \pi_0^d \quad \Longrightarrow \quad \beta^* = \frac{\pi^{d*} - \pi_0^d}{\Pi^* - (\pi_0^d + \pi_0^u)}.$$

The second expression in Equation (S.13) follows from $\Pi^* = \big(p(q^*) - c(q^*)\big)q^* = \big(c'(q^*) - p'(q^*)\big)(q^*)^2$ by (J-FOC).

Monopolistic bargaining. Under (D-FOC), $w = mr(q)$. At $q = q^*$, $w^* = mr(q^*) = mc(q^*)$, so the parties' profits at q^* are identical to the monopsonistic case: $\pi^d(q^*) = \pi^{d*}$ and $\pi^u(q^*) = \pi^{u*}$. Substituting into (B-FOC) yields the same condition (S.14) and hence the same β^*.

Conditions for $\beta^* \in (0,1)$. For $\beta^* > 0$: the condition $\pi_0^d < \pi^{d*}$ makes the numerator positive, and together with $\pi_0^u < \pi^{u*}$ it makes the denominator positive (since $\pi_0^d + \pi_0^u < \pi^{d*} + \pi^{u*} = \Pi^*$). For $\beta^* < 1$: the numerator is smaller than the denominator if and only if $\pi_0^u < \pi^{u*}$.

Uniqueness. The quantity q^* is the unique solution to $mr(q) = mc(q)$ (as mr is strictly decreasing and mc strictly increasing), and Equation (S.14) maps q^* to a unique β^*. □

Remark. When $\pi_0^d \geq \pi^{d*}$ or $\pi_0^u \geq \pi^{u*}$, the joint-profit maximizing quantity is not achievable under a linear price contract, because at q^* at least one party would earn less than its disagreement payoff; in that case $q^* \notin \mathcal{Q}^{ms} \cup \mathcal{Q}^{mp}$ and we adopt the corner convention $\beta^* = 0$ (when $\pi_0^d \geq \pi^{d*}$) or $\beta^* = 1$ (when $\pi_0^u \geq \pi^{u*}$), as in the sequential analysis.

Corollary S.3 (Properties of β^*)**.** *Under the conditions of Proposition S.2:*

(i) Holding q^* fixed, β^* weakly decreases as the upstream marginal cost curve becomes steeper ($c'(q^*)$ increases) and weakly increases as the downstream inverse demand curve becomes steeper ($-p'(q^*)$ increases).

(ii) β^* decreases with the downstream disagreement payoff and increases with the upstream disagreement payoff:

$$\frac{\partial \beta^*}{\partial \pi_0^d} = \frac{-\left(\pi^{u*} - \pi_0^u\right)}{\left[\Pi^* - (\pi_0^d + \pi_0^u)\right]^2} < 0, \qquad \frac{\partial \beta^*}{\partial \pi_0^u} = \frac{\pi^{d*} - \pi_0^d}{\left[\Pi^* - (\pi_0^d + \pi_0^u)\right]^2} > 0.$$

Proof. Write $\beta^* = N/D$ with $N = \pi^{d*} - \pi_0^d = -p'(q^*)(q^*)^2 - \pi_0^d$ and $D = \Pi^* - (\pi_0^d + \pi_0^u) = \left(c'(q^*) - p'(q^*)\right)(q^*)^2 - (\pi_0^d + \pi_0^u)$, with $N, D > 0$ and $N < D$ by Proposition S.2.

Part (i): An increase in $c'(q^*)$, holding q^* and $p'(q^*)$ fixed, increases D and leaves N unchanged, so β^* weakly decreases. For $-p'(q^*)$, write $1/\beta^* = 1 + \left(c'(q^*)(q^*)^2 - \pi_0^u\right)/\left(-p'(q^*)(q^*)^2 - \pi_0^d\right)$: an increase in $-p'(q^*)$ increases the denominator of the second term, so $1/\beta^*$ decreases and β^* increases.

Part (ii): By the quotient rule:

$$\frac{\partial \beta^*}{\partial \pi_0^d} = \frac{(-1)\,D - N\,(-1)}{D^2} = \frac{N - D}{D^2} = \frac{-\left(\pi^{u*} - \pi_0^u\right)}{D^2} < 0, \qquad \frac{\partial \beta^*}{\partial \pi_0^u} = \frac{0 \cdot D - N\,(-1)}{D^2} = \frac{N}{D^2} > 0,$$

where the signs use $\pi_0^u < \pi^{u*}$ and $\pi_0^d < \pi^{d*}$. □

S.8 Determining the Source of Vertical Distortions

As in the main text, we impose that the equilibrium markup and markdown are nonnegative directly in the bargaining problem.

Nonnegative Markup/Markdown Constraint. *The parties bargain over w subject to the constraint that the equilibrium markup and markdown are nonnegative. Under each conduct model,* *the constrained simultaneous bargaining problem is:*

$$\text{Monopolistic:} \quad \begin{cases} \max_{w} \ \left(\pi^d - \pi_0^d\right)^\beta \left(\pi^u - \pi_0^u\right)^{1-\beta} \quad s.t. \quad w \geq mc(q) \\ mr(q) = w \end{cases}$$

$$\text{Monopsonistic:} \quad \begin{cases} \max_{w} \ \left(\pi^d - \pi_0^d\right)^\beta \left(\pi^u - \pi_0^u\right)^{1-\beta} \quad s.t. \quad w \leq mr(q) \\ mc(q) = w \end{cases}$$

Under each conduct model, one of the two inequalities holds with equality through the conduct FOC, and the other reduces to the same condition: under monopolistic bargaining $w = mr(q)$, so the nontrivial constraint is $w \geq mc(q)$, i.e., $mr(q) \geq mc(q)$; under monopsonistic bargaining $w = mc(q)$, so the nontrivial constraint is $w \leq mr(q)$, i.e., $mc(q) \leq mr(q)$. In both cases the constraint is equivalent to $q \leq q^*$.

Theorem S.2 (Conduct Selection Under Simultaneous Timing)**.** *Suppose the conditions of Theorem S.1 hold and $\pi_0^d < \pi^{d*}$, $\pi_0^u < \pi^{u*}$, so that $\beta^* \in (0,1)$. Under the nonnegative markup/markdown constraint, at any $\beta \in [0,1]$, the vertical distortion exists only under monopolistic conduct if $\beta < \beta^*$ and only under monopsonistic conduct if $\beta > \beta^*$.*

Proof. Monopolistic conduct. The D-FOC gives $w = mr(q)$, so $\Delta^d = 0$ automatically, and the nontrivial constraint is $q \leq q^*$.

- If $\beta \leq \beta^*$: By Theorem S.1(ii), $q^{mp}(\beta)$ is increasing in β, and by Proposition S.2, $q^{mp}(\beta^*) = q^*$. Hence $q^{mp}(\beta) \leq q^*$, so $w = mr(q^{mp}) \geq mr(q^*) = mc(q^*) \geq mc(q^{mp})$: the constraint is slack, and the unconstrained equilibrium—whenever it exists, which by Lemma S.2 is guaranteed for $\beta \in (\underline{\beta}^{mp}, \beta^*]$—prevails with a vertical distortion ($\mu^u > 0$ for $\beta < \beta^*$). For β below the existence range, monopolistic bargaining has no equilibrium, with or without the constraint.

- If $\beta > \beta^*$: By Lemma S.2, the unconstrained FOC solution exists for every $\beta \in (\beta^*, 1)$ and satisfies $q^{mp}(\beta) > q^*$ by the same monotonicity, implying $w = mr(q^{mp}) < mr(q^*) = mc(q^*) < mc(q^{mp})$, which violates $w \geq mc(q)$. The constraint therefore binds: $w = mc(q)$ together with $w = mr(q)$ forces $q = q^*$ and $w = w^*$. Both parties earn strictly above their disagreement payoffs at (w^*, q^*) since $\pi^{d*} > \pi_0^d$ and $\pi^{u*} > \pi_0^u$. No distortion: $\mu^u = \Delta^d = 0$.

Monopsonistic conduct. The U-FOC gives $w = mc(q)$, so $\mu^u = 0$ automatically, and the nontrivial constraint is again $q \leq q^*$.

- If $\beta \geq \beta^*$: By Theorem S.1(i), $q^{ms}(\beta)$ is decreasing in β, and $q^{ms}(\beta^*) = q^*$. Hence $q^{ms}(\beta) \leq q^*$, so $w = mc(q^{ms}) \leq mc(q^*) = mr(q^*) \leq mr(q^{ms})$: the constraint is slack, and the unconstrained equilibrium—whenever it exists, which by Lemma S.1 is guaranteed for $\beta \in [\beta^*, \bar{\beta}^{ms})$—prevails with a vertical distortion ($\Delta^d > 0$ for $\beta > \beta^*$). For β above the existence range, monopsonistic bargaining has no equilibrium, with or without the constraint.

- If $\beta < \beta^*$: By Lemma S.1, the unconstrained FOC solution exists for every $\beta \in (0, \beta^*)$ and satisfies $q^{ms}(\beta) > q^*$, implying $w = mc(q^{ms}) > mc(q^*) = mr(q^*) > mr(q^{ms})$, which violates $w \leq mr(q)$. The constraint binds: $w = mr(q)$ together with $w = mc(q)$ forces $q = q^*$ and $w = w^*$, with both participation constraints satisfied strictly. No distortion.

In sum, for $\beta < \beta^*$ monopolistic conduct operates unconstrained (with distortion $\mu^u > 0$) while monopsonistic conduct is constrained to the joint-profit maximizing outcome; for $\beta > \beta^*$ the roles are reversed; and at $\beta = \beta^*$ both models yield q^* with no distortion. □

Remark. When $\pi_0^d \geq \pi^{d*}$ or $\pi_0^u \geq \pi^{u*}$, the corner convention $\beta^* = 0$ or $\beta^* = 1$ applies. In these cases, the conduct model whose constraint would bind cannot reach (w^*, q^*), because at q^* one party earns less than its disagreement payoff; that conduct model then has no equilibrium under the nonnegative markup/markdown constraint, while the other conduct model operates unconstrained at every β, in line with the statement of Theorem S.2 evaluated at the corner value of β^*.

Proposition S.3 (Conduct Identification via the Input Price)**.** *Under the conditions of Theorem S.2:*

(i) Equilibrium output satisfies $q \leq q^*$ under both conduct models, with equality if and only if the markup/markdown constraint binds or $\beta = \beta^*$.

(ii) Under monopolistic bargaining, $w \geq w^*$, with equality if and only if $q = q^*$.

(iii) Under monopsonistic bargaining, $w \leq w^*$, with equality if and only if $q = q^*$.

Therefore, vertical conduct is identified by the sign of $w - w^*$: input prices above w^* are consistent only with monopolistic conduct, and input prices below w^* only with monopsonistic conduct.

Proof. Part (i): When the constraint is slack, the unconstrained equilibrium satisfies $q^{mp}(\beta) \leq q^*$ for $\beta \leq \beta^*$ and $q^{ms}(\beta) \leq q^*$ for $\beta \geq \beta^*$, by the monotonicity of output in

β (Theorem S.1) and $q^{mp}(\beta^*) = q^{ms}(\beta^*) = q^*$ (Proposition S.2). When the constraint binds, $q = q^*$ exactly (Theorem S.2).

Part (ii): Under monopolistic bargaining, $w = mr(q^{mp})$. Since $q^{mp} \leq q^*$ by part (i) and $mr'(q) < 0$:

$$w = mr(q^{mp}) \geq mr(q^*) = w^*,$$

with equality if and only if $q^{mp} = q^*$.

Part (iii): Under monopsonistic bargaining, $w = mc(q^{ms})$. Since $q^{ms} \leq q^*$ and $mc'(q) > 0$:

$$w = mc(q^{ms}) \leq mc(q^*) = w^*,$$

with equality if and only if $q^{ms} = q^*$. □

Remark. As in the sequential analysis, $|w - w^*|$ measures the severity of the vertical distortion: under monopolistic bargaining $w - w^* = mr(q^{mp}) - mr(q^*)$ is strictly increasing in the output distortion $q^* - q^{mp}$, and symmetrically under monopsonistic bargaining. Part (i) also provides the testable implication $q \leq q^*$ of the nonnegative markup/markdown constraint under simultaneous timing.

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